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DETAIL SPECIFICATION SHEET

TECHNICAL CONTENT DEVELOPMENT REQUIREMENTS FOR COMBAT SYSTEM TECHNICAL OPERATIONS MANUALS (CSTOMS); HULL, MECHANICAL, AND ELECTRICAL (HM&E) SYSTEM AND EQUIPMENT MANUALS; ELECTRONIC (INCLUDING SERVICE TEST ELECTRONIC, EXPERIMENTAL ELECTRONIC, AND INTERIOR COMMUNICATION [IC]) SYSTEM AND EQUIPMENT MANUALS; AND WEAPON SYSTEMS AND WEAPON EQUIPMENT MANUALS

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product specified herein shall consist of this specification sheet and MIL-DTL-24784.

1. SCOPE

1.1 Scope. This specification sets forth content requirements for the preparation of technical manuals (TMs) covering description, operation, maintenance, installation, and parts support for Combat System Technical Operations Manuals (CSTOMs); Hull, Mechanical, and Electrical (HM&E) Systems and Equipment Manuals; Electronic (including Service Test Electronic, Experimental Electronic, and Interior Communications [IC]) System and Equipment Manuals; and Weapon Systems and Weapon Equipment Manuals. The requirements are applicable for the output of paper or for the display of page-oriented, linear, and non-linear TMs on an Electronic Display System (EDS).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC 9832 AREA TMSS

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24784	-	Manuals, Technical: General Acquisition and Development Requirements, General Specification for
MIL-DTL-24784/23	-	Illustrated Parts Breakdown (IPB) Requirements

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

OP 1700	-	Standard Fire Control Symbols
SE000-01-IMB-010	-	Navy Installation and Maintenance Book
ST000-AB-GYD-010	-	Portable Electrical/Electronic Test Equipment (PEETE); Stowage Guide

(Copies of these documents are available online via Technical Data Management Information System [TDMIS] at https://mercury.tdmis.navy.mil/ by searching for the publication number. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial [805] 228-0669. These documents are available for ordering via the Naval Logistics Library (NLL) at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

NAVAL SURFACE WARFARE CENTER (NSWC) CORONA DIVISION

NAVSEA Test, Measurement, and Diagnostic Equipment (TMDE) Index

(Copies of this document are available from NSWC Corona Division, Detachment Seal Beach, Strategic Systems Evaluation Division [MS50], Test Systems Support Branch MS532, Bldg, 112, 800 Seal Beach Blvd., Seal Beach, CA 90740-5000; by calling 562-626-7959 or DSN 873-7959; or by FAX [562] 626-7831; or contact the Navy Distance Support [NAVY311] by calling 1-855-NAVY311 [1-855-628-9311]/DSN 510-NAVY311 [510-628-9311]; on the web at http://www.navy311.navy.mil (unclassified) or https://www.navy311.navy.smil.mil (classified); or via email at Navy311@navy.mil or Navy311@navy.smil.mil(NAVY311).)

NAVAL WARFARE DEVELOPMENT COMMAND (NWDC)

Fleet Exercise Publication FXP2	-	Antiair Warfare (AAW) Exercises
Fleet Exercise Publication FXP3	-	Strike Warfare (STW), Surface Warfare (SUW), Intelligence (INT), Command and Control Warfare (C2W), and Command, Control and Communications (CCC) Exercises

(Copies of these documents are available on the Navy Doctrine Command SIPRNET site at http://ndc.navy.smil.mil.)

OPNAV INSTRUCTIONS

OPNAVINST 4790.4	-	Ships' Maintenance and Material Management System Policy

(Copies of this document are available online at http://doni.daps.dla.mil/default.aspx.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS, INC. (IEEE)

IEEE 991 - Logic Circuit Diagrams

(Copies of this document are available online at www.ieee.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Development. This specification provides the technical content data requirements necessary for the development of NAVSEA TMs described in 1.1 (see 6.2). This specification contains some technical content requirements that may not be applicable to all of the TMs described in 1.1. Selective application and tailoring of requirements shall be accomplished as necessary. The applicability of some requirements is also designated by one of the following statements: unless otherwise specified by the acquiring activity; or as/when specified by the acquiring activity.

3.2 Maintenance coverage. Unless otherwise specified by the acquiring activity (see 6.2), each type of TM shall contain, in detail, the maintenance coverage prescribed for the applicable maintenance level(s) based on the maintenance concept or approved maintenance plan.

3.3 Technical content requirements. Technical content may be organized in chapter format or in modular format.

3.3.1 Chapter format. For chapter format TMs, technical content shall be arranged as follows.

3.3.1.1 System TMs.

a. Chapter 1 - General information and safety precautions
b. Chapter 2 - Systems functional description
c. Chapter 3 - Conditions of readiness
d. Chapter 4 - Operation
e. Chapter 5 - Maintenance
f. Chapter 6 - System/equipment installation (when included)
g. Chapter 7 - Operational checkout and troubleshooting
h. Chapter 8 - Illustrated parts breakdown (IPB) (when included)

3.3.1.2 Equipment TMs.

a. Chapter 1 - General information and safety precautions
b. Chapter 2 - Equipment functional description
c. Chapter 3 - Operation
d. Chapter 4 - Maintenance
e. Chapter 5 - System/equipment installation (when included)
f. Chapter 6 - Operational checkout and troubleshooting
g. Chapter 7 - IPB

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3.3.1.3 CSTOMs.

a. Chapter 1 - General information and safety precautions

b. Chapter 2 - Combat system introduction

c. Chapter 3 - Combat system description

d. Chapter 4 - Combat system operational description

e. Chapter 5 - Combat system readiness assessment

f. Chapter 6 - Fault detection and impact evaluation

g. Chapter 7 - Fault isolation

3.3.2 Modular format information. For modular format TMs, technical content information shall be organized into modules that are grouped according to the type of information being documented.

3.3.2.1 Technical content categories. Categories of technical content requirements shall be grouped as follows:

a. Descriptive information consists of the following:

(1) General.

(2) Supporting.

(3) Descriptive.

(4) Functional description.

b. Procedural information consists of the following:

(1) Operation.

(2) Maintenance.

(3) System/equipment installation.

(4) Operational checkout and troubleshooting.

c. IPB.

d. CSTOM information consists of the following:

(1) Combat system introduction and description.

(2) Combat system description.

(3) Combat system operational description.

(4) Combat system readiness assessment.

(5) Fault detection and impact evaluation.

(6) Fault isolation.

3.3.2.2 General content requirements. Each module shall include the following as applicable to the technical content being developed:

a. Reference publications list.

b. Field and factory changes list.

c. Safety information.

d. Additionally, procedural modules shall include the following as applicable to the technical content being developed:

(1) Special tools list.

(2) Materials list.

(3) Required personnel information.

(4) Required conditions list.

(5) Environmental conditions list.

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3.4 General information. The information developed shall be so that command level, supervisory personnel, and other users having a general interest in the equipment can easily and rapidly determine the purpose, physical and functional characteristics, interface requirements, and the operational capabilities of the system/equipment. As applicable, the general information shall contain a general introduction and model differences information.

3.4.1 General introduction. An introduction shall be developed for all TMs. The primary purpose of the introduction is to provide information necessary to use the data provided in the TM as effectively as possible. The introduction shall include the following information:

a. Purpose and scope. The purpose and scope of the manual, including the subject matter being covered and any superseding data, if applicable.

b. Description and designated nomenclature. The designated nomenclature and a brief description of the end item. The introduction may include an illustration (frontispiece) (see 3.7.8.2.1 of MIL-DTL-24784) of the equipment. Applicability (including ship and shore) of the TM including capabilities, subsystem or equipment mission, subsystem or equipment interface description, models, serial numbers, and configuration covered shall also be included. As applicable, a consolidated list of the ship hull numbers (by ship class, if applicable), model numbers, serial numbers, and part numbers covered by the TM shall be included and identified as the configuration identification list.

c. Model differences. When applicable, the differences between models of the equipment shall be briefly delineated. Descriptions of major physical differences shall be supported by reference to appropriate illustrations. These major differences shall also be reflected in the operating instructions, maintenance instructions, and parts list. Multiple model differences that involve a number of equipment sections or subsections shall be presented in tabular form. The method for handling differences between models in other portions of the TM shall be defined here.

d. Effectivities. If applicable, an explanation of the effectivities used throughout the TM.

e. Manufacturer Commercial and Government Entity (CAGE) codes. The introduction shall include the following statement: "Manufacturer CAGE codes. Additional information regarding CAGE codes including manufacturer name and address can be found at http://www.dlis.dla.mil/cage_welcome.asp."

f. Multi-volume TMs. The first volume of a multi-volume TM shall contain general information that is applicable to all volumes and shall list all volumes of the TM set.

g. Distance support and Technical Manual Deficiency/Evaluation Report (TMDER). For chapter format TMs, this information is covered in the Foreword (see A.2.3.2.1.3 of MIL-DTL-24784). For modular format TMs, the text in figure 1 of MIL-DTL-24784 shall be included verbatim.

h. Warranty information. As applicable, warranty and guarantee information shall be included as stated in the contract (see 6.2) and shall be in accordance with the Federal Acquisition Regulations (FAR), including hardware modifications such as ship alterations, engineering change proposals, machinery alterations, ordnance alterations, field changes, and so forth.

i. Other specific introductory information. Additional introductory information related to a specific type of TM shall be included, such as the interface relationship of the TM to other referenced publications and the relationship of the equipment to referenced systems or other equipment.

3.5 Supporting information. The following shall be included as supporting information, as applicable:

a. Safety summary.

b. Reference publications.

c. Special tools and test equipment.

d. Materials required.

e. Equipment modification.

f. Equipment, accessories, and documents supplied.

g. Equipment, accessories, and documents not supplied.

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3.5.1 Safety summary. All TMs containing dangers, warnings, or cautions shall have a safety summary. See 3.7.5 of MIL-DTL-24784 for safety and health requirements. This information shall describe all safety precautions necessary for the protection of personnel and the ship including the hazards associated with system operation and maintenance. The following information shall be included, as applicable.

3.5.1.1 Introduction. This description shall orient system supervisory personnel and shall include the purpose, scope, and organization of the system safety instructions and provide basic safety concepts and responsibilities for safety. Inclusion of general guidance does not preclude the need for a DANGER, WARNING, or CAUTION within the TM. An excessively long safety summary is discouraged. See 3.7.5.10 of MIL-DTL-24784 for information to be included. Figure 1 shows sample content in modular (Information Module/Information Package [IM/IP]) format. For chapter format TMs, this information is covered in the front matter (see A.2.3.2.1.7 of MIL-DTL-24784).

3.5.1.2 Electromagnetic radiation hazards and precautions. If applicable, describe the radiation hazards to topside personnel and the precautions to be taken. The description shall include discussions of the following:

a. Locations of topside and in-board radiation hazardous areas.

b. Minimum safe distances on the axis of beam radiation.

c. Precautions to be taken when entering areas of radiation hazard such as the wearing of copper-screen goggles.

d. The effect of radiation on flammable or explosive material such as induction of radio frequency (RF) currents in metals, sparking, and the consequent possibility of igniting flammables or detonating explosives. The text shall be supported by one or more illustrations identifying the areas of radiation hazards, the location of antennas, and so forth.

3.5.1.3 Hazards to divers. When applicable, the description of hazards to divers shall include, but shall not necessarily be limited to, the following:

a. The nature and intensity of sound energy in water as related to divers.

b. Precautions to be taken when working near operating equipment.

c. Illustrations identifying the hazardous areas and locations of sonar domes.

3.5.1.4 System hazards and precautions. Descriptions of system hazards and precautions shall be included, addressed to system personnel, and referenced to particular system equipment. The descriptions shall be organized to be consistent with the operation of the system. The descriptions shall supplement and extend equipment safety instructions to the system level by warning of potential hazards that can be caused during operation or maintenance.

3.5.1.5 Operational safety summary. A summary shall be included that emphasizes the proper use of equipment controls; describes the hazards to operators, or as applicable, the hazards to persons in areas remote from the operation; and recommends precautions.

3.5.1.6 Maintenance safety summary. A maintenance safety summary shall emphasize the proper use of controls, describe the hazards to maintenance personnel, potential damage to the equipment, and recommend precautions.

3.5.1.7 Hazardous components. Hazardous components, including radioactive devices and elements used with the system, shall be identified and briefly described.. The general handling precautions for such components shall be summarized. The description of a hazardous component shall include brief statements as to the purpose, manner of functioning, nature of built-in safety devices, and nature of the hazardous element. It shall also indicate the relative sensitivity of the component to mechanical shock, vibration, electromagnetic and radioactive radiation, and electrostatic charges.

3.5.2 Reference publications. A tabular listing of the manuals that pertain to system and subsystem equipment and other documents of interest or use to operating or maintenance personnel, such as basic training manuals, special directives, OP 1700, and manuals for associated systems equipment, shall be included. The list of publications shall include the title and publication number of the referenced publications.

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3.5.3 Special tools and test equipment. A tabular listing of all special tools, tool kits, test equipment, miscellaneous parts, and Government-furnished items that form a part of, or are supplied (or not supplied) with the system or equipment shall be included. Items shall be listed alphabetically by their official name or nomenclature. Notes explaining the use of the tables and stock code numbering system shall precede the first table for which a particular note is applicable. For each listed item, the table shall include:

a. Item name consisting of the official name or nomenclature of the listed item.

b. If applicable, a description of its general use and a reference to the illustration contained in the IPB.

c. Quantity of the items required for maintenance of the equipment.

d. Manufacturer's part number consisting of the manufacturer's or the contracting activity's assigned part number as it appears on the item.

e. The CAGE code consisting of the assigned CAGE code of the manufacturer or contracting activity of the listed item.

f. The stock code number consisting of an assigned number that indicates the procurement status of the listed item as follows:

(1) Item is supplied with the equipment.

(2) Item is supplied by the manufacturer.

(3) Item is available from the manufacturer.

3.5.3.1 Equipment and publications required but not supplied (electronic systems and equipment and CSTOM only). A tabular listing of all test equipment and publications required but not supplied with the equipment or system shall be included. The NAVSEA Test, Measurement, and Diagnostic Equipment (TMDE) Index shall be used as a guide for the selection and application of test equipment required for shipboard use. If a measurement requirement is established that cannot be satisfied by the subcategory (SCAT) codes listed in this guide, or alternate test procedures cannot be developed utilizing the listed SCAT, request for assistance should be directed to:

Commander
Naval Sea Systems Command
ATTN: SEA 05S
1333 Isaac Hull Avenue, SE
Stop 5160
Washington Navy Yard, DC 20376-5160

The listing shall contain:

a. SCAT code. When applicable, this column shall contain the four-digit group of numbers, which is included in the NAVSEA TMDE Index, used to identify a measurement requirement.

b. Test equipment category. This column shall contain the item name of each item of equipment required.

c. Representative test equipment model number. This column shall contain the model number of the standard or substitute standard general purpose electronic test equipment, listed in the NAVSEA TMDE Index, as applicable, which can be used to satisfy the measurement parameters.

d. Equipment test parameters. This column shall specify the range of test parameters that shall be satisfied by this test equipment item (it shall not define the ranges of the test equipment model).

e. Application. This column shall indicate the intended use of the test equipment (e.g., troubleshooting, maintenance, or installation).

f. Publication number and title. This column shall specify the publication by number and title.

3.5.4 Materials required. When applicable, this information shall provide a list of all materials (consumable materials and expendable items) required to perform maintenance type procedures. Items shall be listed in alphabetical sequence by noun nomenclature. The materials required list shall be prepared in accordance with the following guidelines:

a. Each material entry in the list shall be identified by "Nomenclature" and "Specification, Part Number, or Not Specified" as applicable. If the materials are authorized for use at only certain maintenance level(s), the restriction shall be indicated by the use of an O, I, or D.

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b. Unless alternate identification is approved by the acquiring activity, materials shall be listed by Government specification in parentheses following the item nomenclature.

c. When more than one of the same item is required, the quantity shall follow the nomenclature in parentheses.

d. Parts that require mandatory replacement in the procedure, such as preformed packing, shall be listed.

3.5.5 Equipment modification. Information shall be developed to include all equipment modification change data for the system/equipment covered by the TM. Field changes, factory changes, engineering changes or notices, modifications, and so forth shall be identified and included as follows:

a. Change number. This column shall list the change number for each field and factory change considered and included in the preparation of the TM.

b. Nomenclature. This column shall list the equipment nomenclature and Government serial numbers of equipment affected by the change.

c. Description. This column shall contain a brief statement identifying the change and its purpose.

3.5.6 Equipment, accessories, and documents supplied. As applicable, a tabular listing of all equipment, accessories, and other documents supplied shall be provided. List the equipment (other than test equipment), its units, and its accessories. The table shall include the following:

a. Quantity. This column shall contain the quantity of each unit and accessories supplied with the equipment.

b. Item name or nomenclature. This column shall contain the official name (e.g., pump, winch) or nomenclature (name and designation) of each component, unit, or accessory. For documents, list the document title.

c. Part identification number (PIN) or component identification number (CID), repairable identification code (RIC), or unit number. This column shall contain the PIN or CID, RIC, or unit number of each equipment unit or accessory. For documents, list the document identification number.

d. Overall dimensions. This column shall contain the crated (if available) and uncrated height, width, and depth in inches (or inches and centimeters) of each unit, accessory, or document.

e. Weight and volume. This column shall contain the crated (if available) and uncrated weight and volume in cubic feet of each unit, accessory, and document.

3.5.7 Equipment, accessories, and documents not supplied. As applicable, information shall be provided describing equipment and accessories (such as controls, foundations mounts, couplings, and accessories) and related publications, which are not furnished with the basic equipment covered by the TM, but that attach or relate importantly to the basic equipment. The data shall be in sufficient detail to establish the correlation of the equipment with respect to physical and functional interfaces. If applicable, a reference should be made to the shipbuilder's power plant TM if it contains additional applicable information.

3.6 Descriptive information. Descriptive information shall be developed and divided as described in 3.6.1 through 3.6.5, as applicable.

3.6.1 Physical arrangement (HM&E systems, weapon systems, and electronic systems only). For systems, all system areas and compartments shall be described and the system equipment and units contained in the areas shall be listed. The physical arrangement description shall be supported by the following illustrations:

a. An inboard profile drawing of the ship or stations showing compartment locations and identifying topside equipment and equipment units comprising the system (see figure 2).

b. Separate illustrations of each compartment and area identifying the listed system equipment (see figure 3). Other equipment, which is installed in the subject system compartments and areas, need not be listed in the text or called out in the illustrations if they do not directly affect the operation or maintenance of the subject system.

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3.6.2 System description (HM&E systems, weapon systems, electronic systems, and weapon equipment only). System description information shall describe the system and its relationship to associated systems, subsystems, and equipment. For other than weapon systems and subsystems or weapon equipment, the descriptions shall combine the physical and general aspects of the system and associated systems contained therein and their interdependencies and interactions. Each of the equipment comprising the system shall be identified and described. Descriptions of operator-attended equipment shall include general statements as to the nature and purpose of units and indicators. The text shall be supported by illustrations. All equipment shall be shown, whenever possible, in relative-scale proportion. Equipment may be separately illustrated with significant features called out, if such details are necessary for proper support of the text. When required, descriptions and illustrations of associated system equipment shall be limited to the major units thereof. The descriptions of associated system equipment shall be more condensed than those of the subject system equipment; otherwise, the same requirements are applicable. In the descriptions, emphasis shall be placed on those associated system equipment that constitute operational or functional interfaces with the subject system and shall be included in the illustrations.

a. For weapon systems and subsystems, the general description shall identify each subsystem group of weapon system equipment and each unit equipment in a group by composite illustrations of the subsystem group of the first ship of the class, for example the fire control subsystem. The text shall only describe significant features of a particular component of an equipment in a group. Significant features would comprise such items as console panels, sub-panels, and control and indicator groups. Each component of an equipment in a group shall be called out as shown on figure 4. Associated systems equipment shall be shown in separate group illustrations similar to figure 4 except that those major components that directly relate to the weapon system operation or maintenance shall be more detailed than the other components. Descriptions shall be limited to major components that constitute operational and functional interfaces with the weapon system. Examples of such components include search radar distribution switchboard and radar order switches that control the availability of weapon system planned-position indicator (PPI) displays and units of the naval tactical data system that control the transmission of target data to and from the weapon system.

b. For weapon equipment, if the equipment operates as part of a system, a system description shall be included. This shall be a non-technical orientation of the subject equipment with other equipment in the system. The system description shall discuss only those components of the system necessary to introduce the subject equipment's system-contributing capabilities. An illustration shall be used to clarify the relationships (figure 5 is a typical equipment/system integration illustration).

3.6.3 Equipment and component description (HM&E equipment and electronic equipment only). This information shall identify and describe the equipment and components. The description shall include the intended use (why, where, when, how, and with what), capabilities, and limitations of the equipment. Descriptions of operator-attended equipment shall include general statements as to the nature and purpose of units and indicators. The text shall be supported by illustrations. Text covering physical description or structural arrangement shall be brief, with special attention given to avoiding unnecessary or repetitious details that are easily illustrated. All equipment shall be shown, whenever possible, in relative-scale proportion. Equipment may be separately illustrated with significant features called out if such details are necessary for proper support of the text. Assembly and detail drawings as appropriate may be referenced. If the TM covers more than one equipment configuration, a table defining the differences shall be included.

3.6.4 Software description. This information shall be developed to provide descriptions for programming software used for systems, equipment, and test equipment, when applicable. Block diagrams, test setup diagrams, and pictorials of display readouts shall be included when necessary to support the descriptive text.

3.6.5 System/equipment characteristics/capabilities. When specified by the acquiring activity (see 6.2), information shall be prepared containing the leading particulars or technical characteristics (reference data) to present the physical, mechanical, and electrical characteristics of the system, subsystem, equipment, or major functional components. The following shall be included:

a. Official nameplate designations and nomenclature (such as manufacturer, type, model, part number, CID, joint electronics type designation, RIC, and allowance parts list [APL]) of all equipment, major units, and support equipment. This list shall include common names and quantities required per unit, if available.

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b. Detailed operating characteristics and significant equipment capabilities and limitations such as pounds of thrust, knots, turning radius, minimum and maximum ranges, degree of coverage, resolution, accuracy, wattages, voltages, horsepower, gallons per minute, input and output signals, load limits, frequency, testable items, resolution, memory capacity, reaction time, transfers per hour, boom capacity, data handling capability, and number of channels.

c. Power input requirements and other externally supplied inputs such as data and control signals, pressurized air, and coolant water shall be listed, along with identity of the source of each, such as ships supply, specific associated equipment, gyrocompass, power requirements, horsepower, pressure, capacity, modes of operation, power output, frequency, pulse characteristics, sensitivity, and selectivity, including tolerances, where applicable.

d. Environmental characteristics such as ambient temperatures, heat dissipation per unit, humidity limits, and coolants (e.g., airflow, water or oil flow rate, chilled and demineralized sea water). Measurements provided shall match the installed equipment.

e. In addition, for weapon systems, a description of the system capabilities and how these enable the ship to perform the primary mission. Subsystem and equipment capabilities shall be described only to the extent necessary to describe system capabilities. Capabilities of associated systems shall be described as applicable. Data such as search radar detection ranges, fire control radar acquisition and tracking ranges, number of targets that can be simultaneously evaluated, number of targets that can be simultaneously engaged, and minimum and maximum effective ranges of missiles and guns shall be included. Envelope drawing(s) shall be developed to illustrate system capabilities (see figure 6). Such drawing(s) shall meet the requirements for training documents and shall be technically accurate for use as a guide for system evaluation training exercises.

3.7 Functional description. Functional description information shall be developed and divided as described in 3.7.1 through 3.7.6, as applicable.

3.7.1 HM&E systems functional description. Functional description for HM&E systems shall describe how the components or equipment comprising the system jointly perform major operations and functions and how associated systems contribute to the performance of these major functions. The description of the entire system shall be supported by a basic block diagram of the system (see figure 7). Each major function shall then be discussed separately at progressively increasing levels of detail supported with overall and functional block diagrams (FBDs). Where information can be better presented pictorially than by text, additional diagrams and other illustrations shall be used. Reference may be made to diagrams provided for troubleshooting. Equipment or associated system interfaces shall be described only as necessary to identify the sources or destinations of system inputs and outputs. Functional description data may be divided as necessary to facilitate usability of the information.

3.7.1.1 Introduction. This information shall describe the general approach that is used in the functional description. The introduction shall also describe briefly the interrelationship between the system and associated systems.

3.7.1.2 System function directory. This information shall include a system function directory tabulating the operation control functions and the signal data described in the detailed level of functional description (see 3.7.1.4). The tabulation shall include the following information, as applicable:

a. Official name of the function, common name, and symbol.

b. Type of control or signal (alternating current frequency and voltage, direct current polarity and voltage, hydraulic pressure, mechanical motion, synchro, and so forth).

c. The origin and termination of the control or signal.

d. Identify equipment (e.g., relay transmitters, coordinate converters, distribution boxes, switches) between the origin and termination of the output control or signal.

e. A reference to supporting illustrations.

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3.7.1.3 Simplified functional description. Information shall be developed for system simplified functional descriptions. The simplified functional description shall be confined to data such as origin of the major functions at an equipment control or sensor; transmission of the signal via intermediate equipment such as switchboards, relay transmitters, coordinate converters, and distribution boxes; and presentation of the function at terminal equipment. Control functions essential to the development of a signal shall be introduced and briefly described in their relationship to the signal. The description shall be supported by electronic, electrical, and mechanical system type block diagrams, as applicable (see MIL-DTL-24784 for examples of block diagrams). Functions involving computations may be explained in mathematical terms, but at a level no higher than high school mathematics; that is, algebra, plane geometry, and trigonometry (through sine, cosine, and tangent).

3.7.1.4 Detailed functional description. Information shall be developed for system detailed functional descriptions. The detailed-level description shall explain the system major functions in terms directly related to the diagrams provided for troubleshooting. Data and control functions shall be described down to the level of an equipment or an equipment group.

3.7.2 HM&E equipment functional description. Functional description information for HM&E equipment describing how the equipment operates shall be provided. Major assemblies of the equipment and interaction on major subassemblies shall be described. A detailed mechanical and electrical functional description, including descriptions of power distribution, power supplies, and regulators, shall be provided. Simplified electrical and electronic schematic diagrams, piping diagrams, and mechanical schematic diagrams shall support the text, as applicable. The functional description of digital equipment shall be similar to that for conventional equipment. However, the functional and hardware makeup of digital equipment requires a variation in the method of presentation and descriptive requirements. An overall description of the functional relationship of the logic sections, units, and assemblies comprising the equipment shall be provided. The functional description information shall be presented in successive levels of increasing detail as described in 3.7.2.2 through 3.7.2.4.

3.7.2.1 Introduction. This information shall describe the general approach that is used in the functional description. The introduction shall also briefly describe the interrelationship between the equipment and associated components.

3.7.2.2 Simplified functional description. This information shall provide narrative text that supports and refers to the overall FBD. All major functions such as transmit, receive, control, display, power distribution, cooling, and so forth shall be described at this level.

3.7.2.3 Detailed functional description. This information shall describe the development of each major function at the level of detail shown on its related signal flow diagram, logic diagram, or other diagram provided in troubleshooting. The text shall support and refer directly to the diagrams. The following shall be included, as applicable:

a. An introduction to and description of the signal characteristics:

(1) Signal levels or bi-stable states utilized by the equipment; that is, true (1) and false (0), and their relative voltage levels.

(2) Interpretation of the word-code bit structure; that is, address, instruction, or data bit codes.

(3) Signals identified by their operational accomplishment.

b. Overall and FBDs, and associated descriptions.

c. Logic principles shall be described beginning with an introduction to the basic digital logic symbology used in the manual. Each logic function shall be described and supported by Boolean equations, truth tables, simplified logic diagrams, and timing diagrams.

d. Functional description of power distribution, power supplies, and regulators.

3.7.2.4 Integrated circuits and micro-miniature capsules. This information shall provide a brief description of the overall functional operation for circuits within different types of non-repairable integrated circuits and micro-miniature capsules. Non-repairable circuits are not required for functional understandings of theory of operation. Simplified schematics in accordance with SE000-01-IMB-010 or logic diagrams showing input and output connections, truth tables, and waveforms shall support the text, as applicable.

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3.7.3 Weapon systems functional description. Major weapon system functions shall be identified and described in functional descriptions. The highest threat level shall be considered the primary mode of weapon system operation. Other modes shall be treated as secondary and, insofar as feasible, as modifications of the primary mode. Thus, functions that are common to all modes shall be described under the primary mode and referred to under the secondary mode, with such modifications as may be necessary. Each function shall be analyzed by describing the origin of the function and by stating how each pertinent equipment contributes to the purpose of the overall system function. Functions shall be covered in terms directly related to the system troubleshooting procedures and diagrams. Each functional analysis shall be presented in two levels, simplified and detailed, so that personnel training based on this approach can support the concept. Descriptions shall be supported by illustrations, which may include block diagrams, geometrical diagrams, and FBDs.

3.7.3.1 Introduction. This information shall briefly describe the system interfaces and explain interrelationships between the subject system and associated systems and equipment. The major system functions shall be introduced briefly and the general approach used in the description and analysis of the major function shall be explained. The introduction shall introduce all major system functions and physical interfaces. Subsequent information shall contain a description and analysis of major system functions. The introduction shall inform the reader that each major function is separately analyzed in two levels: simplified and detailed.

3.7.3.2 Weapon system interfaces. This information shall include descriptions of weapon system interface relationships to associated systems and equipment and shall be supported by interface FBDs that illustrate system integration (see figure 8).

3.7.3.3 Major system functions. All major system functions shall be listed, defined, and included. Major functions shall reflect the operational phases or events of the primary mode of system operation. Secondary system modes, integration with associated systems, multiple-radar synchronizing systems, evaluation instrumentation systems, and other such miscellaneous but primary subjects shall be treated separately as related functions. Each major function shall be thoroughly analyzed from generation to output on the basis of concurrent or sequential sub-functions such as data and control functions. The analytic levels in order of increasing complexity are from simplified to detailed. The detailed level shall be an expansion of, and correlate to, concepts introduced at the simplified level. Each level shall be complete within itself. The selected levels of presentation shall indoctrinate the reader in weapon system technology.

a. Simplified. The objective of the simplified level shall be to describe the major functions in terms understandable by a person possessing fundamental technical knowledge and thus, provide this person with a basis for better understanding of the detailed analysis. The simplified functional analysis shall explain the generation, transformation, interrelation, and use of the data and control functions that constitute the major function. The analysis shall cover essentials such as origin of the function at an equipment control or sensor; transmission of the function via intermediate equipment such as switchboards, relay transmitters, coordinate converters, and distribution boxes; and manifestation of the function at the terminal equipment. Data and control circuitry that are essential and contribute to the operation of a major function shall be introduced and briefly described. Primary emphasis shall be placed on the development of functional concepts in the geometrical reference frame provided by OP 1700 or other manuals provided by the acquiring activity. Only secondary consideration need be given to the mechanization of functions. Functions involving computations may be explained in mathematical terms, but at a level no higher than high school mathematics; that is, algebra, plane geometry, and trigonometry (through sine, cosine, and tangent). Explanations requiring advanced mathematics higher than that expressed on a high school level shall not be presented. The simplified analysis shall be supported by one or more simplified functional diagrams, as appropriate, for the nature and complexity of the function (see figures 9 and 10).

b. Detailed. The objective of the detailed level shall be to describe the major functions in terms directly related to the system troubleshooting procedures and diagrams in the operational checkout and troubleshooting portion of the TM. Accordingly, a detailed functional analysis shall examine the mechanization of data and control functions down to the level of particular and essential circuit components or functional modules. Essential circuit components and functional modules are those that transform data or control the transmission of data and orders. Nonessential circuit components, such as in-line amplifiers, are a practical necessity but do not require detail analysis. Illustration support shall be provided by reference to system FBDs.

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3.7.4 Weapon equipment functional description. Functional descriptions shall provide a detailed analysis of the principles of operations of the overall equipment and its functions. The development of the equipment functions in every mode of operation shall be described. The structure and organization of this data shall parallel the troubleshooting data contained in the TM. The text shall refer to and support the troubleshooting. Simplified and detailed principles of operation shall be provided. The continuity of the simplified and detailed principles of operation shall be such that summarizing is not required for understanding combined functions of the equipment as operated in the intended weapon system application. In instances where the various functional units may, by multiple switching, assume a variety of configurations, coverage shall be accomplished by assuming each of the various possible conditions and describing only the modified portion of the signal path, action, or information flow. Illustrations of alternate configurations shall be limited to that required to integrate signal modification and control.

3.7.4.1 Introduction. This information shall describe the general approach that is used in the functional description. The introduction shall also describe briefly the interrelationship between the equipment and associated components.

3.7.4.2 Simplified functional description. The simplified functional description shall provide an overall explanation of the functional operation of the equipment in relationship to the system of which it is a part. This explanation shall provide an understanding of the way in which the equipment performs functions by modes of operation, placing emphasis on the primary modes, without going into mechanical or electrical circuit details. To ensure continuity, this description shall bridge the gap between the introductory description and the more detailed descriptions to follow. First order equipment-to-system interfaces such as target data accumulation, processing and display, target designation, types of guidance, launcher assignment, and pointing characteristics shall be described. The description shall be supported by an FBD of equipment-to-system interface showing only main data flow from the equipment to related equipment within the system. Figure 11 depicts a typical relationship between equipment and system.

3.7.4.2.1 Simplified functional description data flow. The simplified functional description data flow shall be described in groups of signals between major components within the equipment. Text for subordinate modes of operation shall reference applicable portions of normal mode description. Unique or not commonly known features, engineering concepts, or specialized applications of existing concepts shall be explained only when necessary for an understanding of the functional operation of the equipment. This provision shall apply, for example, to unique features and concepts of self-checking and automatic test programming features of equipment. The text shall be supported with block diagrams. The number of diagrams required for clarity depends upon the variety of outputs and the changes imposed by different operational modes. To avoid redundancy, appropriate references shall be made to diagrams provided for troubleshooting.

3.7.4.3 Detailed functional description. The detailed functional description shall begin with a detailed block diagram presentation of each major functional block presented in the general functional description. The presentation shall clearly indicate what group of signals covered in the general functional description is being expanded in the detailed functional description. For example, data flow on the simplified FBD shall be expanded into individual signals and the generation path of the individual signals further expanded into a detailed block diagram (see MIL-DTL-24784). Outputs shall be identified by names or fire control symbols. Diagrams that are redundant with troubleshooting diagrams contained in the manual, in total or in part, shall not be included. The troubleshooting diagrams in the manual shall be referenced to the detailed functional description as the basic supporting illustrations. Diagram titles shall be synonymous with the heading titles.

a. The description shall first briefly describe the composition of the output by tracing the output back to its origin. This shall illustrate functional dependency of the circuits or mechanisms and shall familiarize the technician with the way in which the signal flow diagrams, when required, are to be used for fault-isolation. It also provides the technician with a summary of the total generation path as background for the more detailed description that shall follow.

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b. Next, the text shall, in effect, “talk through” each of the signal flow diagrams from source to output. These presentations shall be supplemented, as required for completeness, by reference to second-order supporting diagrams contained in the manual. As required for clarity, additional supporting illustrations shall be provided. These may include timing diagrams or a block diagram giving a more condensed view of a particularly complex signal flow diagram (see MIL-DTL-24784). Such additional illustrations shall be consistent with the troubleshooting diagrams with respect to nomenclature, symbology, and reference designations. These diagrams shall clearly label and define source and destination input and output signals and mechanical occurrences, and the time and phase relationships.

c. A detailed analysis shall be provided for areas of new, unusual, or complex design. Descriptions of conventional circuits or devices that are covered in Navy Class A training shall be included only to the depth required to integrate them into the overall analysis. Detailed analysis text shall be supported as appropriate by detailed logic diagrams and partial electrical, electronic, pneumatic, hydraulic, and mechanical schematics (see MIL-DTL-24784). Detailed logic diagrams shall include the level of detail specified in IEEE 991. Detailed mechanical analysis descriptions shall be supported as appropriate by mechanical schematics, gearing, and linkage diagrams. Cutaway views shall show internal parts of complex mechanisms in the detail necessary to ensure comprehension of a detailed mechanical analysis. Callouts shall be used to locate each functionally significant part.

d. Mathematical analysis shall be provided for equipment elements that perform mathematical functions, such as fire control, guidance, and computer equipment, in sufficient detail that shall provide the technician or trainee with a complete understanding of the mechanics of the equipment performing the function. Mathematics shall be limited to the level of high school algebra, geometry, and trigonometry. However, if a higher level of mathematics is required for a valid definition, it shall be presented in the simplest possible terms, preferably including an approximate equivalent expressed in a simpler form. Mathematics shall not be used to justify design or to describe basic circuit elements.

3.7.4.4 Functional elements. This information shall describe the functional elements consisting of electrical, electronic, pneumatic, hydraulic, mechanical, and optical units. The elements shall be described in sufficient detail to provide the technician with a complete understanding of how these elements accomplish their functions. Descriptions of conventional circuits or devices that are covered in Navy Class A training generally shall not be extended below the level of single function stages or individual active elements (e.g., amplifier stage, rectifier, synchro, motor transducer, and digital logic block). However, the learning objectives of the Class A training should be the determining factor for the depth of coverage.

3.7.4.4.1 Detailed functional elements. Depth of coverage, which may be defined as detailed circuit analysis or mechanical analysis, shall be restricted to areas of new or unusual design. When functional elements are new, unusual, or complex in design and; therefore, require more detailed coverage, such coverage may be presented at one time and not in a straightforward sequence with respect to each individual signal flow diagram. In such instances, specific paragraph references shall be provided in logical functional sequence with respect to each individual signal flow diagram. In such instances, specific paragraph references shall be provided in logical functional sequence for all out-of-sequence text material. Typical examples are as follows:

a. Unusually lengthy descriptions of complex functional elements (circuitry or mechanism) should be deferred as amplifying information if such descriptions would tend to obscure the continuity of an overall description.

b. Complex functional elements having identical multiple applications, or one application contributing to two or more output functions, shall be fully described only once and otherwise covered by references to that point in the text.

3.7.5 Electronic systems functional description. The functional description for electronic systems shall describe how the components or equipment comprising the system jointly perform major operations and functions and how associated systems contribute to the performance of these major functions. The description of the entire system shall be supported by a basic block diagram of the system (see figure 7). Each major function shall then be discussed separately at progressively increasing levels of detail supported with overall and FBDs. Where information can be better presented pictorially than by text, additional diagrams and other illustrations shall be used. Reference may be made to troubleshooting diagrams. Equipment or associated system interfaces shall be described only as necessary to identify the sources or destinations of system inputs and outputs.

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3.7.5.1 Introduction. This information shall describe the general approach that is used in the functional description. The introduction shall also briefly describe the interrelationship between the system and associated systems.

3.7.5.2 System function directory. This information shall include a system function directory tabulating the operation control functions and the signal data described in the detailed level of functional description. The tabulation shall include the following information, as applicable:

a. Official name of the function, common name, and symbol.

b. Type of control or signal (alternating current [AC] frequency and voltage, direct current [DC] polarity and voltage, hydraulic pressure, mechanical motion, synchro, and so forth).

c. The origin and termination of the control or signal.

d. Identify equipment (e.g., relay transmitters, coordinate converters, distribution boxes, switches) between the origin and termination of the output control or signal.

e. A reference to where the function is illustrated, including the troubleshooting diagrams.

3.7.5.3 Simplified functional description. The simplified functional description shall be confined to data, such as origin of the function at an equipment control or sensor; transmission of the signal via intermediate equipment such as switchboards, relay transmitters, coordinate converters, and distribution boxes; and presentation of the function at terminal equipment. Control functions essential to the development of a signal shall be introduced and briefly described in their relationship to the signal. The description shall be supported by an FBD. Functions involving computations may be explained in mathematical terms, but at a level no higher than high school mathematics.

3.7.5.4 Detailed functional description. The detailed functional description shall explain the system major functions in terms directly related to the diagrams developed for troubleshooting. Data and control functions shall be described down to the level of an equipment or an equipment group.

3.7.6 Electronic equipment functional description. Information describing the detailed analysis of the functional descriptions of the overall equipment and its functions shall be provided. The development of the equipment functions in every mode of operation shall be described and supported with FBDs. The text shall support and reference the functional troubleshooting diagrams. The functional description of digital equipment shall be similar to that for conventional equipment. However, the functional and hardware makeup of digital equipment requires a variation in the method of presentation and descriptive requirements. An overall description of the functional relationship of the logic sections, units, and assemblies comprising the equipment shall be provided. The functional descriptions shall be presented in successive levels of increasing detail as described in 3.7.6.2 and 3.7.6.3.

3.7.6.1 Introduction. This information shall describe the general approach that is used in the functional description. The introduction shall also briefly describe the interrelationship between the equipment and associated components.

3.7.6.2 Simplified functional description. This information shall provide narrative text that supports and refers to the overall FBD. All major functions such as transmit, receive, control, display, power distribution, cooling, and so forth shall be described at this level.

3.7.6.3 Detailed functional description. This information shall describe the development of each major function at the level of detail shown on its related signal flow diagram, logic diagram, or other diagram provided in troubleshooting. The text shall support and refer directly to the diagrams. The following shall be included, as applicable.

a. An introduction to and description of the signal characteristics:

(1) Signal levels or bi-stable states utilized by the equipment; that is, true (1) and false (0), and their relative voltage levels.

(2) Interpretation of the word-code bit structure; that is, address, instruction, or data bit codes.

(3) Signals, identified by their operational accomplishment.

b. Overall and FBDs and associated descriptions.

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c. Logic principles shall be described beginning with an introduction to the basic digital logic symbology used in the manual. Each logic function shall be described and supported by Boolean equations, truth tables, simplified logic diagrams, and timing diagrams.

d. Functional description of power distribution, power supplies, and regulators shall be in accordance with 3.7.6.4.

3.7.6.4 Detailed circuit analysis. This information shall briefly describe conventional electronic circuits found in SE000-01-IMB-010 and refer to the maintenance schematic diagrams provided in troubleshooting. Detail circuits not covered in SE000-01-IMB-010 shall be described and supported by simplified schematic diagrams. AC and DC power distribution shall be described in detail and supported by reference to the power distribution diagrams provided in the TM troubleshooting. Mechanical devices, cooling systems, and so forth shall be described and supported by reference to mechanical schematic, overall, and FBDs.

3.7.6.5 Integrated circuits and micro-miniature capsules. This information shall include a brief description of overall functional operation for circuits within different types of non-repairable integrated circuits and micro-miniature capsules. Non-repairable circuits are not required for functional understandings of theory of operation. Simplified schematics in accordance with SE000-01-IMB-010 or logic diagrams showing input and output connections, truth tables, and waveforms as applicable shall support the text.

3.7.6.6 Equipment function directory. This information shall include an equipment function directory tabulating operation control functions and the signal data described in the detailed level of functional description. The tabulation shall include the following information, as applicable:

a. Official name of the function, common name, and symbol.

b. Type of control or signal (AC frequency and voltage, DC polarity and voltage, hydraulic pressure, mechanical motion, synchro, and so forth).

c. The origin and termination of the control or signal.

d. Identify components (e.g., relay transmitters, coordinate converters, distribution boxes, switches) between the origin and termination of the output control or signal.

e. A reference to where the function is illustrated, including the fault isolation diagrams.

3.8 Procedural information. Task-oriented procedural data such as operating, maintenance, operational checkout and troubleshooting, and installation data shall be divided into individual, stand-alone units of procedural oriented information. Procedures that can be hazardous to personnel or equipment shall be emphasized by DANGERS, WARNINGS, or CAUTIONS, as applicable (see MIL-DTL-24784).

3.8.1 Operation. The operation information shall be subdivided and developed to contain operating instructions including all the procedures necessary to enable operating personnel to efficiently and effectively use the system or equipment in accomplishing its designated task. These operating instructions shall be in sufficient detail to allow operators, having previous experience in the operation of similar or related equipment, to independently and safely operate the equipment without additional training or explanation. When applicable, at the beginning of each operating procedure, instructions shall include safety precautions to be observed and tools, parts, materials, and test equipment required (see 3.5.3). Procedures that can be hazardous to personnel or equipment shall be emphasized by dangers, warnings, or cautions, as applicable. Operating procedures may have an introduction. In a system TM, equipment-operating instructions shall be general in nature since the detailed operating instructions in the equipment manuals are required to be complete in every available mode or condition (both system interfacing and independent).

3.8.1.1 Introduction. The introduction shall describe each operator's relationship to the equipment and each other and shall identify those units having controls and indicators that the operator is expected to use in the performance of his duties. The intended function and application of the equipment shall be fully explained so that the operators will know exactly what he or they should expect to accomplish with the equipment.

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3.8.1.2 Controls and indicators. Data for the system/equipment hardware controls and indicators shall be provided and shall include a description of all operator controls, indicators, protective devices, and jacks. The following information shall be included in tabular format, as applicable:

a. Names of panel designations as marked on the equipment.

b. Positions and operating functions for each control and the normal operating condition of each indicator in each of the operating functions.

c. The text shall be supported by detailed illustrations (see figure 12).

d. When more than one operator is required to operate the equipment, his designated position and function and the relationship to his controls and indicators shall be specified.

3.8.1.3 Displays and alerts. Data for the system/equipment software displays and alerts shall be provided and shall include a description of all operator monitor displays and audio alerts. The following information shall be included in tabular format, as applicable:

a. Index no. reference, if used.

b. Display/Alert nomenclature.

c. Function.

The purpose of the table is to provide the required data for each display and alert. The table will indicate the index number (when used) referenced from the illustration, the associated display or alert and its nomenclature, including the reference designator, if applicable, and the function of the display or alert.

3.8.1.4 System operating procedures. The system operating procedures shall be prepared in accordance with 3.8.1.4.1 through 3.8.1.4.5.

3.8.1.4.1 Preoperational conditions and setup. Specific preoperational conditions presumed to be in effect prior to system operation shall be established. A system readiness check-off list of significant switch positions and indicator status shall be tabulated. For indicators such as dials, where a band of readings are possible, upper and lower limits shall be delineated. The initial conditions of associated system equipment that directly affect system operation shall be treated in a similar manner. In addition, for weapon systems, loading instructions shall be provided as follows:

a. Initial loading instructions: Provide instructions for the initial loading of the computer program into the computer if the system includes a general-purpose digital computer. All required instructions covering operation of the peripheral input equipment and controls on the computer maintenance panel shall be included to ensure that the applicable program and routine are correctly loaded.

b. Unique parametric instructions: The instructions and data required to load or to set any specific parametric values, modes, or sequences into the equipment that are unique to the facility and that are not included in the computer program shall be included. This material may include the setup of items such as program constants and parallels, into specific memory locations, and the setting of controls that modify equipment interfaces in readiness for tactical system operation at the specific facility.

3.8.1.4.2 Operating modes. Beginning with the primary mode, each available mode of operation shall be described in the normal and characteristic sequence of major phases, events, tactical options, supervisory commands, and responsive actions. Operating modes that are common to all modes (e.g., launcher loading and assignment) shall be detailed under the primary mode and referred to under the secondary modes with such modifications of procedure as may be necessary. The primary mode operating description shall be supported by either a block or logic illustration of operational sequence. The block diagram form shall be used when the operator has no choice of action. However, when an operational phase involves any operator's judgment, the description shall be supported by an operational sequence logic diagram that indicates the factors that must be favorable prior to operator's action, or if unfavorable, indicates an alternate optional action.

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3.8.1.4.3 Normal operation. The duties of system operators shall be described in terms of general responsibility and specific step-by-step procedures for operating the system in all of the primary modes. Descriptive words (such as switch, button, dial, or indicator) may be added to clarify the type of control involved (for example: "Press ACCESS button and observe channel spot."). If system controls and indicators or displays and alerts must be observed or used during normal operation, a reference to the controls and indicators information (see 3.8.1.2) and displays and alerts information (see 3.8.1.3) may be used to support the procedures. Controls and indicators or displays and alerts provided only for maintenance and non-system application shall not be called out.

3.8.1.4.4 Emergency operation. Step-by-step procedures shall be provided for emergency operation of the system. If specially designated controls have been provided for emergencies, a short statement shall be included describing how they modify or otherwise affect normal system operation. When applicable, emergency operational routines shall be included that emphasize the controls that permit immediate braking or de-energizing of the system. Emergency procedures shall be supported by illustrations.

3.8.1.4.5 Special operation. Special operations such as test, checkout, training, or evaluation exercises shall be described. Illustration support shall include block diagrams and pictorial diagrams.

3.8.1.5 Equipment operating procedures. Equipment operating procedures shall include the following information:

a. Procedures:

(1) Operator turn-on: All steps necessary to bring the equipment from OFF through STANDBY condition to full operation.

(2) Modes of operation: Procedures for each mode of operation (e.g., manual, automatic, local, remote, and so forth). The use and relative advantage of each mode shall also be described.

(3) Operation under interfering conditions: Describe the equipment anti-jamming and interference reduction features, the advantages of each feature, and the operating procedures to be followed in all possible situations. Supporting illustrations such as indicator displays, wave forms, and so forth shall be included that provide typical observations of jamming and interference for evaluation by the operator.

(4) Operator turn-off: This procedure shall include all steps necessary to bring the equipment from full operation through STANDBY to OFF condition.

(5) Battle-short or emergency operation: This procedure shall cover operating the equipment during emergency conditions (e.g., control failure, air failure, lube oil failure, loss of cooling water, and so forth). Emergency operator instructions shall be included. Provide a warning or caution to return the equipment to proper operation when the emergency is over.

(6) Emergency turn-off: This procedure shall cover turning the equipment off during an emergency (e.g., fire, water, smoke, hazard to personnel, loss of coolant, normal power, and so forth).

(7) Electromagnetic interference (EMI): This procedure shall include information pertaining to the recognition of EMI, such as sights, sounds, lack of response, or other irregularities, and operating procedures to cope with them during operations.

b. Method of presentation. Operating procedures shall be presented in concise, simply-worded, step-by-step procedures and shall include the following:

(1) A short explanation of the operation to be performed.

(2) Initial safety requirements (e.g., actions, inspections, and reference to emergency turn-off procedures).

(3) Connection of any accessory equipment not permanently connected.

(4) Instructions for obtaining or confirming the presence of all critical inputs such as power, coolant, air, signal, air conditioning, and so forth.

(5) Procedures for setting controls and making adjustments that shall be accomplished by the operator prior to equipment turn-on.

(6) Procedures for determining operational readiness and the acceptable indications expected from built-in indicators such as meters, lamps, gauges, and recorder readouts.

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(7) Milestones in the operational status of the equipment shall be identified and included by brief statements such as “the generator is now in STANDBY”.

(8) Visual or audio observations that occur as a result of an operator action such as boom lowering, sweep rotation, blower motor running, and so forth.

(9) Operator’s checks and adjustments in proper sequence.

(10) Operating controls and indicating devices as well as normal in-use positions or indications shall be included. A reference to the controls and indicators information (see 3.8.1.2) or displays and indicators information (see 3.8.1.3) may be used to support the procedures.

(11) Operator’s maintenance actions and schedules.

3.8.1.6 Non-tactical operation (weapon systems only). This information shall include, but not be limited to such non-tactical operations as training exercises using dummy missiles and simulated targets, training and evaluation exercises using telemetering missiles, and recording equipment. The procedures shall not duplicate information contained in a subsystem or equipment manual.

3.8.1.7 Conditions of readiness (HM&E systems, electronic systems, and weapon systems). This information shall be provided for guidance purposes and shall represent engineering considerations. Conditions of readiness requirements shall be listed prior to the operation of the system. Activities or conditions within compartments or areas outside the system shall be described as necessary. When it is impractical or illogical to separate the conditions of readiness information from operational information, a request shall be made with justification to combine these requirements with the system operation requirements. Do not duplicate information that is contained in the Engineering Operational Sequencing System (EOSS) or Combat System Operational Sequencing System (CSOSS) manuals.

3.8.1.7.1 Conditions of coverage. An orientation of system personnel shall be included.

a. The conditions of readiness shall be included as follows:

(1) Watch condition. The condition of all systems equipment and the personnel situation (how many of what class or type of personnel to be stationed where) for watch condition shall be described.

(2) General quarters condition. The condition of all systems equipment and the personnel situation shall be described for general quarters condition.

(3) Emergency condition. The condition of all systems equipment and the personnel situation for emergency condition shall be described.

b. Coverage shall include, but not be limited to, the following:

(1) A block diagram showing signal paths of the conditions of readiness (see figure 13).

(2) Block diagrams and descriptions of IC links between compartments and areas, including sound-powered telephone circuits, announcing systems, and closed-circuit television.

(3) Presumptions relative to system status at specified times (e.g., from watch condition to general quarters condition, or from general quarters condition to emergency condition).

(4) The operating console and panel positions shall be illustrated.

3.8.2 Maintenance. Scheduled and corrective maintenance procedures shall be developed and provided. These maintenance procedures shall be logically subdivided into task-oriented segments or modules to enable a technician to receive, process, test, and maintain the system or equipment to the lowest level replaceable component or part in accordance with the established maintenance concept or approved maintenance plan.

3.8.2.1 Introduction. The introduction shall provide a brief explanation of the purpose, scope, and arrangement of the corrective maintenance data sufficient for an understanding of its use and relationship to the rest of the maintenance documentation.

a. When specified by the acquiring activity (see 6.2), the introduction for weapon equipment shall also include and explain the use of a table of established values or an adjustment effectivity list.

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(1) Table of established values: Any functional parameters with center values dependent upon specific parts or assemblies in the equipment, which have sufficient accuracy or long-term stability requirements with respect to established center values, shall be listed in a table of established values. Such parameters shall be listed in alphabetical order of symbols or descriptive nomenclature, or by appropriate functional groups, as applicable. Adequate space shall be provided for entries of the actual values pertaining to individual sets of the equipment. In addition, the table shall reference the establishing procedure for each datum value to be entered. Usually, this will be an adjustment procedure, but may be the replacement procedure itself if the installed item is to be monitored for long-term stability although no compensating adjustments are required. This table is to be included in the manual and the values are to be entered by the technician using the material. Figure 14 is an example of a table of established values.

(2) Adjustment effectivity list: A cross-reference list shall be provided to relate equipment adjustment and alignment procedures to output functions. All adjustment and alignment procedures for the subject equipment shall be listed in numerical order. For each procedure, the outputs that are subject to change when that procedure is accomplished shall be referenced. When more than one output is affected by a given procedure, the output that is most critical or most sensitive to the procedure shall be flagged, underlined, or marked by asterisks to identify the priority setting. If two or more outputs are equally critical to one procedure, then only one shall be flagged arbitrarily unless the several outputs are related to different specific portions of the procedure. In the latter situation, the applicable step numbers shall be identified as a reference guide to verification of restored operability following any equipment adjustments resulting from indications other than specific test faults. Figure 15 is a sample of an adjustment effectivity list.

b. For weapon systems, the introduction shall describe the principles of test-based system alignment in terms of its vital contribution to system effectiveness, the subsystem interdependencies involved, and the test and troubleshooting documentation required to maintain system alignment. To promote a better understanding of system alignment, the description shall point out the distinction between alignments and adjustments. The alignment interdependencies among the various equipment shall be illustrated by a classification chart arranged in a diagrammatic sequence to fully depict the dependency relationships of all alignments affected or related.

3.8.2.2 Scheduled maintenance. Scheduled maintenance shall not be included in the TM. Scheduled maintenance instructions are furnished in the Planned Maintenance System (PMS). However, the following information shall be provided, verbatim:

"INTRODUCTION.

Scheduled maintenance instructions are furnished in the Planned Maintenance System (PMS). When conflicts exist between this manual and the PMS, the PMS documentation shall take precedence. Such conflicts should be reported immediately, using the Technical Manual Deficiency/Evaluation Report (TMDER) form.

PLANNED MAINTENANCE SYSTEM.

Recommended preventive maintenance procedures to be performed on a scheduled basis are provided in PMS documentation. OPNAVINST 4790.4 describes the PMS and also covers departmental and work center record keeping, as well as the Maintenance Index Page (MIP) and Maintenance Requirements Cards (MRCs). The MRCs cover scheduled inspection and lubrication procedures for the (insert name of system or equipment being covered here). The extensive and comprehensive scheduled maintenance information provided by the MRCs precludes the need for the detailed coverage within this manual. Specific corrective maintenance requirements (adjustments, alignment, repair, etc.) are covered as part of this manual."

3.8.2.3 System corrective maintenance procedure synopsis (weapon systems only). This information shall provide coverage for a synoptic description of each system alignment. Each synopsis shall state the objectives of the alignment and list the initial requirements and the significant parameters or functions to be considered. The synopsis shall highlight major alignment procedures to provide maintenance personnel with the background information necessary for understanding the alignment and what is to be achieved by a satisfactory conclusion of alignment procedures. The text shall identify any additional system alignment procedures to be accomplished prior to, in conjunction with, or subsequent to the subject system alignment.

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3.8.2.4 Maintenance information. Maintenance information shall be functionally divided as necessary to convey the maintenance tasks described below in a logical and useable manner. Maintenance information may contain an introduction. The information shall provide all the corrective maintenance information required to adjust and align the equipment and remove, clean, inspect, repair, reinstall, and align repairable parts, modules, subassemblies, and assemblies. As applicable, at the beginning of each segment or module, instructions shall include safety precautions to be observed and tools, parts, materials, and test equipment required (see 3.8.3). Procedures that can be hazardous to personnel or equipment shall be emphasized by DANGERS, WARNINGS, or CAUTIONS, as applicable. Instructions shall identify the action to be accomplished; preliminary control settings; test equipment setup instructions; and step-by-step instructions, with supporting illustrations, to accomplish the maintenance task. Corrective maintenance instructions shall be provided for all items designated repairable irrespective of the maintenance concept unless this information is included in another TM and can be referenced.

3.8.2.4.1 Maintenance tasks. Information shall be developed in a logical order for the maintenance tasks listed below, as applicable. For usability and clarity, the maintenance tasks listed below may be combined in a single segment or module or provided in individual segments or modules as necessary.

a. Preliminary setup.
b. Maintenance turn-on (test).
c. Adjust/align.
d. Repair.
e. Remove.
f. Disassembly.
g. Inspect.
h. Reassembly.
i. Adjust.
j. Install.
k. Calibrate.
l. Checkout.
m. Cleaning.
n. Lubrication.
o. Storage.
p. Grooming.
q. Unit break-in.
r. Packaging.
s. Transportation.
t. Handling.
u. Special preservation, storage, and transportation.
v. Special handling and packaging.
w. Refurbishment.
x. Examination and acceptance criteria.
y. Test and inspection.

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3.8.2.4.2 Adjustment and alignment procedures. Step-by-step procedures required to perform alignment and adjustment of systems/equipment or to restore electrical and mechanical alignment and adjustment between various system equipment shall be provided. Alignments and adjustments for all repairable assemblies and parts shall also be provided. All values and tolerances shall be included. The alignment shall be cross-referenced to respective fault-isolation procedures and diagrams provided in the operational checkout and troubleshooting information. Alignment procedures shall include references to equipment publications where further procedures are required at the equipment. Adjustments and alignments shall include, but not be limited to:

a. Non-operator type adjustments.

b. Alignments requiring external jigs, test equipment, or bench setups.

c. Alignments that are accomplished after a repair or replacement of a part or module.

d. Test equipment setup and other illustrations necessary to support the procedures.

3.8.2.4.3 Special preservation, storage, and transportation requirements. The requirements for special preservation, storage, and transportation of the repairable items shall be identified. For example, preservation for storage when the repair item is not to be installed (Fleet use) for an extended period of time and the methods used for preservation and storage of the repairable items are critical.

3.8.2.4.4 Special handling procedures. The procedures required for using the special crates, boxes, containers, transportation vehicles, and other facilities for repairable items handling shall be described.

3.8.2.4.5 Special packaging procedures. The procedures required for special packaging of the repairable items shall be described.

3.8.2.5 Yard or tender corrective maintenance (weapon equipment only). When specified by the acquiring activity (see 6.2), this information shall provide procedures and instructions for performing maintenance that is beyond the capabilities of the ship's force. The procedures shall be presented in step-by-step narrative form and shall be supported by selected illustrations and by reference to appropriate engineering drawings, specifications, ordnance directives, and similar data. An introduction may be included as necessary. The presentation format for yard or tender procedures shall be based upon recommendations submitted to and approved by the Government. General overhaul and repair practices shall be presented first. A series of information shall cover the major physical subdivisions of the equipment. For each item requiring replacement, procedures shall be given for removal, installation, and test. If repair is practicable below the level of replacement parts, procedures shall be given. If repair or replacement requires readjustment or alignment, references shall be made to MRCs when appropriate, otherwise the required procedure shall be included. Procedures that are obvious upon inspection of equipment shall be omitted. Block diagrams illustrating possible sequences of disassembly and reassembly shall be provided.

3.8.3 System/equipment installation. When specified by the acquiring activity (see 6.2), drawings and information concerning system/equipment installation shall be provided. The following type of information shall be included: site selection or installation location guide lines (such as moisture precautions and maximum temperature allowed, as appropriate), special tools and materials requirements, unpacking, and handling (if unusual procedures or precautions are required), preparation of foundations, mechanical assembly procedures, mounting instructions, bolting diagrams, safety precautions, grounding and bonding, clearances for access, ventilation, fluid cooling requirements, clearances for motion under shock, and recommendations for reduction of electrical and EMI and other interface requirements, as applicable. In addition, coverage shall contain tests and test procedures required to demonstrate that the equipment after installation is capable of satisfying operational requirements. Also, it shall include information relating to the electromagnetic compatibility (EMC) measures taken in the original design or subsequently incorporated to maintain EMC integrity of the equipment at all times in its life cycle. The measures shall include, as a minimum, shielding, filtering, grounding, bonding, and routing. In addition, for system TMs, installation drawings and information not contained in the equipment TM necessary to install and checkout the system shall be included.

3.8.3.1 System cable interconnection check (all systems). This information shall provide a checklist with procedures to verify the proper installation of all system cables. The checks shall be conducted with all power off and all equipment completely shut down.

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3.8.3.2 Site or installation location. Data supplemental to the installation drawings shall be provided. If all site information is contained on the installation drawings, reference shall be made to the applicable drawings.

3.8.3.3 Unpacking and repacking. Unpacking and repacking data supplemental to the installation drawings shall be provided. Include step-by-step procedures to prevent damage to the equipment or injury to personnel. Supporting illustrations shall be provided to clarify procedures. When packing for reshipment is required, step-by-step procedures for packing shall be included and illustrated. Any special environmental conditions required for storage shall be provided. Instructions shall be included for items in the following categories:

a. De-preservation procedures required at time of installation.
b. Re-preservation packaging required prior to repacking for storage or shipment.
c. Intricate mounting, blocking, or bracing.
d. Special cushion inserts.
e. Repairable items.
f. Sensitive or fragile components.
g. Items held in special cradles.
h. Items furnished in reusable containers.
i. Special environmental conditions required for storage.
j. Special handling procedures required.
k. Container storage or disposition instructions, as applicable.

3.8.3.4 Preparation of foundations. Preparation of foundation data supplemental to the installation drawings shall be provided. If all information is contained on the installation drawings, reference shall be made to the applicable drawings.

3.8.3.5 Input requirements. A summary of the input data contained on the installation drawings shall be provided. Parameters with tolerances, if applicable, shall be included with each of the inputs listed. Examples of inputs are as follows:

a. Power.
b. Ventilation.
c. Dry air (waveguide pressure).
d. Ship gyro information.
e. Fluid cooling.
f. Steam.
g. Refrigerant.

3.8.3.6 Utilities list (HM&E and electronic systems only). This information shall provide a utilities list that presents all utilities required in tabular form, and the quantities of each, in each system, compartment, and area, such as air, water, power, steam, and refrigerant.

3.8.3.7 Active system tests (HM&E and electronic systems only). This information shall include all active system test procedures required to verify the proper installation and operation of the system. Reference may be made to applicable tests and procedures in corrective maintenance. Procedures for complete setup, testing, shut down, and data analysis shall also be provided.

3.8.3.8 Installation procedures. As applicable, installation procedures shall be provided. For HM&E, weapon, and electronic systems, complete step-by-step instructions shall be provided for installation of system components not covered in any of the equipment manuals for the equipment comprising the system. Illustrations shall be included, as applicable. Step-by-step procedures shall be developed for the following when not provided on the installation drawings:

a. Instructions required to assemble units/components.

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b. Instructions required to mount units/components. Include bolting and bracing diagrams and data on shock mounts.

c. Instructions for making electrical, waveguide, plumbing, transmission line, and all other interface connections (external) to the equipment.

d. Instructions for interconnecting units comprising the equipment.

e. Servicing procedures such as initial lubrication and adjustments.

f. Instructions for bonding and grounding.

3.8.3.8.1 Installation drawings. Reproduction of appropriate installation control drawings shall consist of the following, as applicable:

a. Drawing list.

b. Block diagrams.

c. Outline and installation drawing.

d. Auxiliary cooling diagram.

e. Auxiliary dry gas diagram.

f. Cable running sheets.

g. Summary list of installation material.

h. RF transmission line diagram.

i. Hydraulic fluid piping diagram.

3.8.3.9 Installation checkout. Step-by-step procedures to demonstrate that the system/equipment operates correctly and within tolerances shall be provided. These procedures shall provide for equipment checkout in three test phases as follows:

a. Phase 1 - installation inspection and pre-energizing procedures. Inspection procedures shall be provided in the form of checklists to verify the following:

(1) That all units of the equipment and required auxiliary equipment have been installed and that their location and orientation is proper; that all cables, filters, antennas, waveguides, transmission lines, dehydrators, coolant lines, piping and so forth have been installed in accordance with plans and specifications; and that continuity exists in all interconnections.

(2) That the test equipment listed in the special tools and test equipment list (see 3.5.3) is onboard, operating satisfactorily, has been calibrated, and has necessary storage in accordance with ST000-AB-GYD-010.

(3) That all alterations such as field changes and SHIPALTS have been accomplished and that all installation and reference standards are entered on the installation standards summary sheet.

(4) That all rotating devices are free from obstruction.

(5) That there is access to the equipment for maintenance.

(6) That all pre-energizing servicing procedures, including lubrication, have been accomplished.

(7) That it is safe to turn on the equipment.

b. Phase 2 - initial turn-on and preliminary test. Procedures shall be included for energizing the equipment for the first time. This may be accomplished by reference to the applicable portions of operation. Step-by-step procedures shall be included for testing electrical, steam and fluid supply, supply circuits, distribution panels, switches, breakers, relief valves, and interlocks, as applicable. Procedures shall be included for testing piping, electrical cables, wire rope, stays, proper installation of transmission lines and waveguides, hangar spacing, torqueing of connectors, pressure testing, flow rates, standing wave ratio and attenuation checks, and so forth.

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c. Phase 3 - installation verification test. Complete instructions shall be included for testing the equipment in all modes of operation. Where applicable, refer to the scheduled performance tests in scheduled maintenance. Procedures shall cover checking gauges, meters, alarms, and other sensing devices for proper operation and calibration. The tests shall verify that all inputs are in tolerance. Where applicable, include voltage standing wave ratio and insertion loss tests to verify the proper installation of antenna-to-equipment waveguide runs and transducer impedance and source level checks to verify proper installation of transducers, domes, and cables. In addition, for HM&E systems, also include dynamic and static load tests for hoists, winches, etc. Preliminary setup data shall be included in each procedure. When it is required that an alignment be accomplished prior to performing a test, the alignment shall be included or referenced in the procedure.

(1) Test procedures for systems. Testing procedures shall be presented in a logical order as follows:

(a) Energize the equipment.

(b) When test results are within the required tolerance, include reference to the next logical test.

(c) When test results are out of tolerance, include a reference to the corrective maintenance or troubleshooting data. Reference shall be made to troubleshooting diagrams except where probable causes of failure can be predicted, in which case reference may be made directly to an alignment or repair procedure.

(2) Test procedures for equipment. Testing procedures shall be presented in a logical order as follows:

(a) Energize the equipment.

(b) Test the first units (normally power supplied) that must be operating properly.

(c) When test results are within the required tolerance, include reference to the next logical test.

(d) When test results are out of tolerance, include a reference to the corrective maintenance or troubleshooting data.

(3) Installation standards summary sheet. The installation standards summary sheet shall provide spaces for recording the results of all installation verification tests and significant reference standard tests that should be made a matter of record for reference by the technician during troubleshooting and for a standard to be re-established after overhaul (see figure 16). Each space shall be identified by the step or paragraph number or title that provides the instructions for accomplishment. Each space shall contain the respective unit of measurement (for example, amp, dB, or V). When applicable, waveforms shall be included to show the points on the pattern where the measurement is to be taken. In addition, critical installation data shall be included, such as the length of the transmission line.

3.8.4 Operational checkout and troubleshooting. Operational checkout and troubleshooting information shall be developed as described in 3.8.4.1 through 3.8.4.6. The information shall be developed to the extent required to maintain the system or equipment at the authorized maintenance level. The number of interrelated systems, assemblies, subassemblies, and components, types of equipment, and maintenance concept/plan shall be taken into consideration to determine the type and depth of checkout and troubleshooting procedures to be developed. Additional testing and troubleshooting information may be developed as described in 3.8.4.7 through 3.8.4.11.

3.8.4.1 Procedure development assumptions. Procedures shall be developed in accordance with the established maintenance concept and shall be based on the following assumptions:

a. When more than one level of maintenance must be provided for, separate checkout and troubleshooting procedures shall be developed for each maintenance level.

b. Operational checkout, maintenance turn-on, and troubleshooting procedures shall include all elements of pretest setup and post-test shutdown.

c. Maintenance turn-on procedures to energize the equipment from the fully de-energized condition to full operation may be used in lieu of an operational checkout procedure to determine which major function or supporting function is malfunctioning.

d. Successful completion of an operational checkout or testing procedure verifies system/equipment operation.

e. Procedures normally trace one malfunction at a time. However, possible multiple failures shall be anticipated and considered in the testing process.

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f. If several components are suspected, the one most likely to have failed shall be considered first.

g. Whenever a component is replaced, checkout shall be restarted to ensure that the replacement did not introduce a new fault into the system or equipment. This assumption shall not be interpreted as a requirement to repeat an entire system/equipment checkout in all cases.

3.8.4.2 Introduction. The introduction shall describe the general approach that is used in operational checkout and troubleshooting. The introduction shall provide a brief explanation of the purpose, scope, and arrangement of the data, sufficient for an understanding of its use.

3.8.4.3 System fault descriptor (system troubleshooting only). This information shall contain a description of reported malfunctions and related maintenance codes for each system. An introduction explaining the fault descriptor data and how this data is used as an aid in relating reported malfunctions to maintenance codes shall be included. For each system, a table listing all fault descriptions along with the corresponding maintenance action to be taken shall be developed. Fault descriptor data and associated malfunction identification is normally derived from built-in-test using preprogrammed tests or diagnostic software or from electronically controlled sensors within the systems to be tested.

3.8.4.4 Operational checkout. Operational checkout procedures or maintenance turn-on procedures that subject the systems, subsystems, components, accessories, and items of equipment to prescribed conditions to determine that they shall function in accordance with predetermined test parameters shall be developed. The following information shall be included, as applicable:

a. Introduction. An introduction shall be included explaining how the operational checkout procedures are to be used to perform testing and how they relate to the associated troubleshooting procedures.

b. General procedures and precautions. Any general procedures that must be performed prior to checkout and precautions that must be taken during the performance of the checkout procedure shall be included. Alignment/adjustment procedures that must be performed prior to or during the checkout procedures shall also be included.

c. Pretest setup procedures. Procedures for connecting any test and accessory equipment, including cable connections, shall be included. Procedures for the initial setting of controls shall also be provided.

d. Operational checkout procedures. Operational checkout procedures shall consist of a series of numbered steps and sub-steps that lead to an indication or condition. Based on these indications or conditions, a corrective action is provided. This corrective action can either be stated as a specific remedy or can be a reference to a detailed troubleshooting procedure. This process is continued until the entire operational checkout procedure is completed.

e. Post-operational checkout shutdown procedures. Procedures to return the system or equipment to its normal configuration, prior to pretest setup, if required, shall be included. When applicable, procedures shall be included for both normal shutdown and emergency shutdown.

3.8.4.5 Troubleshooting procedures. Troubleshooting procedures for detecting, isolating, and correcting systems, subsystems, and equipment failures and malfunctions shall be developed. The following information shall be included, as applicable.

a. Introduction: An introduction shall be included explaining how the troubleshooting procedures are to be used to perform troubleshooting and how they relate to the associated operational checkout. When applicable, a consolidated list of malfunctions or symptoms that reference to the applicable troubleshooting procedures shall be included.

b. General procedures and precautions: Any general procedures that must be performed prior to troubleshooting or precautions that must be taken during the performance of the procedure shall be included. Alignment/adjustment procedures that must be performed prior to or during the troubleshooting procedures shall also be included.

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c. Troubleshooting procedures: Based on the complexity of the troubleshooting to be performed, troubleshooting procedures can be structured differently and, therefore, shall contain different content. The following two methods shall be used to prepare troubleshooting procedures.

(1) Method A - text-logic: Troubleshooting procedures for specific fault symptoms shall combine text and logic and consist of a series of steps and sub-steps that lead to an indication or condition (usually stated in the form of a question). Based on these indications or conditions, a “YES” or “NO” response is provided to guide the technician to either the next step or a series of steps, or to a corrective action that may consist of a malfunction or action followed by a reference to the module or paragraph that contains the data to perform the corrective action. This process is continued until the entire troubleshooting procedure is completed.

(2) Method B - text: Troubleshooting procedures shall consist of an all-inclusive series of specific fault symptoms for the system/equipment being troubleshot. For each fault symptom, the probable malfunction or series of malfunctions that may have caused the fault shall be listed. For each probable malfunction identified, a corrective action shall be stated with a reference to the module or paragraph that contains the data to perform the corrective action.

d. Post-troubleshooting shutdown procedures: Procedures to return the system or equipment to its normal configuration, prior to setup, if required, shall be included. When applicable, procedures shall be included for both normal shutdown and emergency shutdown.

3.8.4.6 Operational checkout and troubleshooting procedures. Information shall be developed containing operational checkout and troubleshooting procedures for integrated systems and for each independent system and subsystem or equipment, as applicable. Content and development requirements are provided in 3.8.4.6.1 through 3.8.4.6.2.1.

3.8.4.6.1 Operational checkout and troubleshooting procedures content. Operational checkout and troubleshooting procedures shall guide a technician in as practical a manner as possible in detecting, isolating, and correcting system and equipment failures/malfunctions. Procedures shall ultimately lead to isolating faults to an appropriate adjustment, replaceable parts, interface wires, or mechanical linkage and direct repair or replacement of parts authorized for repair or replacement at the maintenance level covered. As applicable, safety precautions to be observed and tools, parts, materials, and test equipment required (see 3.3.2.2) shall be included at the beginning of each procedure. Procedures shall be accompanied by reference data such as schematics, signal flow diagrams, functional circuit diagrams, sequenced switching diagrams, waveforms, tables, and other illustrations for comprehensive understanding of the procedures. If a large number of schematics or other type diagrams are required, they may be included in a separate segment or module (see 3.8.4.7). The schematics shall integrate fluid, mechanical, electrical, and electronic components. Illustrations may also be included that locate and identify the controls and displays used to perform the testing and troubleshooting procedures. For systems, special attention shall be given to system interface wiring fault isolation procedures. Wiring fault isolation procedures shall include the following types of data, as applicable:

a. Specific wire reading access points and resistances for wiring components (where practical).

b. Wire-to-wire and wire-to-ground criteria for circuit integrity.

c. Where required, special wire definition (including interconnecting criteria for proper sealing or terminal application) and special notations where wire harnesses should be completely replaced and not repaired.

3.8.4.6.2 Operational checkout and troubleshooting procedure development. Based on the following factors and as agreed to by the Government, operational checkout and troubleshooting procedures may be combined into a single segment or module or may be developed in a separate operational checkout segment or module and a separate troubleshooting segment or module.

a. Complexity of the system/equipment.

b. The type of test equipment used.

c. System/equipment self-test or built-in-test/built-in-test-equipment (BIT/BITE) capability.

d. Complexity of the test and troubleshooting procedures as determined by the task analysis.

e. Clarity and usability.

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3.8.4.6.2.1 Combined operational checkout and troubleshooting. Depending on the complexity and the intended use, operational checkout and troubleshooting procedures may be combined and included in a single segment or module. The operational checkout and troubleshooting information may be combined into a single procedure (see d[1] below), or when necessary, developed as individual checkout and troubleshooting procedures within the same segment or module (see d[2] and [3] below). The following information shall be included, as applicable:

a. Introduction. An introduction shall be included explaining how the operational checkout and troubleshooting procedures are to be used to perform checkout and troubleshooting and how they relate to the associated information that included the corrective actions that returns the equipment to proper operation.

b. General procedures and precautions. Any general procedures that must be performed prior to checkout and precautions to be followed during the performance of the checkout procedure shall be included. Alignment/adjustment procedures that must be performed prior to or during the checkout and troubleshooting procedures shall also be included.

c. Pretest setup procedures. Procedures for connecting any test and accessory equipment, including cable connections, shall be included. Procedures for the initial setting of controls shall also be provided.

d. Operational checkout and troubleshooting procedures. Depending on their complexity, the operational checkout and troubleshooting procedures may be combined in a single procedure or may be prepared as separate procedures within a single segment or module.

(1) Combined operational checkout and troubleshooting procedures: Combined operational checkout and troubleshooting procedures shall consist of a series of test procedures (steps and sub-steps) that lead to an indication or condition. When a normal indication is obtained, the operational checkout continues until the entire checkout is completed or until an abnormal condition or indication is observed. When the test procedure results in an abnormal indication or condition, a malfunction or a series of malfunctions is provided. For each malfunction, the possible corrective actions shall be provided. When required, the corrective action may include a reference to the module or paragraph that contains the data to perform the corrective action.

(2) Separate operational checkout procedures: When it is determined that the operational checkout procedures shall be separated from the troubleshooting procedures, the operational checkout procedures shall be included under the heading “OPERATIONAL CHECKOUT”. Operational checkout procedures shall be developed in accordance with 3.8.4.4.d.

(3) Separate troubleshooting procedures: When it is determined that the troubleshooting procedures shall be separated from the operational checkout procedures, the troubleshooting procedures shall be included under the heading “TROUBLESHOOTING”. Troubleshooting procedures shall be developed in accordance with 3.8.4.5.c.

e. Post-operational shutdown procedures. If required, procedures to return the system or equipment to its normal configuration, prior to operational checkout or troubleshooting setup shall be included. When applicable, procedures shall be included for both normal shutdown and emergency shutdown.

3.8.4.7 System/equipment testing and troubleshooting reference. As determined in paragraph 3.8.4.6.1, reference data that is required to support operational checkout and troubleshooting shall be developed. Reference information may include an introduction. Reference data may include:

a. Electronic, electrical, mechanical, and maintenance schematic diagrams.

b. Piping diagrams.

c. Cabling, wiring, or signal tracing diagrams.

d. System control function diagrams.

e. System data function diagrams.

f. System functional diagrams (SFDs).

g. Digital SFDs.

h. Power distribution diagrams.

See 3.7.8 of MIL-DTL-24784 for specific development requirements.

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3.8.4.8 Relay coil, switch, and lamp indices. These indices shall be prepared for all relay coils, switches, and indicator lamps. The relay and lamp indices shall include the item reference designation, the functional name, energizing voltage, and a reference to the troubleshooting diagram(s). Any components that are part of a circuit card assembly that is a lowest replaceable unit shall not be listed on the component indices.

3.8.4.9 Protective device index. This index shall list all protective devices, such as fuses, circuit breakers, and so forth. The index shall include the item reference designation, front panel marking of the device, trip-out value of the circuit breaker and rating of fuses, name of the circuit protected, and a reference to troubleshooting diagram(s).

3.8.4.10 Redundant pluggable electronic components. When applicable, a list of redundant (identical) pluggable electronic components shall be provided to facilitate the interchange of parts in use within the system/equipment. The list of components shall appear in tabular form with items grouped by part number and arranged in alphanumeric part number order. The list shall include reference designators for each part number listed. Reference designators shall be arranged in ascending order with the least critical component appearing first. Non-pluggable items and items having minor degrading impact on the system/equipment shall not be listed (e.g., indicator lamps and light bulbs). This table shall be titled “Component Interchangeability”.

3.8.4.11 Maintenance assistance modules (MAMs). When applicable, a list of MAMs and other redundant and swappable components for the system/equipment shall be provided. This table shall be titled “Special Purpose Test Equipment - MAMs”. These MAMs enable maintainers to swap or replace suspected faulty parts identified by troubleshooting routines.

3.9 IPB. When specified by the acquiring activity (see 6.2), IPB information may be developed and delivered as indicated below:

a. Provided in a separate TM.

b. Developed as IPB information and included in the applicable maintenance TM.

c. Developed as a parts information database and included in the applicable maintenance TM.

See MIL-DTL-24784/23 for detailed requirements.

3.10 CSTOM. Combat system description information shall be developed and divided into the following, as applicable.

3.10.1 Combat system introduction and description. Coverage shall consist of information for combat system personnel that summarizes ship physical and functional characteristics, missions, capabilities, and the interrelationships between combat system elements. The information shall introduce the CSTOM and provide a physical and functional description of the ship. An overview of the ship mission and capabilities shall be included, and shall identify the elements comprising the combat system and the physical location of selected elements.

3.10.1.1 Types of data level descriptions. Descriptive portions of CSTOMs shall identify principal combat system elements with a textual synopsis for each. A reference to the element documentation shall be made for detailed information. The data in the CSTOM shall consist of levels of narrative and illustration coverage as follows:

a. Level 1 detailed introduction. Narrative text explaining combat system and interfaces and operational or testing dependencies or capabilities. The text shall be supported with block or pictorial drawings.

b. Level 2 functional and operational descriptions. Narrative text explaining principal combat system and element functions, operations, actions, capabilities, or fault isolation methods and tools. The text shall be supported with block or pictorial drawings.

c. Level 3 detailed description. Narrative text explaining signal characteristics and flow or test procedures. The text shall be supported with signal level drawings.

3.10.1.2 Introduction. This information shall briefly describe the purpose, scope, and content of the CSTOM.

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3.10.1.3 Ship and combat system description. This information shall provide a description of ship missions, capabilities, and physical characteristics. Coverage shall include a level 1 detailed introduction of all ship combat system elements and related armament; such as search radar subsystem, combat direction system, electronic warfare system, gun weapon system, underwater weapon system, missile weapon system, light airborne multi-purpose system (LAMPS), external navigation system, external communication system, and support system. The primary ship combat missions of anti-air warfare (AAW), anti-submarine warfare (ASW), and surface and shore bombardment shall be described. An introduction to combat system operation in the manual, semiautomatic, automatic, and casualty modes shall be provided. Major combat system functions of detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment shall be described. A level 1 detailed introduction block diagram, similar to figure 17, shall be provided. The effective engagement envelop for each weapon system shall be illustrated.

3.10.1.4 Location of combat system components. This information shall provide a description of the ship combat system compartments and areas. A ship profile drawing, similar to figure 18, shall pictorially identify external combat system elements and area locations. A three-dimensional plan view and room arrangement diagram, similar to figures 19 and 20, shall depict strategically located important combat system compartments and areas. Except for CSTOMs developed for a single ship configuration, these drawings shall be typical and shall represent the lead ship of a class. Figure references shall be provided and areas shall be shaded to identify compartments and areas for which equipment room arrangement diagrams are provided.

3.10.2 Combat system description. The combat system description shall contain an abridged description of the physical and functional characteristics and, as applicable, a brief description and flow diagrams of operational programs for each element. A pictorial of the combat system, similar to figure 21, shall be provided. Related text shall describe element functions; functions shall be described to the depth necessary to enable the reader to understand each element's role in the combat system. The physical and functional characteristics of the combat system elements shall be described to level 1 detailed introduction and level 2 functional and operational descriptions. Pictorial art, similar to figures 22 and 23, shall be used to depict the physical characteristics and define the functional interface between elements and between equipment of the element. A separate table for each element shall list major equipment. The tables shall include common name or abbreviation, nomenclature, quantity, and location. A bibliography for each element shall list all source documentation. In addition to an introduction, the elements described in 3.10.2.2 through 3.10.2.15 shall be included, as applicable.

3.10.2.1 Introduction. This information shall introduce the elements that comprise the combat system and provide the user with an overview of the content of the element information.

3.10.2.2 Aegis combat system. This information shall describe and illustrate the elements of the Aegis combat system, including, but not limited to, the following:

a. Radar system.
b. Command and decision system.
c. Aegis display system.
d. Weapon control system.
e. Fire control system.
f. Vertical launch system.
g. Surface-to-air guided missile.
h. Operational readiness test system.
i. Aegis combat training system.

3.10.2.3 Search radar system. This information shall describe and illustrate major components of the sea and air search radar system that is not included in the Aegis combat system, including the radar recognition and radar distribution systems. When required for clarity, the radar recognition and radar distribution systems may be covered separately.

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3.10.2.4 Combat direction system. Where the Aegis combat system is not installed, this information shall describe and illustrate all components of the combat direction system. Areas of interface, other than with the ship support system, shall be addressed to the extent necessary for user understanding of the combat direction system role in the ship combat system. Block diagrams, similar to figure 24, shall illustrate the combat direction system and weapon system computer operational module interfaces. When applicable, combat direction system equipment for monitoring system performance and for testing combat system interfaces shall be described. Identification of operating stations for each mode configuration shall be provided.

3.10.2.5 Electronic warfare system. This information shall describe and illustrate all components of the electronic warfare subsystem, including electronic support measures, electronic countermeasures, and ancillary equipment.

3.10.2.6 Gun weapon system. This information shall be developed for each installed gun weapon system including the Close-In Weapon System (CIWS). This information shall describe and illustrate all components of the gun weapon system, including gun ordnance and ancillary equipment.

3.10.2.7 Antisubmarine warfare system. This information shall describe and illustrate all components of the antisubmarine warfare system, including the underwater surveillance and communication system and the underwater weapon system.

3.10.2.8 Missile weapon system. Information shall be developed for each installed missile weapon system other than Aegis such as Harpoon, Tomahawk, and Seasparrow. This information shall describe and illustrate all components of the missile weapon system, including operational and training ordnance and ancillary equipment.

3.10.2.9 LAMPS system. This information shall describe and illustrate all components of the LAMPS system, including all airborne LAMPS equipment, ordnance, and shipboard LAMPS equipment.

3.10.2.10 Underwater countermeasures system. This information shall describe and illustrate all components of the underwater countermeasures system, including ancillary equipment.

3.10.2.11 External communication system. This information shall describe and illustrate all external communications systems and related components. Coverage shall support RF links from extremely low frequency (ELF) to extremely high frequency (EHF), including, but not limited to, voice, video, data networking, and control and management information.

3.10.2.12 Internal communication system. This information shall describe and illustrate all IC systems, including audio and data communication systems, alarm systems, indicating and control systems, and ancillary equipment.

3.10.2.13 Navigation system. This information shall describe and illustrate all components of the navigation system. Coverage shall be provided for, but not limited to, satellite navigation, Radio Direction Finder (RDF), Tactical Air Navigation (TACAN), and ancillary equipment.

3.10.2.14 Combat system support system. This information shall describe and illustrate the support equipment associated with the combat system. Simplified block diagrams, similar to figures 25 and 26, shall depict interfaces between the ship support systems and the combat system. At a minimum, coverage for the following support systems shall be provided. For usability and clarity, each system may be covered individually.

a. Air and nitrogen pressurization systems.
b. Liquid cooling and heating systems.
c. Air conditioning.
d. Bridge ship-control system.
e. Ship parameters.
f. Power distribution.
g. Test equipment repair and calibration facility, when applicable.

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3.10.2.15 Additional systems. Information shall be developed for any additional combat system not covered in the paragraphs above.

3.10.3 Combat system operational description. This information shall contain the description of major combat system functions used in all modes of combat system operations, including combat system functions of detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment. These major functions of the combat system shall be defined and summarized and each function shall be described in detail. Level 1 combat system configurations, including normal, alternate, and backup features available through combat system switching that relate to the major combat system functions shall be described. A level 2 functional flow pictorial, similar to figure 27, shall illustrate each described combat system function or principal sub-function. A bibliography for each function shall list all source documentation. In addition to an introduction, the functions described in 3.10.3.2 through 3.10.3.6 shall be included, as applicable.

3.10.3.1 Introduction. This information shall introduce and define combat system functions and element functions pertinent to the modes of combat system operation and explain data transmission between elements. The general description shall prepare the reader for subsequent more detailed functional descriptions.

3.10.3.2 Detection and entry. This information shall describe the initial track, detection and entry functions, and functional flow between elements performed for all radar, sonar, visual, or remote contacts to include air, surface, or subsurface threats entered manually or automatically into the combat system.

3.10.3.3 Tracking and identification. This information shall describe combat direction system tracking and identification functions and functional flow between elements occurring for any track entered into the combat system, either locally by own-ship sensors or remotely by any participating unit within the operational areas of the Fleet.

3.10.3.4 Threat evaluation and threat-to-weapon pairing. This information shall describe threat evaluation and threat-to-weapon pairing function, including weapon assignment. Data flow between elements to accomplish threat evaluation and weapon pairing for tracks considered hostile to own-ship and to Fleet security shall be described.

3.10.3.5 Engagement and engagement assessment. This information shall describe target engagement and engagement assessment functions for functional flow between elements occurring for targets assigned to the electronics warfare system, gun weapon system, underwater weapon system, missile weapon system, interceptor aircraft, or LAMPS.

3.10.3.6 Additional functions. Information shall be developed for any additional functions not covered in the paragraphs above.

3.10.4 Combat system readiness assessment. This information shall provide guidance to the Ship Electronic Readiness Team (SERT) for evaluating and improving combat system material and operational readiness. Combat system readiness assessment information shall include data for readiness assessment, fault detection and impact evaluation, and fault isolation. The guidance shall primarily address organizational level maintenance within the capability of ship senior maintenance personnel. In addition to an introduction, the readiness assessment, fault detection and impact evaluation, and fault isolation data described in 3.10.4.2 through 3.10.4.7 shall be included, as applicable. A bibliography shall be included listing all source documentation.

3.10.4.1 Introduction. This information shall contain an overview of the combat system readiness assessment, fault detection, fault isolation, and impact evaluation tools. A drawing similar to figure 28 shall be used to introduce and explain the relationship between readiness assessment, fault detection and impact evaluation, and fault isolation. The following introductory information shall be included:

a. Fault detection and impact evaluation overview. The use of impact evaluation tables and operational fault directories provided for fault detection to readiness assessment shall be explained. The relationship of fault detection to readiness assessment shall be stressed. Primary topics shall cover the fault detection role in readiness assessment.

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b. Fault isolation overview. A description of a typical fault isolation pictorial and its associated table of interface references shall be provided. The table of interface references, similar to figure 29, shall be described. The use of a typical fault isolation pictorial, similar to figure 30, for diagnosing combat system faults shall be explained in detail.

3.10.4.2 Readiness assessment information. This information shall provide the principles and objectives of combat system readiness assessment using the SERT concept. The data shall define the means for determining system readiness within ship organizational constraints. Details of the PMS, with reference to the PMS record, shall be discussed. The description shall cover the scope and organization of readiness assessment and identify major factors and processes contributing to combat system readiness. A drawing similar to figure 31 shall illustrate the dependency of readiness assessment on fault detection, fault isolation, and impact evaluation.

3.10.4.3 SERT information. This information shall explain SERT organization and responsibilities. Emphasis shall be given to SERT activities such as test, training, material, and personnel management; preventive and corrective maintenance management; and status reporting. Techniques to generate optimum test schedules based on personnel and material assets and the ship's operating schedule shall be presented. Recommended forms, similar to figures 32 and 33, for preparing mission summary reports and combat system daily fault reports shall be provided and described.

3.10.4.4 PMS. This information shall describe the principles and elements of integrated testing and explain the concepts and documentation available for testing. The major objectives of fault isolation shall be defined, and the various types of fault isolation tools that personnel may select for fault isolation shall be described. Each type of fault isolation tool (i.e., directories, computer programs, fault logic diagrams, and so forth), shall be sufficiently described to facilitate its understanding and use. A figure, similar to figure 34, shall depict the PMS. The integration of combat system, element, and equipment documentation for establishing maintenance requirements, test scheduling, fault isolation, and corrective action verification shall be depicted by a drawing similar to figure 35.

3.10.4.5 Training and maintenance support elements. This information shall identify and describe the training and maintenance support elements available to, or resident in, the applicable ship class and shall evaluate their functions as tools to increase combat system personnel and equipment readiness. Tables similar to figures 36 and 37 shall be developed to:

a. Identify the elements of the combat system challenged by each of the exercises in Fleet Exercise Publications, FXP2 and FXP3.

b. Identify on a system or equipment level all exercises that challenge the element and identify the major combat system functions, such as detection, entry, identification, and tracking, that are challenged by each exercise.

c. Identify the training operational programs and procedures resident on the ship; determine the element challenged by each one, and provide a brief description.

3.10.4.6 Ship test equipment repair facility (STERF). When the ship configuration includes a STERF (phase A-1, A-2, and C-1, C-2), information shall include a list of test equipment calibration requirements that show parameters to be calibrated, calibration periodicity, and references to calibration procedures. When the ship configuration does not include a STERF, documentation shall be specified that prescribes calibration requirements and outlines procedures for obtaining off-ship calibration services.

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3.10.4.7 Readiness assessment diagrams (RAD) and synoptic test description. This information shall identify the top-level combat system, element, and equipment level tests required to determine combat system readiness for AAW, ASW, surface warfare (SUW), and shore missions. A table of tests with fault isolation pictorial and fault impact evaluation references similar to figure 38, and an RAD, similar to figure 39, shall be provided for each combat mission (AAW, ASW, SUW, and shore bombardment). The table and its associated RAD shall be provided for each major combat system function, including detection and entry, tracking and identification, threat evaluation and threat-to-weapon pairing, and engagement and engagement assessment. The RAD shall identify required functional interfaces applicable to the specific mission and major combat system function and provide a reference to the table that lists the highest-level test (combat system level, element level, or equipment level) that verifies each function identified. Detailed, comprehensive user instructions that explain the use of the RAD and its associated table shall be provided. A synoptic description of each combat system, element, and equipment test included on all the tables shall be provided. The test descriptions shall state test objectives and restrictions and specify significant parameters that evaluate readiness.

3.10.5 Fault detection and impact evaluation. This information shall describe methods of detecting material and personnel faults encountered during testing and operation and shall provide guidance to the SERT in evaluating the impact of detected faults on combat system readiness. The purpose of the information is to support the SERT in carrying out their corrective maintenance management responsibilities of coordinating fault isolation efforts and discerning impact evaluation.

3.10.5.1 Operational fault directories. This information shall include operational fault directories for faults observed during combat system operation. Operational fault directories shall identify hardware and software fault isolation tools employed with each element to isolate operationally related fault symptoms. Operational fault directories for each mission and major combat system functions shall be provided (see figure 40 for a typical presentation format). Faults occurring as a result of loss or degradation of inter-system functions and parameters shall be provided. A narrative shall explain the role of operational fault directories in the combat system maintenance concept.

3.10.5.2 Impact evaluation. This information shall provide SERT with the criteria required to determine the impact of specific faults on combat system capabilities. A table, similar to figure 41, shall list equipment functions and the impact the loss of these functions shall have on specific combat system missions. Primary topics of combat system degradation, trade-off considerations, restoration time, fault isolation constraints, and alternative combat system configurations shall be related.

3.10.6 Fault isolation. This information shall identify and discuss available combat system and element level fault isolation tools and the techniques employed to verify and correct inter-subsystem faults. Additional information shall provide fault isolation procedures and diagrams for specific interfaces. Depending on the number of interfaces involved, information may address the interfaces between a pair of elements or the interfaces of one element with all other elements. Where intra-element interface fault isolation is necessary, information shall be developed to cover that element.

3.10.6.1 Introduction. This information shall introduce and summarize the following two segments or modules and provide a listing of the fault isolation diagrams.

3.10.6.2 Fault isolation techniques. This information shall provide the identification and description of fault isolation tools and procedures, and their effective use. This shall include, but not be limited to, installed operational monitoring equipment, built-in test equipment, performance monitoring software, and PMS testing. The techniques used with each combat system element shall be described in detail. Descriptions shall encompass such factors as when and under what circumstances a particular fault isolation tool should be used, and the degree of reduced capabilities incurred as a result of using a particular fault isolation tool.

3.10.6.3 Introduction to combat system functional interface diagrams (CSFIDs). This information shall introduce and describe the content and intended use of the CSFIDs presented. This information shall include and describe the use of the interface function directory and the CSFID symbology table.

3.10.6.3.1 Functional data flow. In CFIDs, functional data flow will normally be depicted from left to right and top to bottom. However, where this logic is not practical, the direction of functional flow will be represented by arrows.

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3.10.6.3.2 Intersystem tabulation. Intersystem tabulation will be provided to identify cable, connector, terminal, and wire information for analog interfaces. Data fields will be tabulated for intersystem digital interfaces and accessible termination or test points for terminal pin connectors within equipments will be provided.

3.10.6.4 Support system fault isolation. When CSOSS documentation is deemed to be necessary for any support system, information shall be developed to address fault isolation techniques for the various systems that comprise the support system. The information shall include only those support systems for which CSOSS is deficient. Descriptions shall be to level 2 or to level 3 (refer to 3.10.1.1), as applicable, and involve only basic methods of fault isolation. Fault isolation material to support this shall include system schematics similar to figures 42 and 43; tables of fault symptoms and probable causes, similar to figure 44; and a table of audiovisual alarms similar to figure 45. References to the applicability of support systems to individual elements of the combat system shall be provided. The table of audiovisual alarms, in addition to specifying alarm and alarm sensor locations, shall prescribe action to be taken when the alarms are activated. The action, in most cases, will involve preliminary checks to verify and assess the abnormal condition, and a reference to a system publication (provided in the bibliography) for detailed fault isolation procedures. The primary topics shall support, as applicable, fault isolation of the following support system elements:

a. Central dry air.
b. High-pressure air.
c. High-pressure nitrogen.
d. Electronic salt-water liquid coolant.
e. Chilled water liquid coolant.
f. Air conditioning.
g. Steam heating.
h. Salt-water fire main.
i. Anti-icing.
j. Power distribution.
k. Ship parameters.

3.10.6.5 Fault isolation. Ideally, information shall be developed for each pair of interfacing combat system elements. Alternately, information may be developed to cover the interfaces involving a central element. In addition, information may be developed to cover the intra-system interfaces for a combat system element where system-level documentation does not exist. When descriptions are required to support the fault isolation combat system elements, the descriptions shall be developed to level 2 or to level 3, as applicable, and involve only basic methods of fault isolation. The introduction shall list the fault isolation diagrams contained in the information.

4. VERIFICATION

4.1 Verification. The verification requirements shall be in accordance with MIL-DTL-24784.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The TMs prepared to this specification are intended to be used for installation, operation, maintenance, repair, and parts support of naval CSTOMs; HM&E systems and equipment; electronic (including service test electronic, experimental electronic, and IC) systems and equipment; and weapon systems and equipment. The TMs should be used as a training document in the classroom and as a source of on-the-job training.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification (or any Technical Manual Contract Requirement [TMCR] referencing this specification).

b. Type and quantity of the manual required (see 3.1).

c. The maintenance coverage in accordance with the technical content requirements provided herein (see 3.2).

d. Warranty and guarantee information (see 3.4.1).

e. If system/equipment characteristics/capabilities information are to be provided (see 3.6.5).

f. The use of a table of established values or an adjustment effectivity list (see 3.8.2.1).

g. If procedures and instructions for performing maintenance that is beyond the capabilities of the ship's force are to be provided (see 3.8.2.5).

h. If drawings and information concerning system/equipment installation are to be provided (see 3.8.3).

i. If an IPB is to be developed (see 3.9).

j. Packaging requirements (see 5.1).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been authorized and assigned an Acquisition Management Systems Control (AMSC) number or a TMCR based on those specifications and standards must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4 Definitions. The words or phrases used throughout this specification are defined in MIL-DTL-24784.

6.5 Acronyms. The acronyms used in this specification sheet are defined in MIL-DTL-24784.

6.6 Subject term (key word) listing.

Descriptive Information

Illustrated Parts Breakdown

Installation

Maintenance

Operation

Operational Checkout

Procedural Information

Troubleshooting

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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SAFETY PRECAUTION INFORMATION PACKAGE
MK 20 MOD 0 RADAR

GENERAL SAFETY PRECAUTIONS

The following are general safety precautions that are not related to any specific procedures and therefore do not appear elsewhere in this manual. These are recommended precautions that personnel must understand and apply during all phases of operation. Should situations arise that are not covered in these safety precautions, the Commanding Officer or other authority will issue orders as necessary to cover the situation.

KEEP AWAY FROM LIVE CIRCUITS

Operational and maintenance personnel must observe all safety regulations. Do not replace components or make adjustments inside equipment with the power turned on. Under certain conditions, dangerous potentials may exist even after power is removed due to charges retained by internal capacitors. To avoid bodily injury, always remove power and discharge a circuit by grounding before touching it.

DO NOT SERVICE OR ADJUST ALONE

Under no circumstances should any person reach inside any cabinet for the purpose of servicing or adjusting the equipment except in the presence of someone who is capable of rendering first aid. Whenever the nature of the operation permits, keep one hand away from equipment to reduce the hazard of current flowing through the body.

FIRST AID

An injury, no matter how slight, shall never go unattended. Always obtain first aid or medical attention immediately, and file an injury report in accordance with OPNAVINST 5102.1 series, subj: Mishap Investigation and Reporting.

RESUSCITATION

Personnel working with or near high voltage shall be familiar with approved methods of resuscitation. Should someone be injured and stop breathing, begin resuscitation immediately. A delay could cost the victim's life. Resuscitation procedures shall be posted in all electrically hazardous areas.

DANGER, WARNING, AND CAUTION STATEMENTS

Dangers, warnings, and cautions appear throughout this manual. It is important that the significance of each be thoroughly understood by personnel using this manual. Definitions are as follows:



DANGER is used to highlight a statement or some other notification about an operating or maintenance procedure, practice, or condition, that if not strictly observed will result in death or serious injury, or threatens the primary mission of the ship.



WARNING is used to highlight a statement or some other notification about an operating or maintenance procedure, practice, or condition, that if not strictly observed could result in death, injury, or long-term health hazards.

FIGURE 1. Example content for safety summary (IM/IP format).

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CAUTION

CAUTION is used to highlight a statement or some other notification about an operating or maintenance procedure, practice, or condition, that if not strictly observed could result in damage to or destruction of equipment, or loss of mission effectiveness.

The following dangers, warnings, and cautions appear in the text of this manual, and are repeated here for emphasis.

⚠ DANGER

115v present on synchro when radar is in the off mode.
006 00 (page 2), 006 00 (page 12), 009 00 (page 1), 022 00 (page 7), 022 00 (page 24)

⚠ WARNING

Ensure that adjustment screw is not grounded with screwdriver.
008 00 (page 6)

⚠ WARNING

Motor unit weight is approximately 170 pounds. Utilize appropriate lift-rigging and support devices to avoid injury to personnel.
009 00 (page 5), 009 00 (page 15)

CAUTION

Use care when disconnecting rigid coax so as not to bend or damage assembly.
009 00 (page 16), 010 00 (page 2), 012 00 (page 20)

FIGURE 1. Example content for safety summary (IM/IP format) – Continued.

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WARNINGS APPLICABLE TO HAZARDOUS MATERIALS

Warnings for hazardous materials listed in this manual are designed to warn personnel of hazards associated with such items when they come in contact with them by actual use. Additional information related to hazardous materials is provided in OPNAVINST 5100.23, Navy Occupational Safety and Health (NAVOSH) Program Manual, NAVSUPINST 5100.27, Navy Hazardous Material Control Program, and the DoD 6050.5, Hazardous Materials Information System (HMIS) series publications. For each hazardous material used within the Navy, a Material Safety Data Sheet (MSDS) must be provided and available for review by users. Consult your local safety and health staff concerning any questions regarding hazardous materials, MSDSs, personal protective equipment requirements, and appropriate handling and emergency procedures and disposal guidance.

Symbols are used within the manual to provide quick recognition of the hazard by the user. The following provides an explanation of the hazard symbols and gives the complete warnings for hazardous material used in this manual.

EXPLANATION OF HAZARDOUS MATERIALS SYMBOLS



CHEMICAL - drops of liquid on hand shows that the material will cause burns or irritation to human skin or tissue.



FIRE - flame shows that a material may ignite and cause burns.



EYE PROTECTION – person with goggles shows that the material will injure the eyes.



VAPOR - human figure in a cloud shows that material vapors present a danger to life or health.

HAZARDOUS MATERIALS WARNINGS

In the text of the manual, the caption WARNING is not used for hazardous material warnings; hazards are cited with appropriate icon(s), and the nomenclature of the hazardous material that relates to the complete warning identified below. Users of hazardous materials shall refer to these complete warnings.



Acrylic Lacquer, MIL-L-81352

Acrylic lacquer, MIL-L-81352, is toxic, flammable, and highly irritating to the eyes. Protection: chemical splash proof goggles, gloves, and good ventilation. Keep container closed; keep sparks, flames, and heat away. Keep lacquer off skin, eyes, and clothes. Do not breathe vapors.

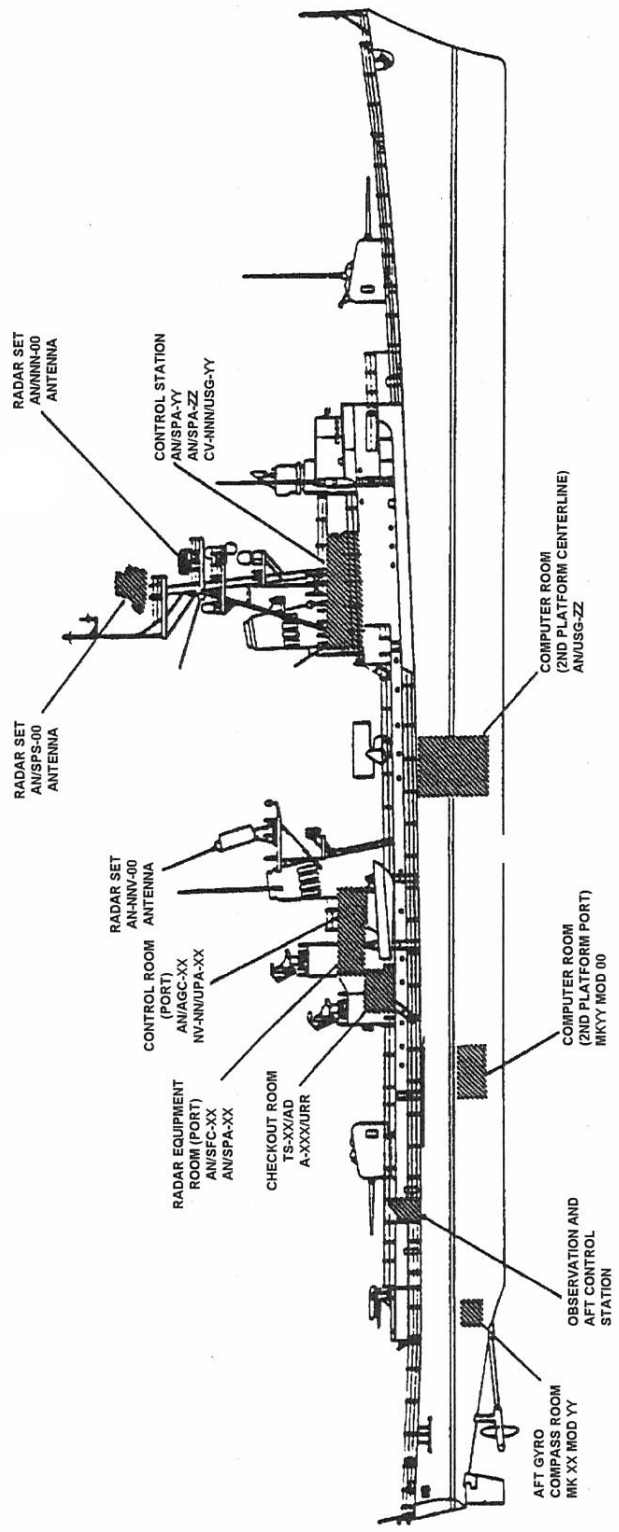


Drycleaning Solvent, P-D-680

Drycleaning solvent, P-D-680, Type II, is toxic and combustible. Protection: chemical splash proof goggles, gloves, and forced ventilation (or respirator). Keep container closed; keep sparks, flames, and heat away. Keep solvent off skin, eyes, and clothes. Do not breathe vapors.

FIGURE 1. Example content for safety summary (IM/IP format) – Continued.

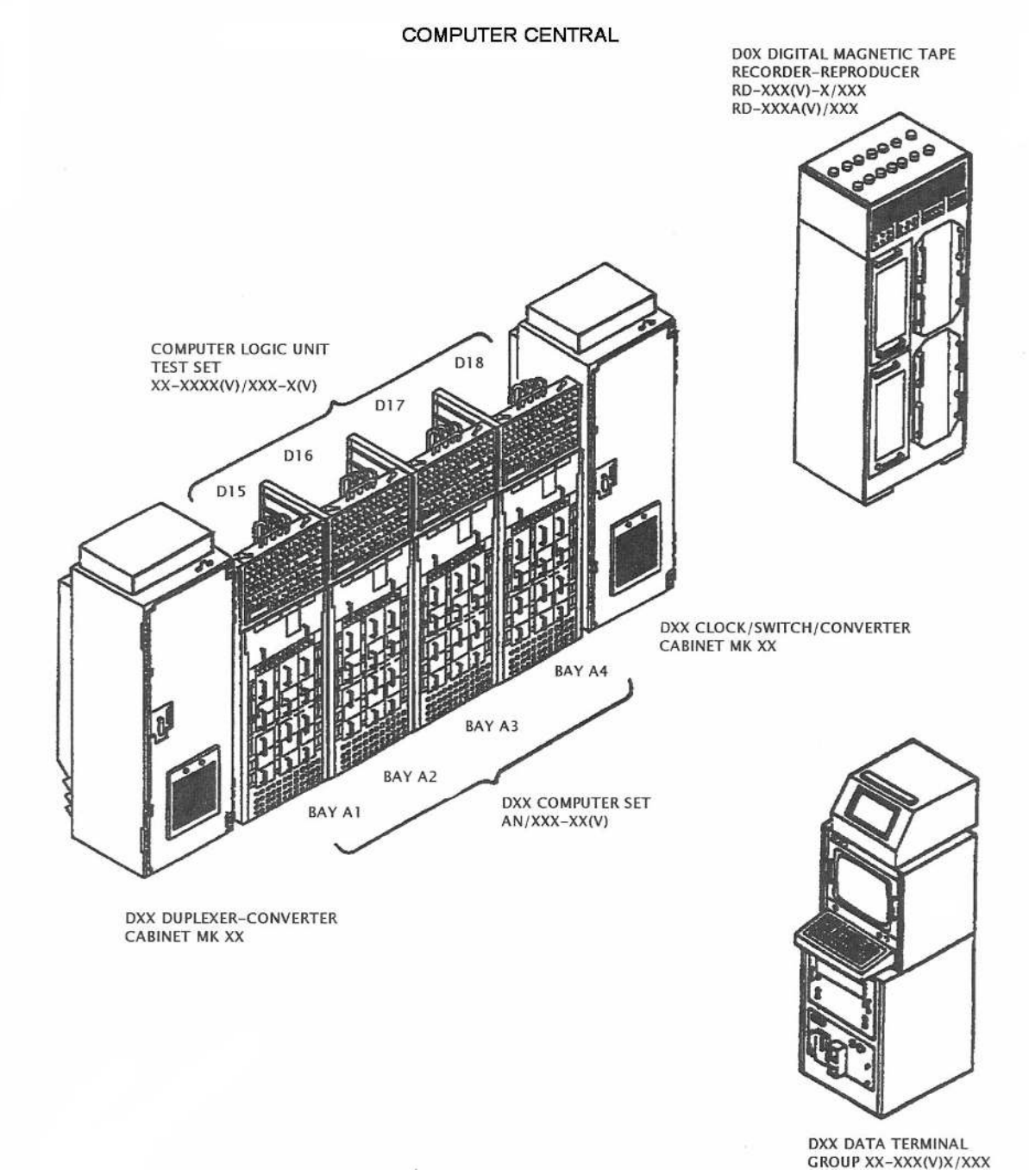
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 2. System compartments.

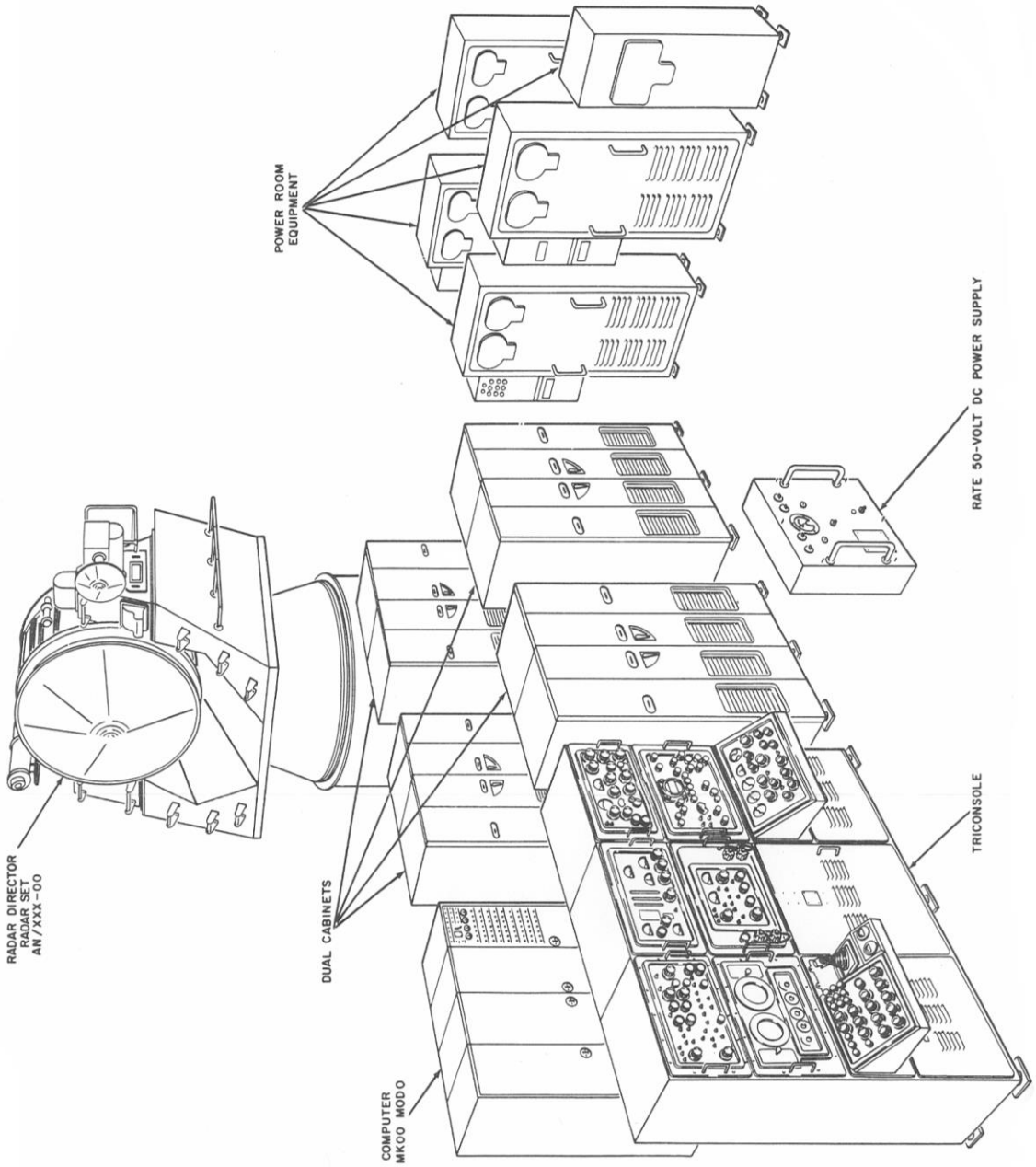
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 3. Compartment areas.

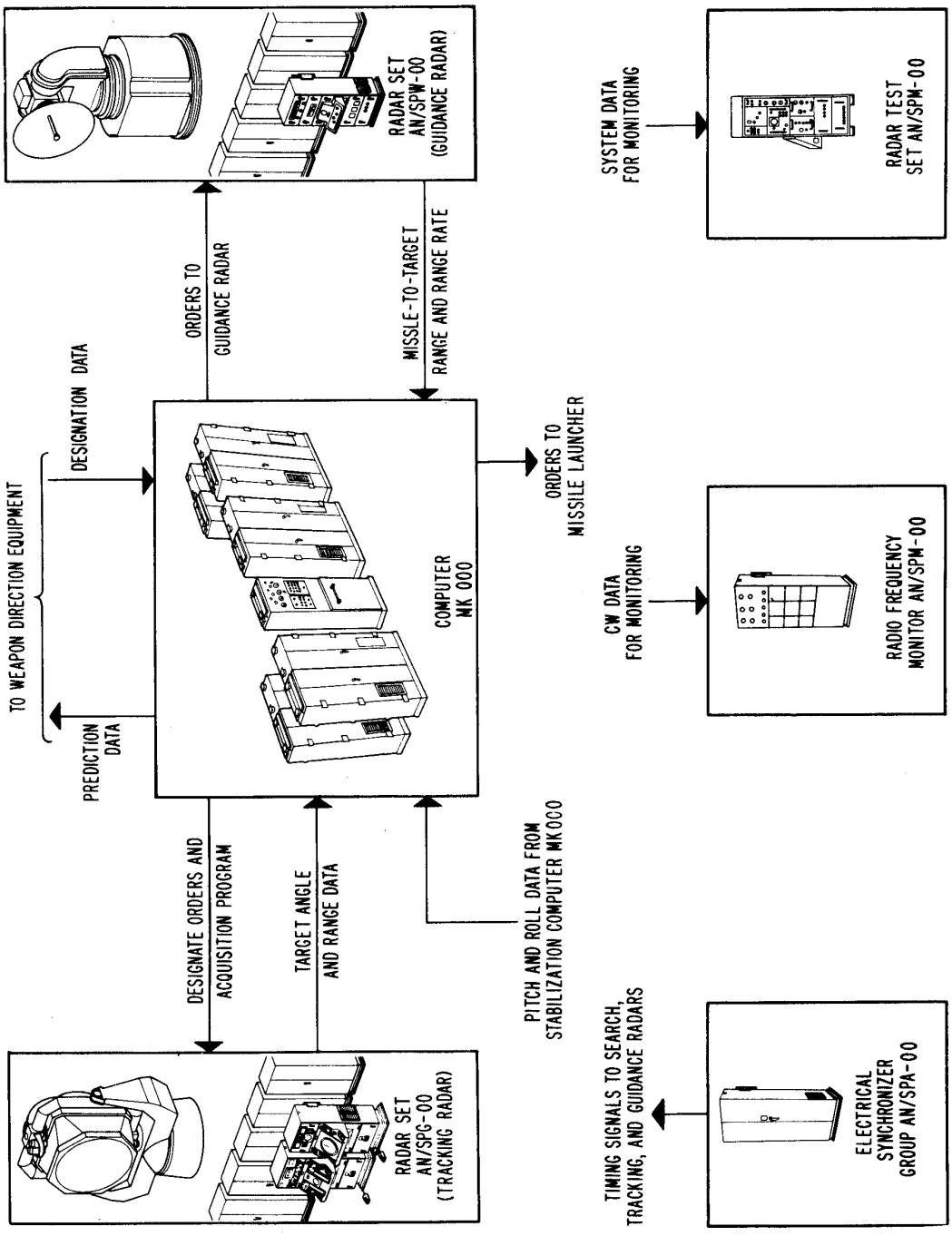
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 4. Weapon system/subsystem/equipment group (typical).

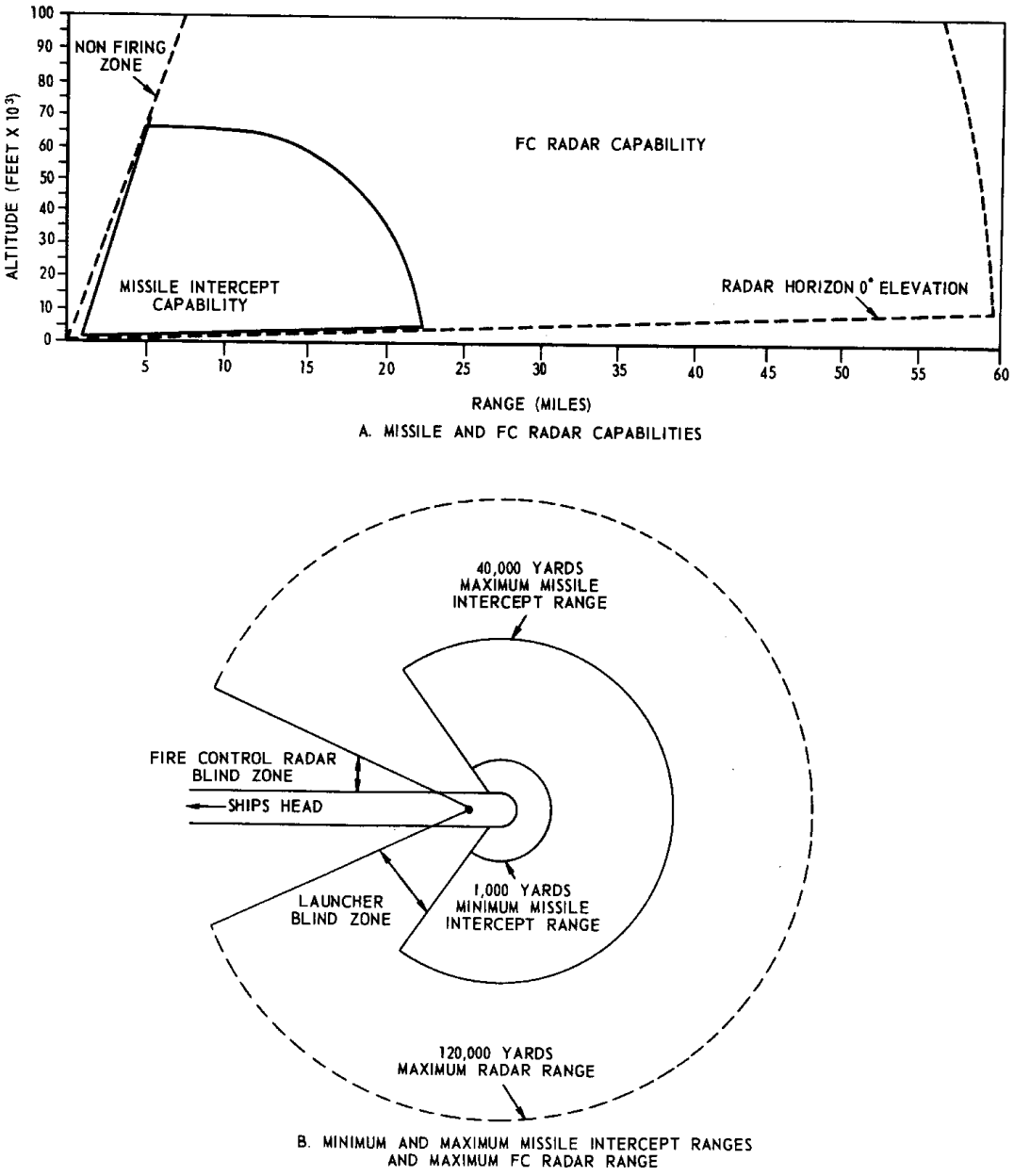
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 5. Equipment/system integration illustration.

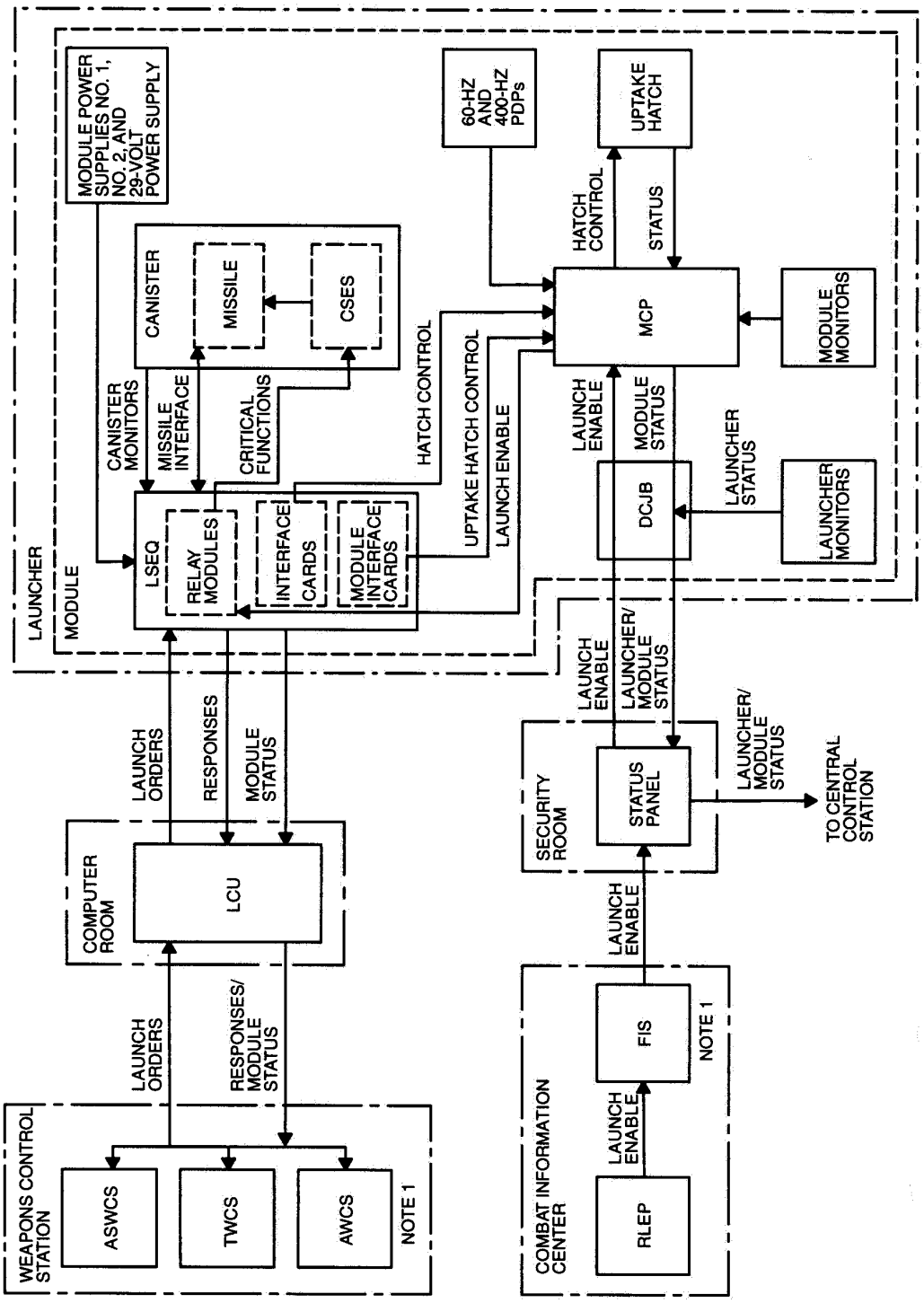
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 6. System capability envelope drawing.

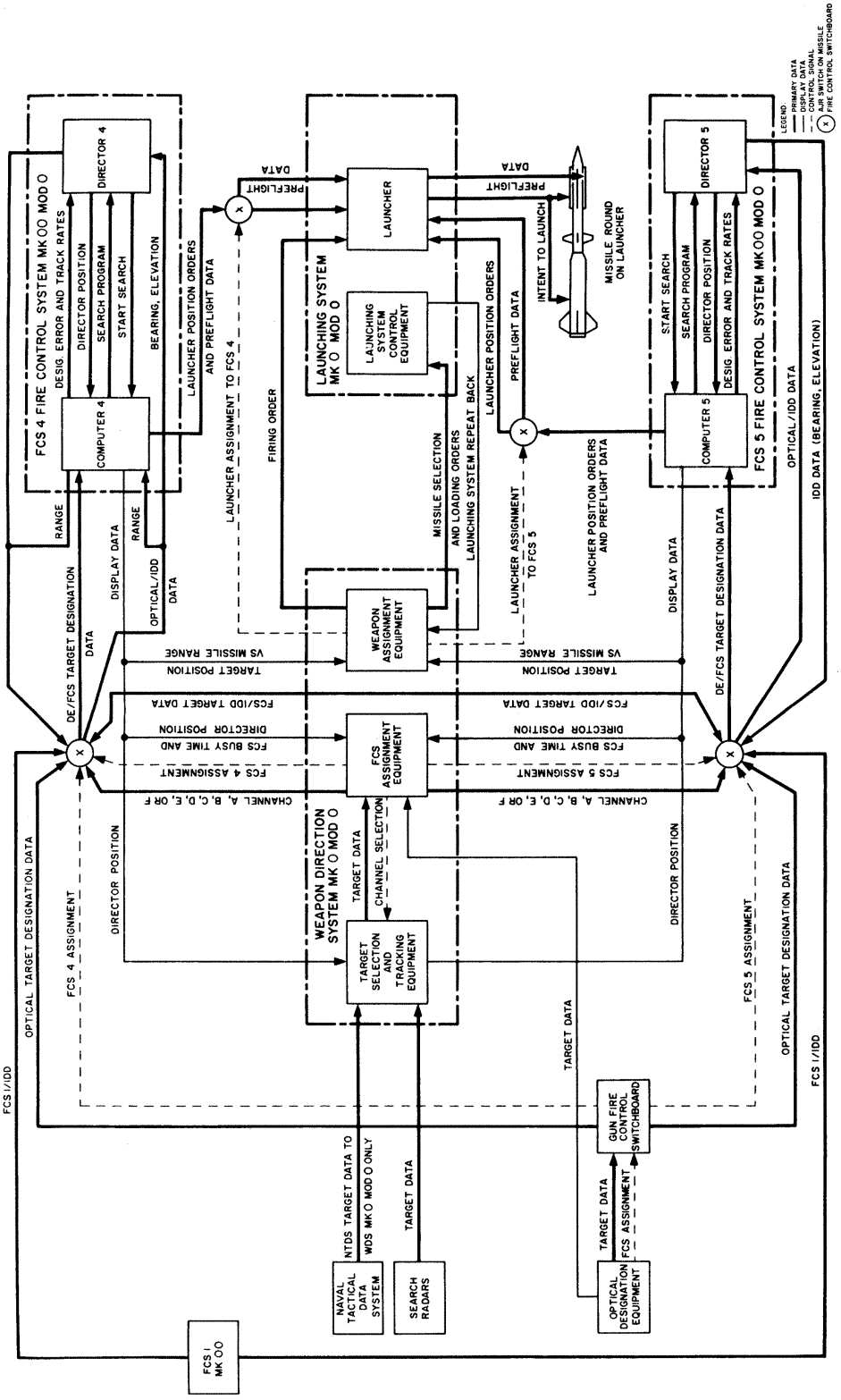
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 7. System block diagram.

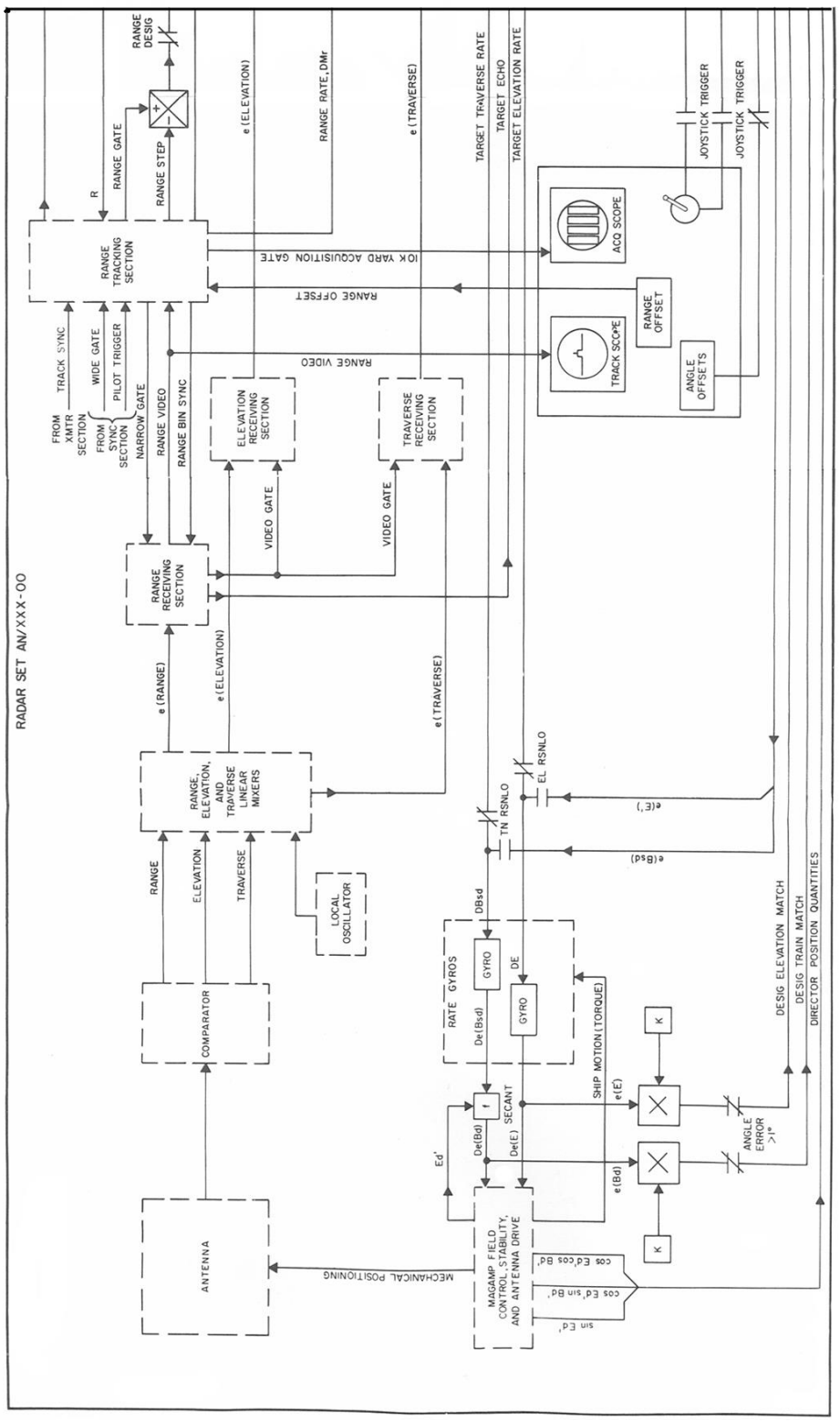
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 8. Functional block diagram, interface.

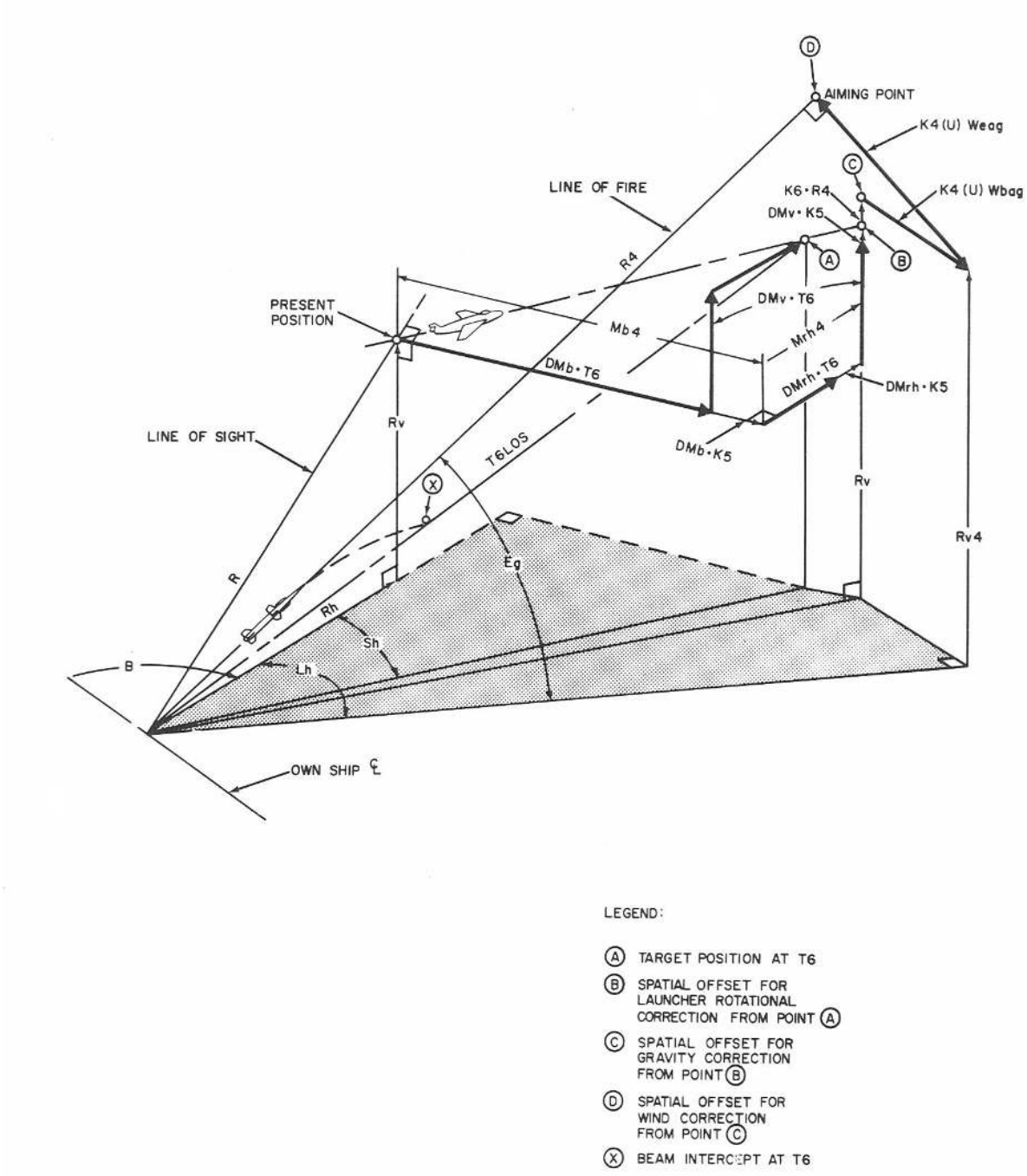
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 9. Functional diagram, simplified.

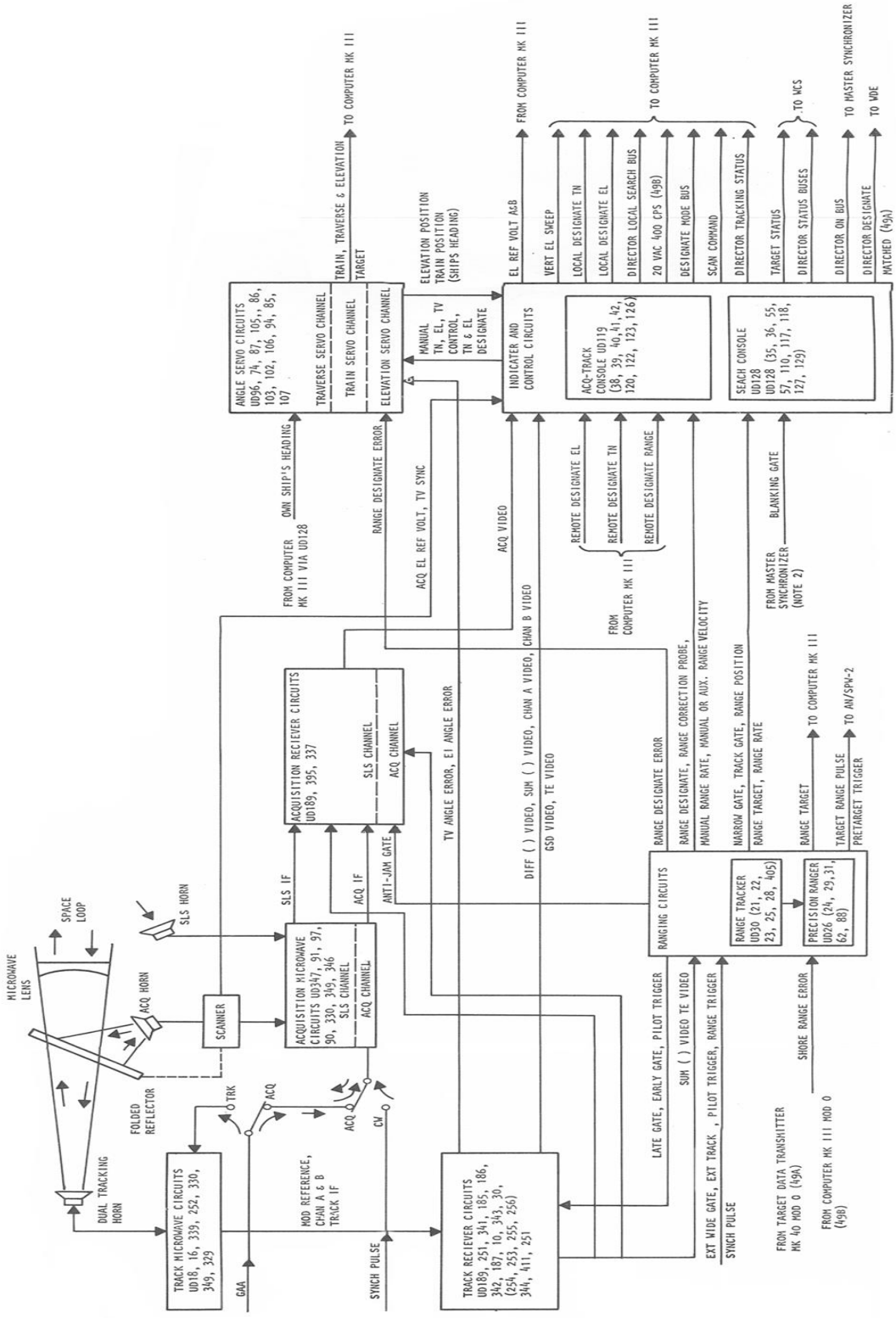
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FIGURE 10. Simplified geometric diagram.

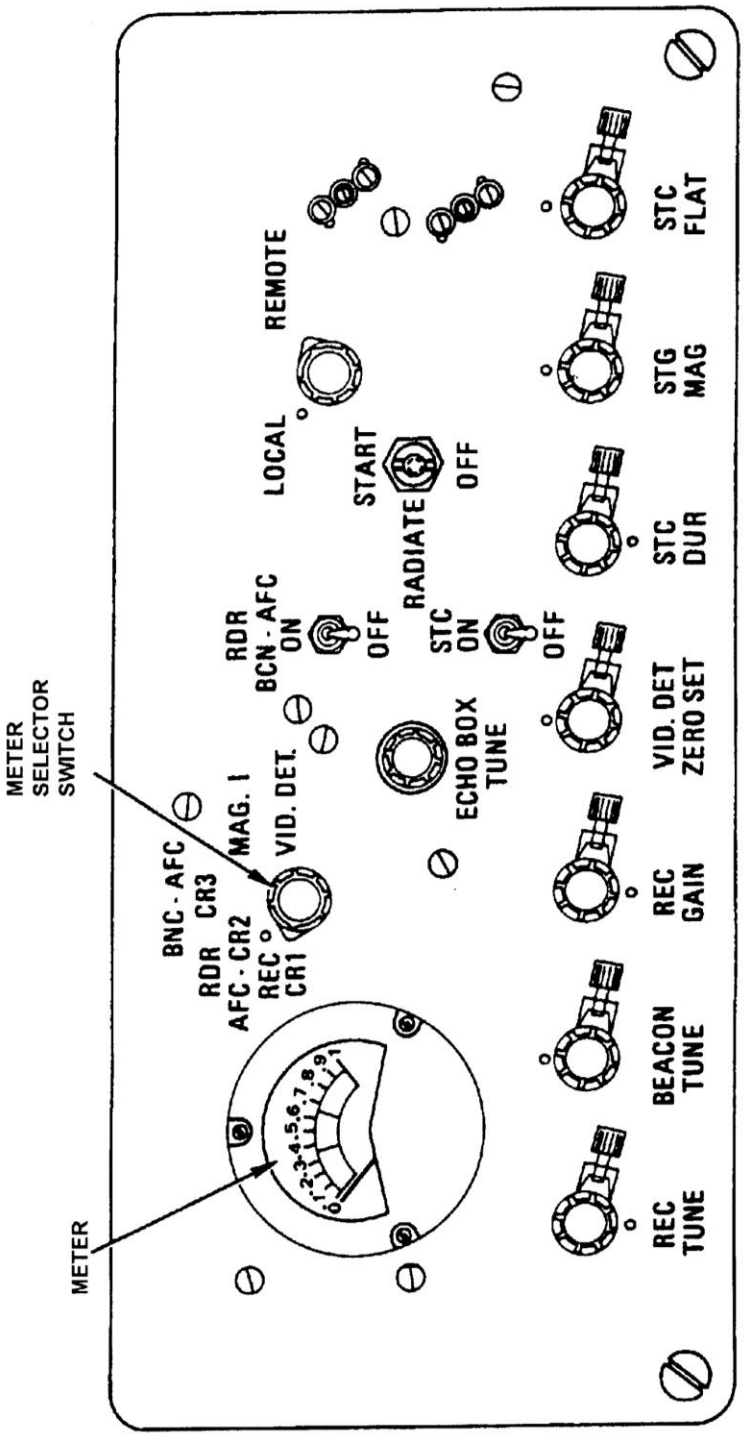
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 11. Equipment/system relationship block diagram.

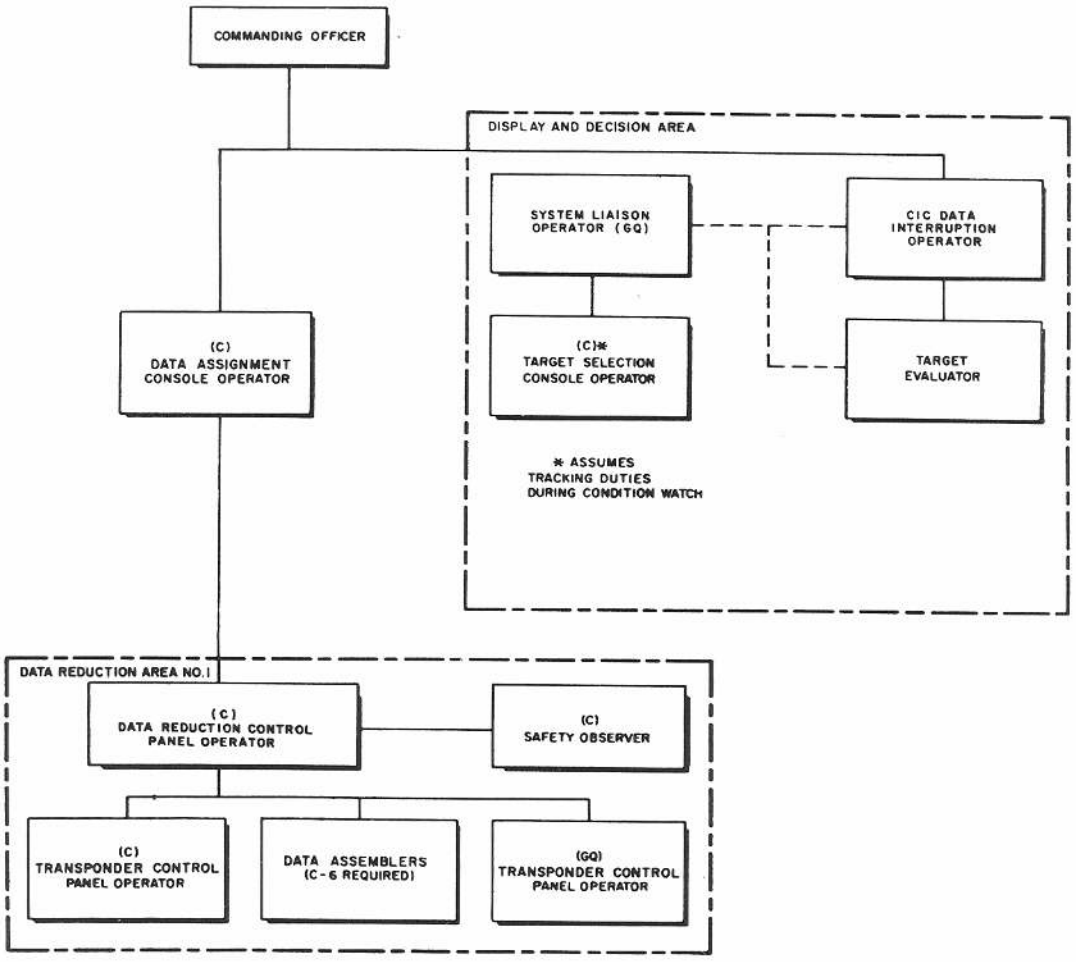
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 12. Equipment controls and indicators.

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FIGURE 13. Conditions of readiness, general quarters conditions.

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CLASSIFICATION
NAVORD OP 0000 (PMS/SMS) VOLUME 0 PART 0

TABLE 0-0. RADAR SET AN/XXX-00, TABLE OF ESTABLISHED VALUES

Parameter	Value		Establishing Procedure
	PC IN	PC OUT	
Absolute Range Calibration			MRC R-12
Precision Range Gear Train Range Dial Slippage	____	____	
Range of 2nd Range Calibrator Return	____	____	
Range of 6th Range Calibrator Return	____	____	
Range of 11th Range Calibrator Return	____	____	
Range of 16th Range Calibrator Return	____	____	
Range of Maximum Range	____	____	
Range Calibrator Return	____	____	
ACQ-SLS Test Fixture Attenuator Setting AT(20)11:			
SLS switch at BLANK	____	____	
SLS switch at CANCEL	____	____	
Acquisition Receiver Sensitivity (Pulse Compression Out)	____	____	MRC R-19
Angle Servo Input (Elevation):			1. Perform MRC R-5 through R-11.
Second Range Calibrator Return:			2. Perform test of figure 5-62, sheet 5, box 19 and record values.
3 Mils Up	____	____	
1 Mil Up	____	____	
1 Mil Down	____	____	
3 Mils Down	____	____	
4:1 Signal-to-Noise Range Calibrator Return:			
3 Mils Up	____	____	
1 Mil Up	____	____	
1 Mil Down	____	____	
3 Mils Down	____	____	
2:1 Signal-to-Noise Range Calibrator Return:			
3 Mils Up	____	____	
1 Mil Up	____	____	
1 Mil Down	____	____	
3 Mils Down	____	____	

UNCLASSIFIED WHEN VALUE COLUMN BLANK
CLASSIFICATION

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 14. Table of established values.

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Table 4-1. Adjustment Effectivity List

MR Number	Adjustment/Alignment Procedures	Output
R-1	1. Adjust VERT. POS. RANGE	Preamplifier output dc reference range.
	2. Adjust 2ND STAGE PLATE BAL. and VAR. ATTEN. BAL.	Preamplifier output dc reference stability.
R-2	1. Adjust GAIN ADJ.	Preamplifier gain calibration.
	2. Adjust attenuator section balance.	Attenuator dc differential balance.
	3. Adjust HF differential balance capacitor.	Preamplifier high frequency differential balance.
R-3	1. Adjust output compensation.	Preamplifier high frequency response.
	2. Adjust input capacitors.	Square-cornered, flat-topped square wave display.
	3. Adjust attenuator capacitors.	Square-cornered, flat-topped square wave display.

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 15. Adjustment effectivity list.

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RADIO RECEIVER R-XXX/URR

NAVSEA

INSTALLATION STANDARDS SUMMARY

Input Voltage _____Vac
Input Frequency _____Hz
(When reference standard tests are made)

Date _____
Serial Number _____
of Model _____
Installed in (ship or station)
_____
Length of transmission line
_____

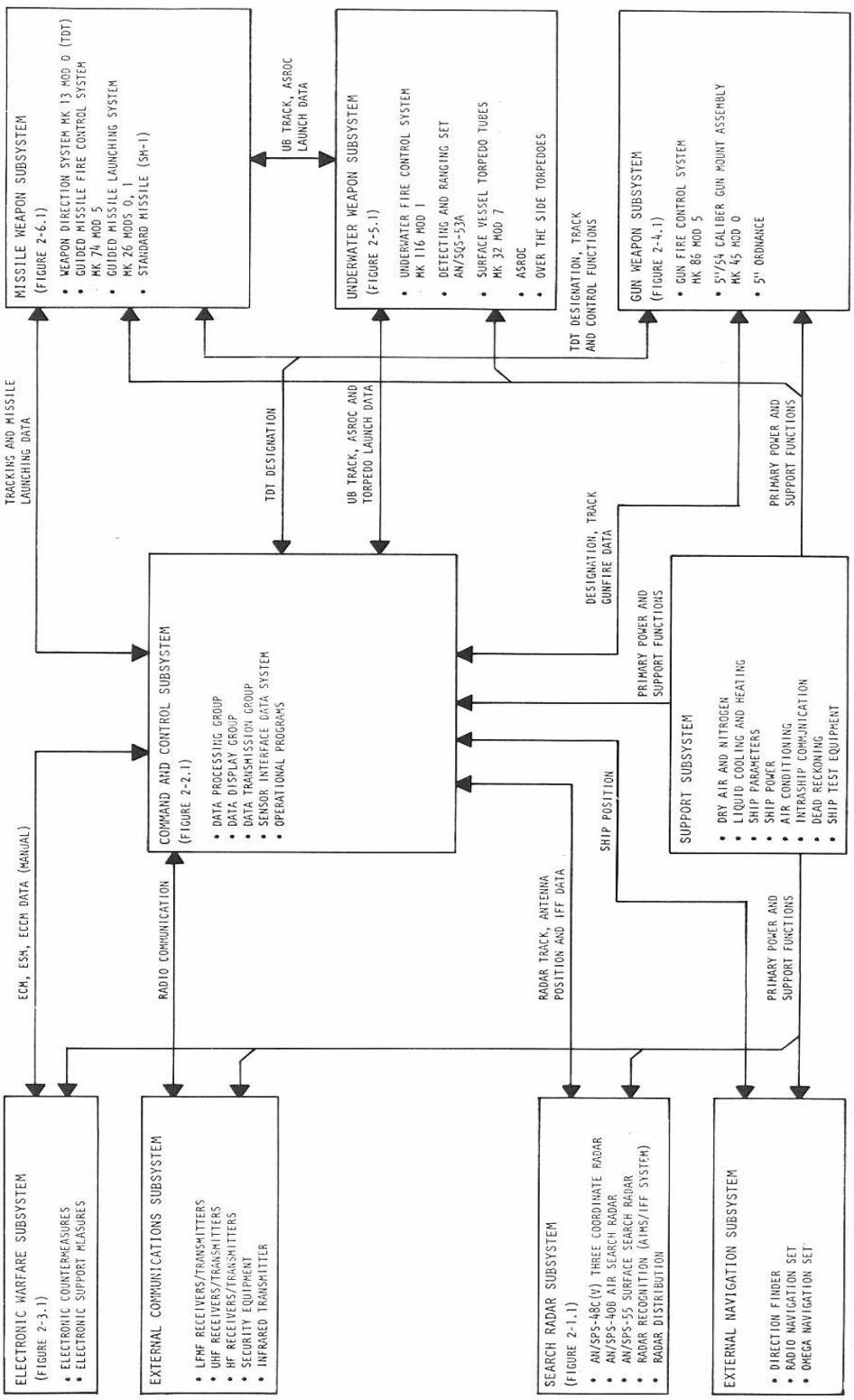
Record on this summary sheet the test indications which have been obtained during the installation verification test.

Paragraph No.	Observed Value	Reference Standard
8-10	a._____ Check	
8-21	a._____ VDC	a. 1.0 to 1.2 VDC
	b._____ VDC	b. 2.35 to 2.36 VDC
	c._____ VDC	c. 24.0 to 24.5 VDC
	d._____ VDC	d. 12.6 to 12.8 VDC
8-33	a._____ VDC	a. 2.5 to 2.7 VDC
	b._____ VDC	b. 11.3 to 11.7 VDC
	c._____ VDC	c. 3.0 to 3.6 VDC
	d._____ VDC	d. 6.6 to 7.0 VDC

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 16. Installation standards summary sheet.

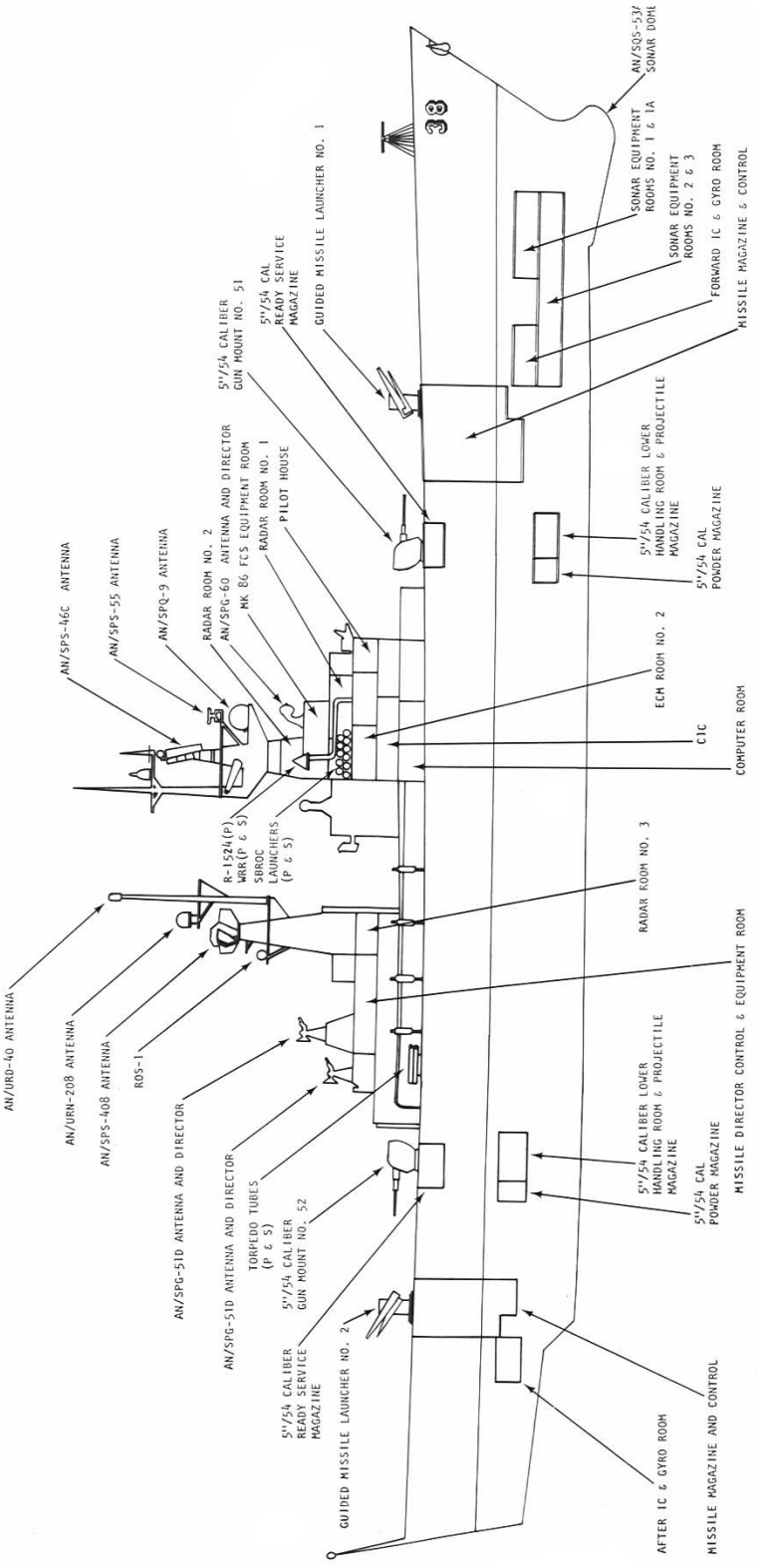
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 17. Combat system level 1 block diagram.

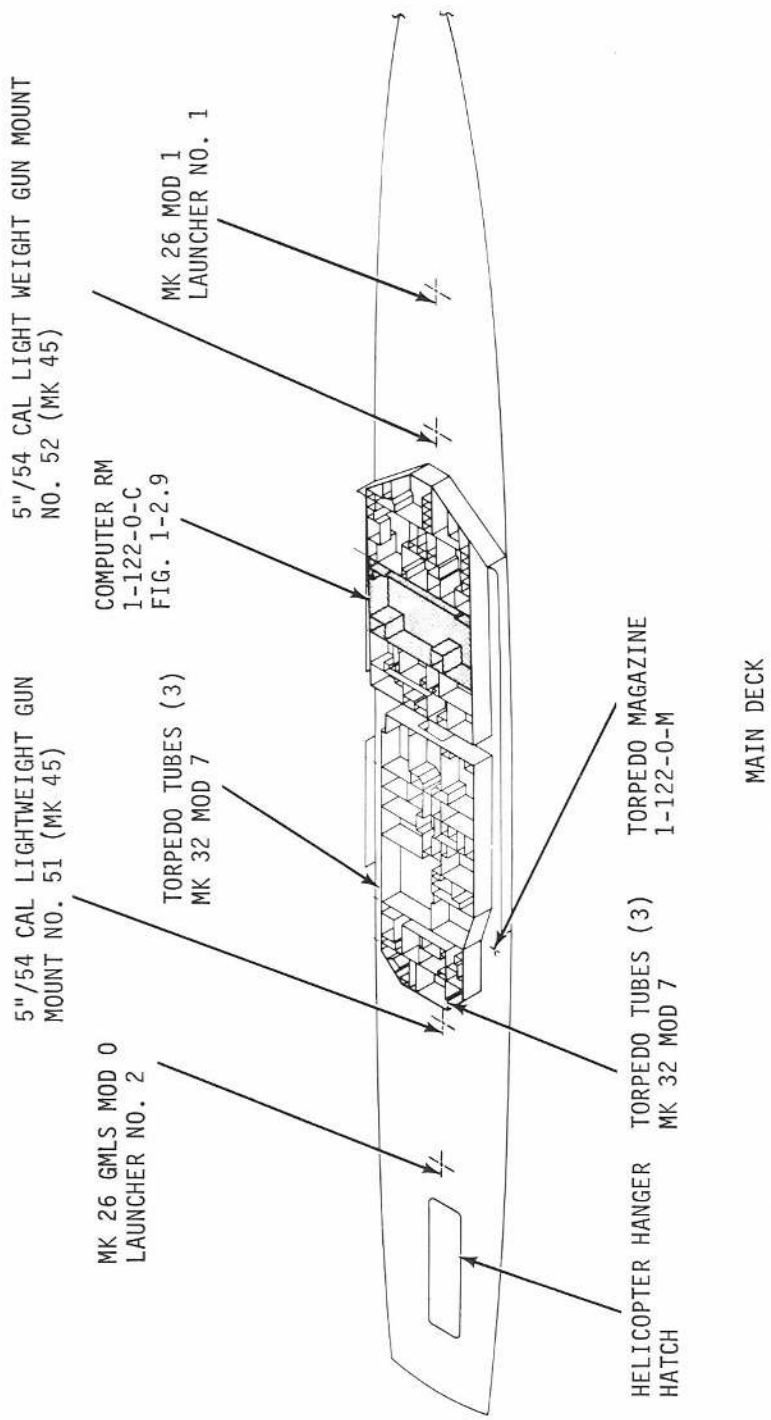
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 18. Ship profile and external components pictorial.

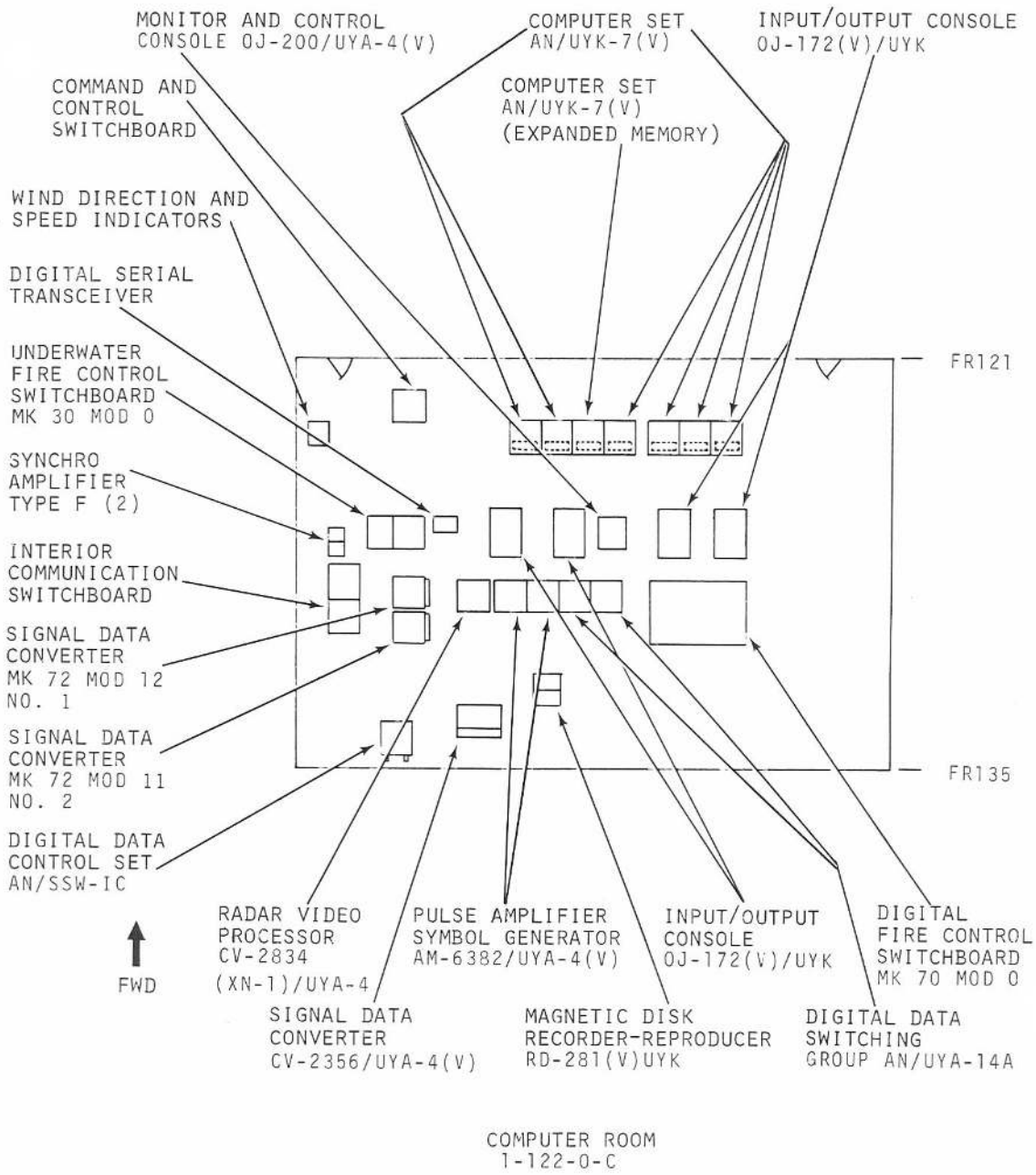
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Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 19. Three dimensional deck and compartments arrangement pictorial.

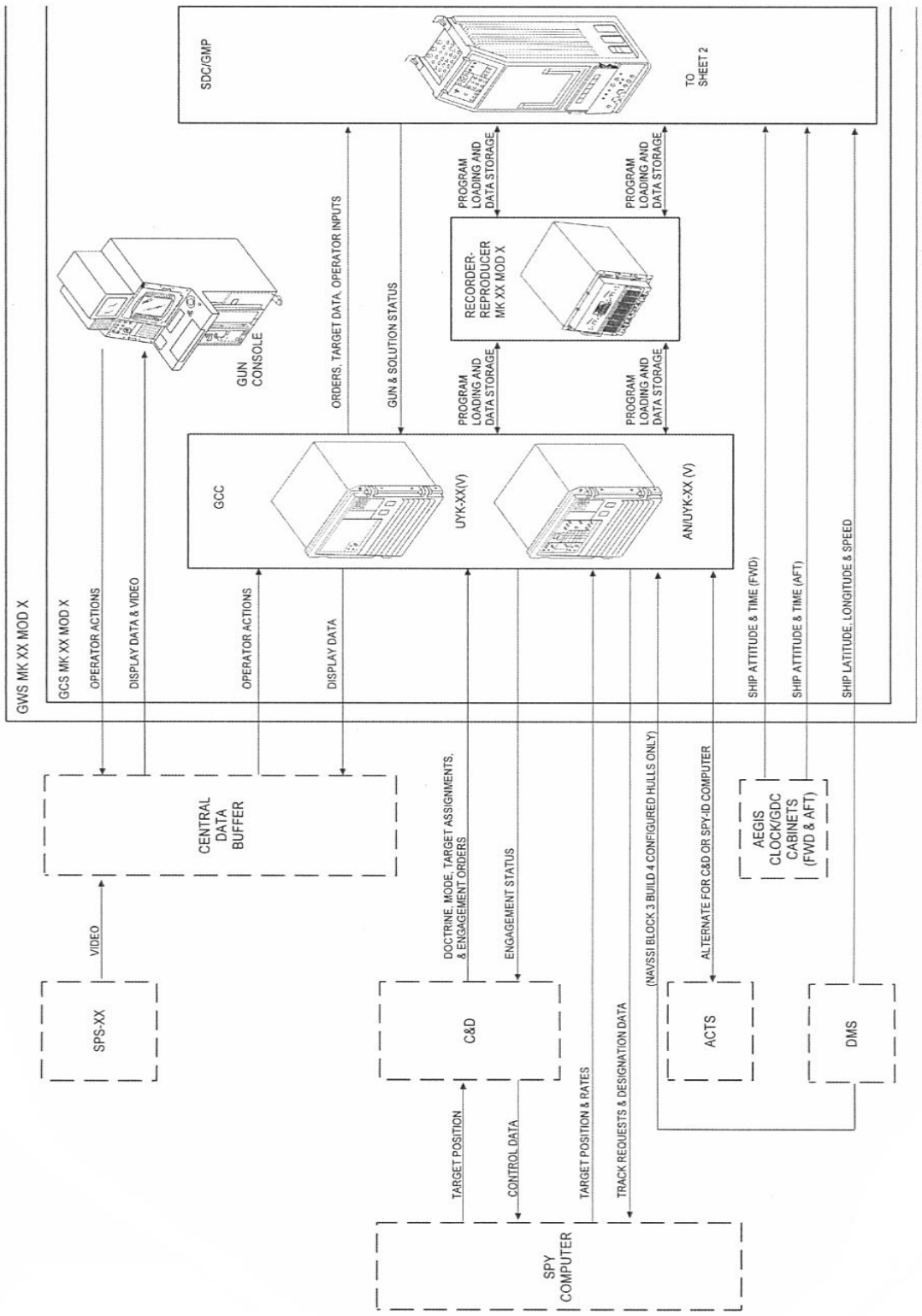
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Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 20. Room arrangement diagram.

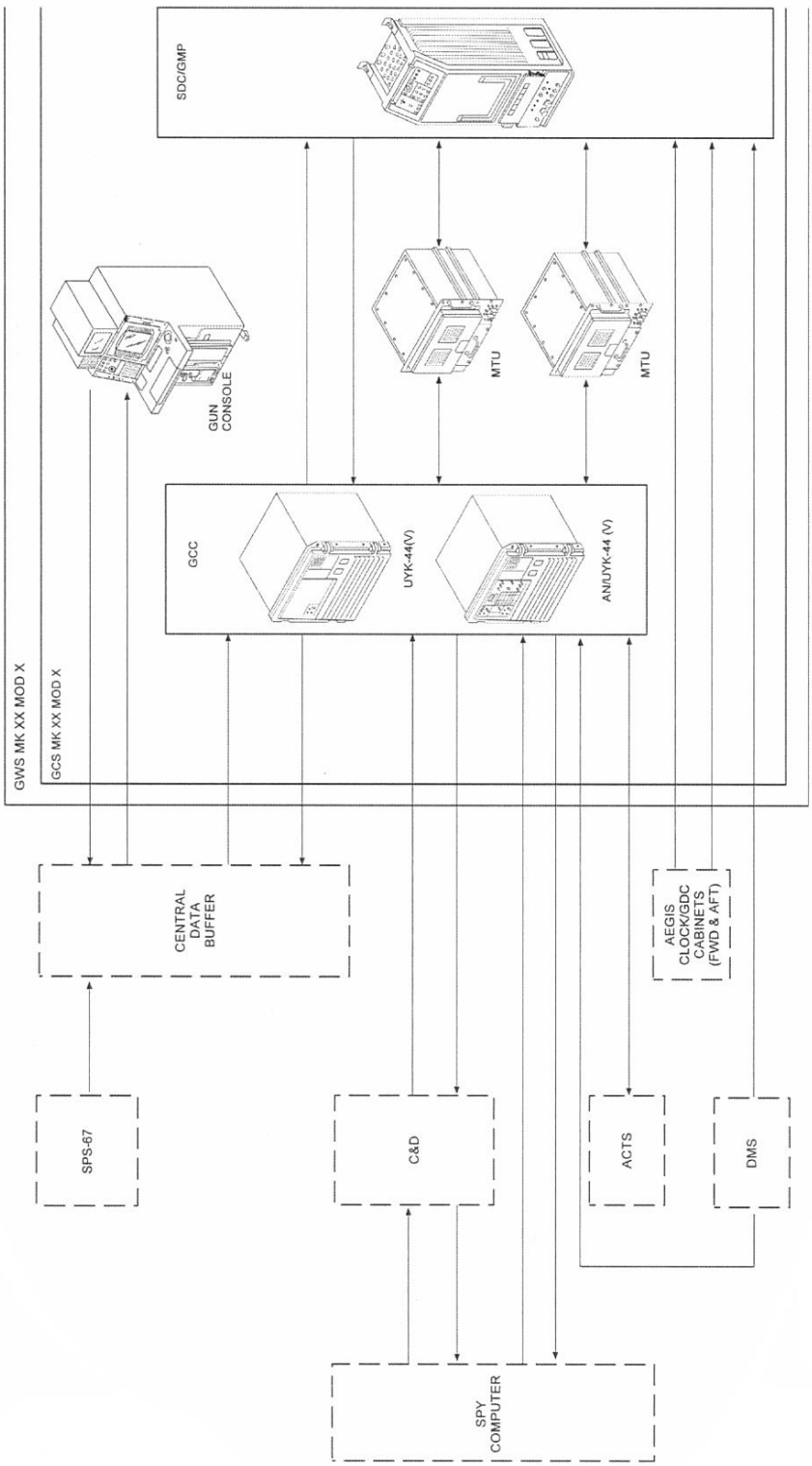
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:34Z
Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 21. Combat system level 1 pictorial.

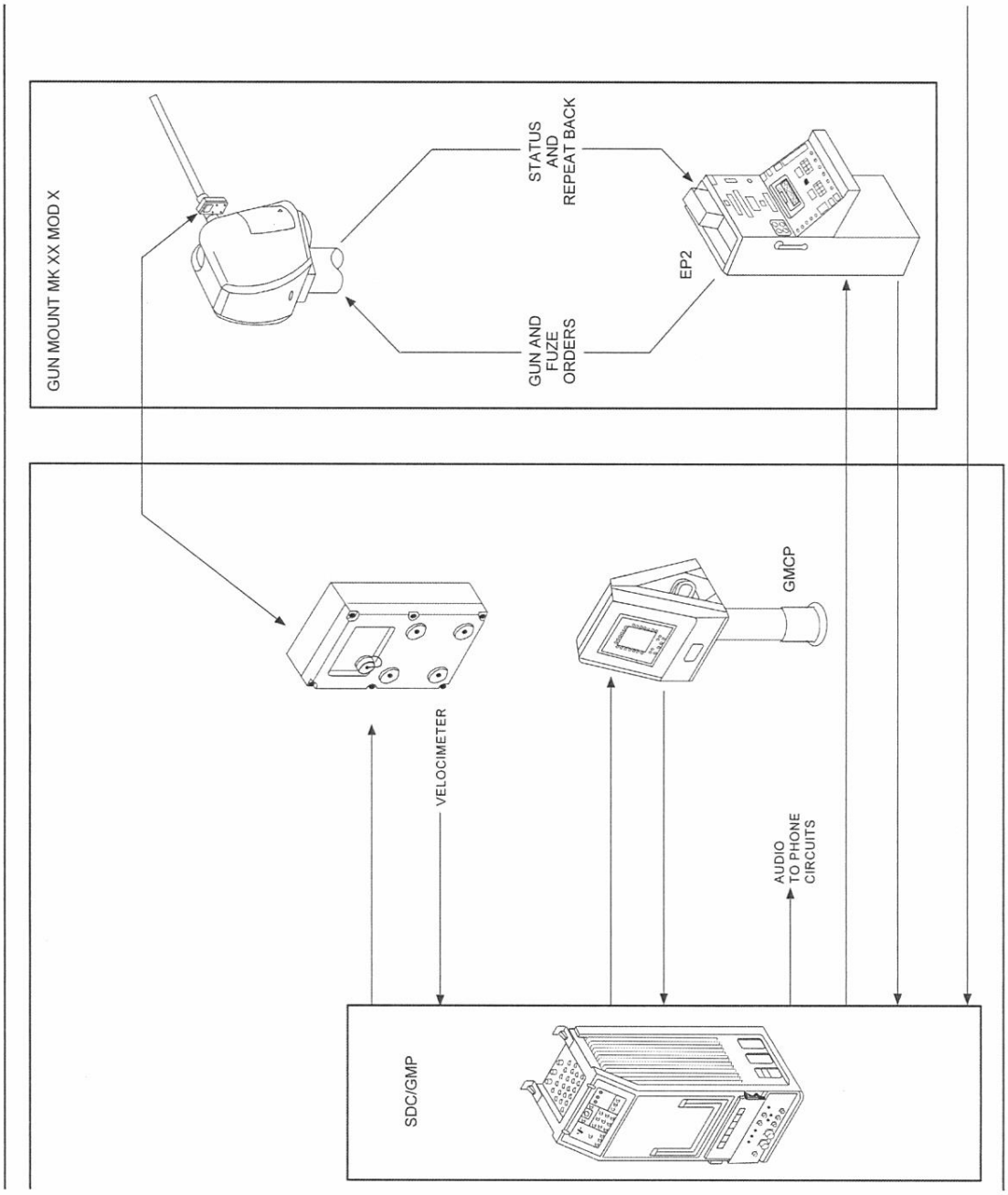
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 22. System level 2 pictorial.

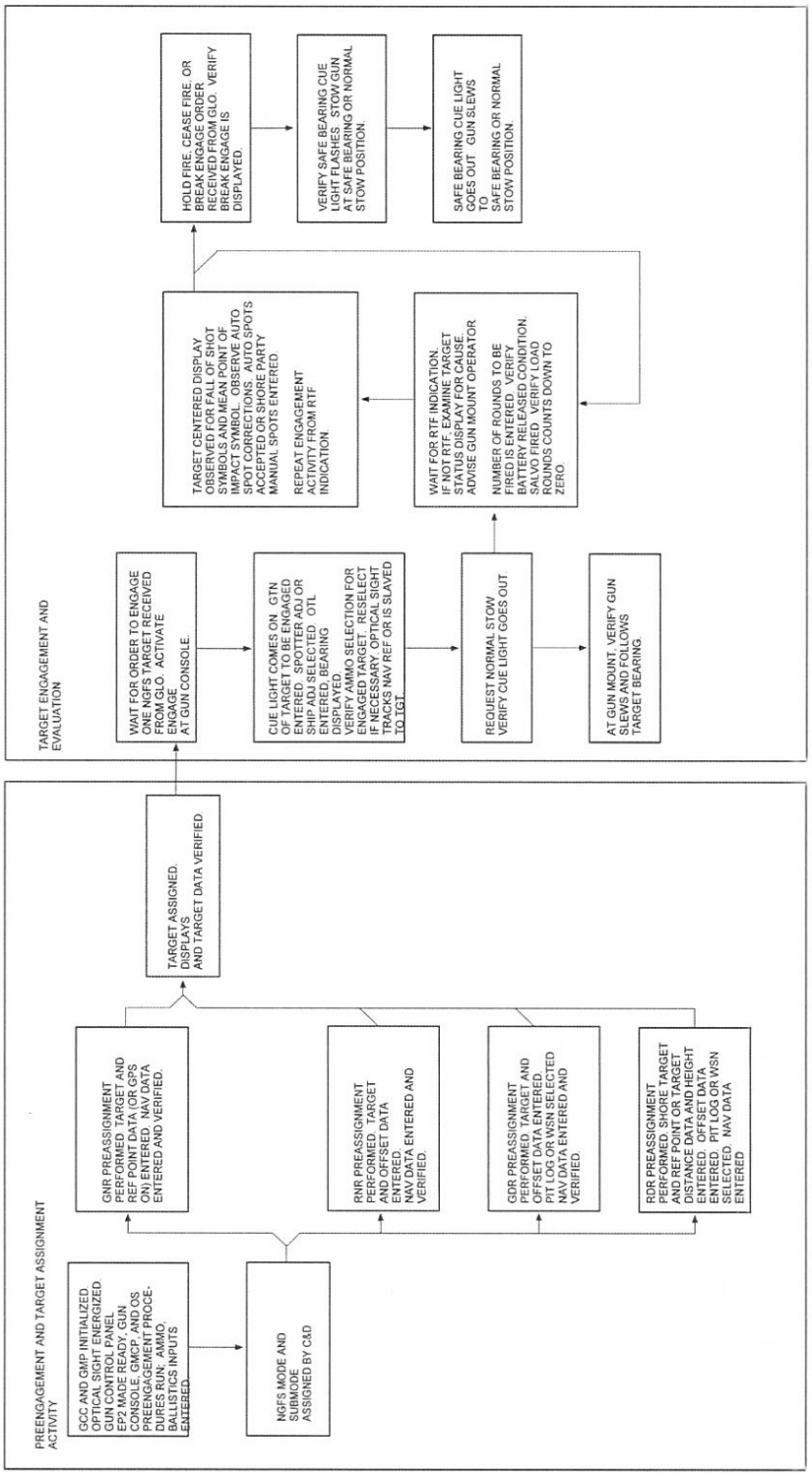
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Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 23. Equipment level 2 pictorial.

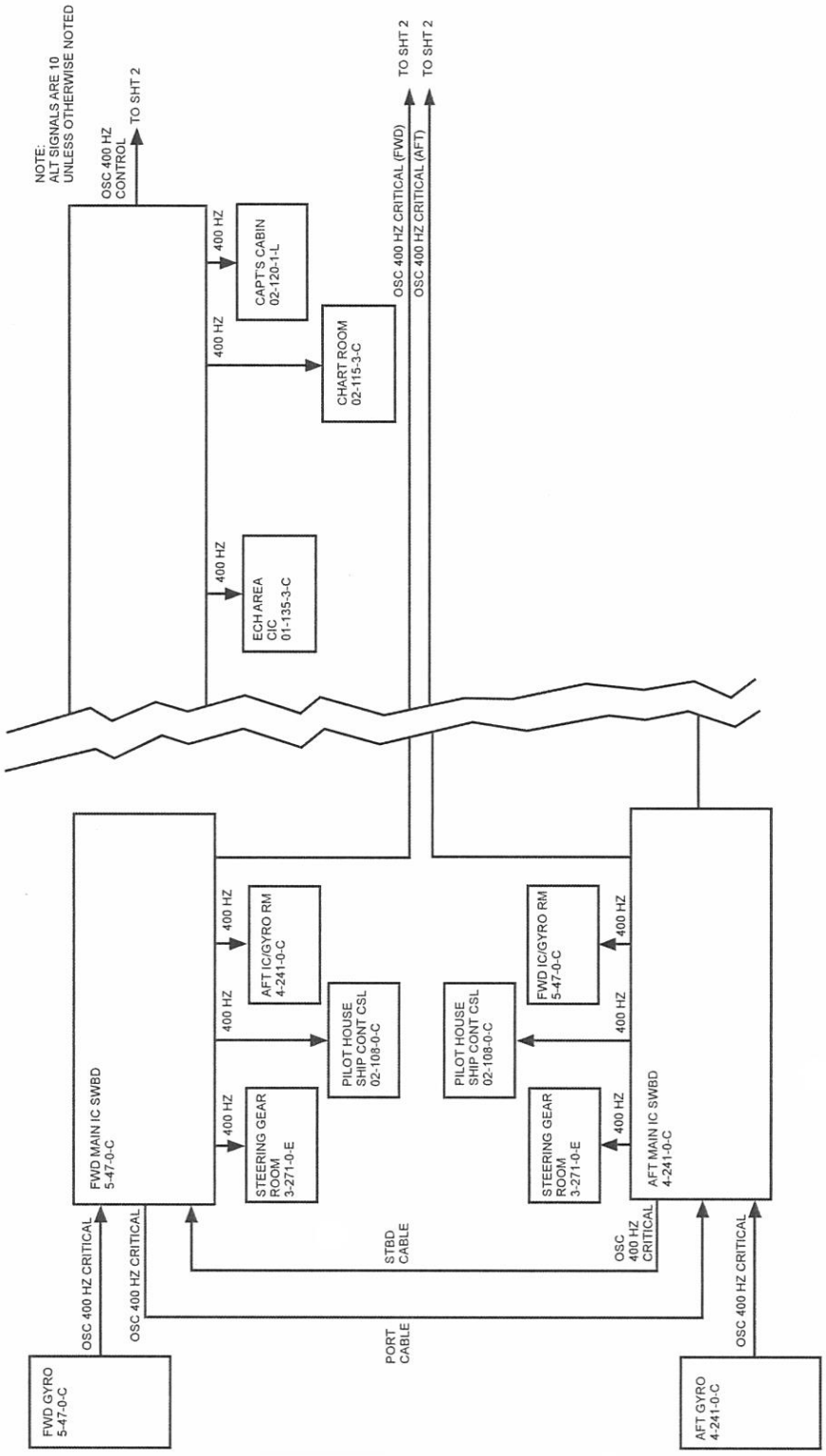
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Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 24. System operational program block diagram.

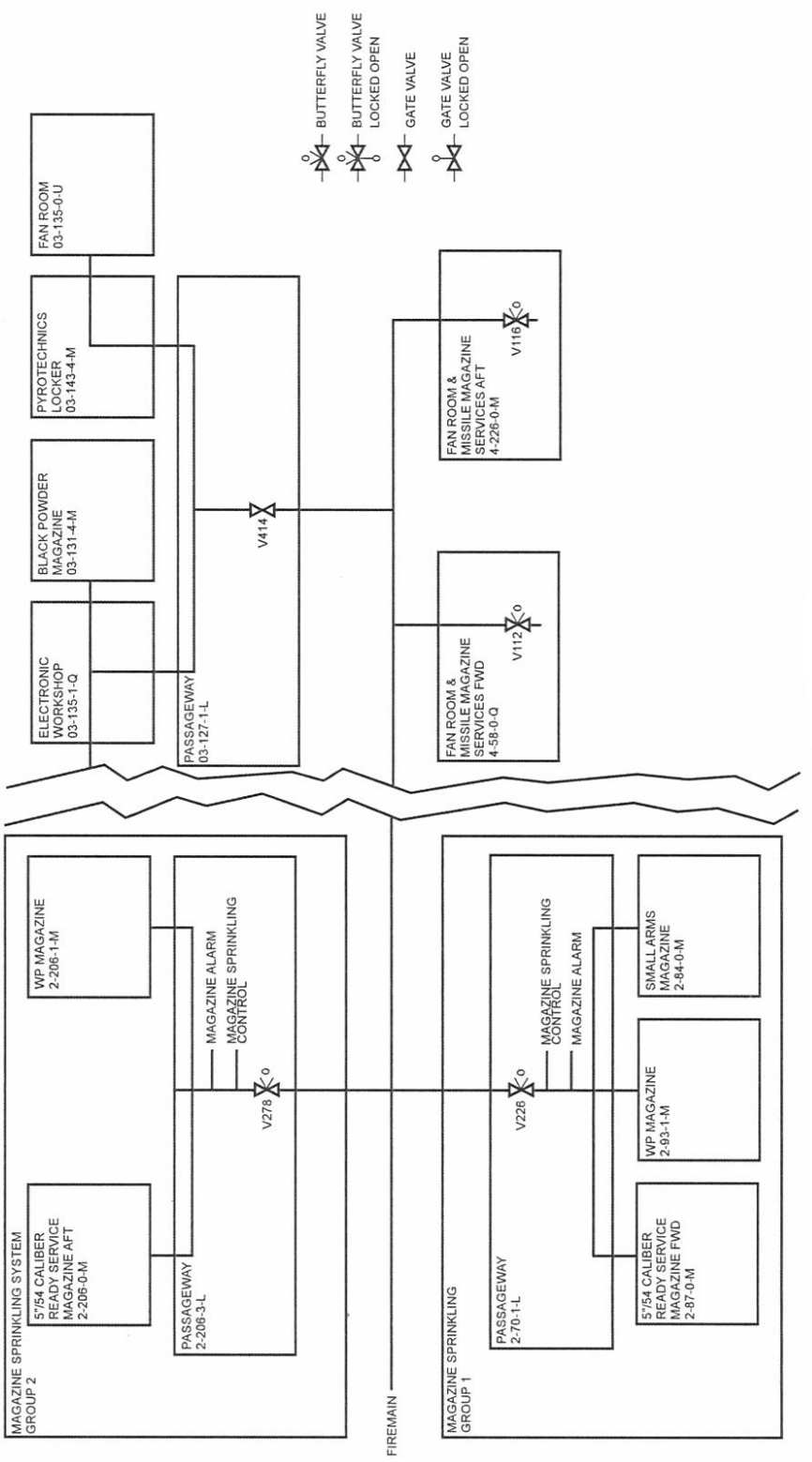
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 25. Support system (own ship heading distribution) block diagram.

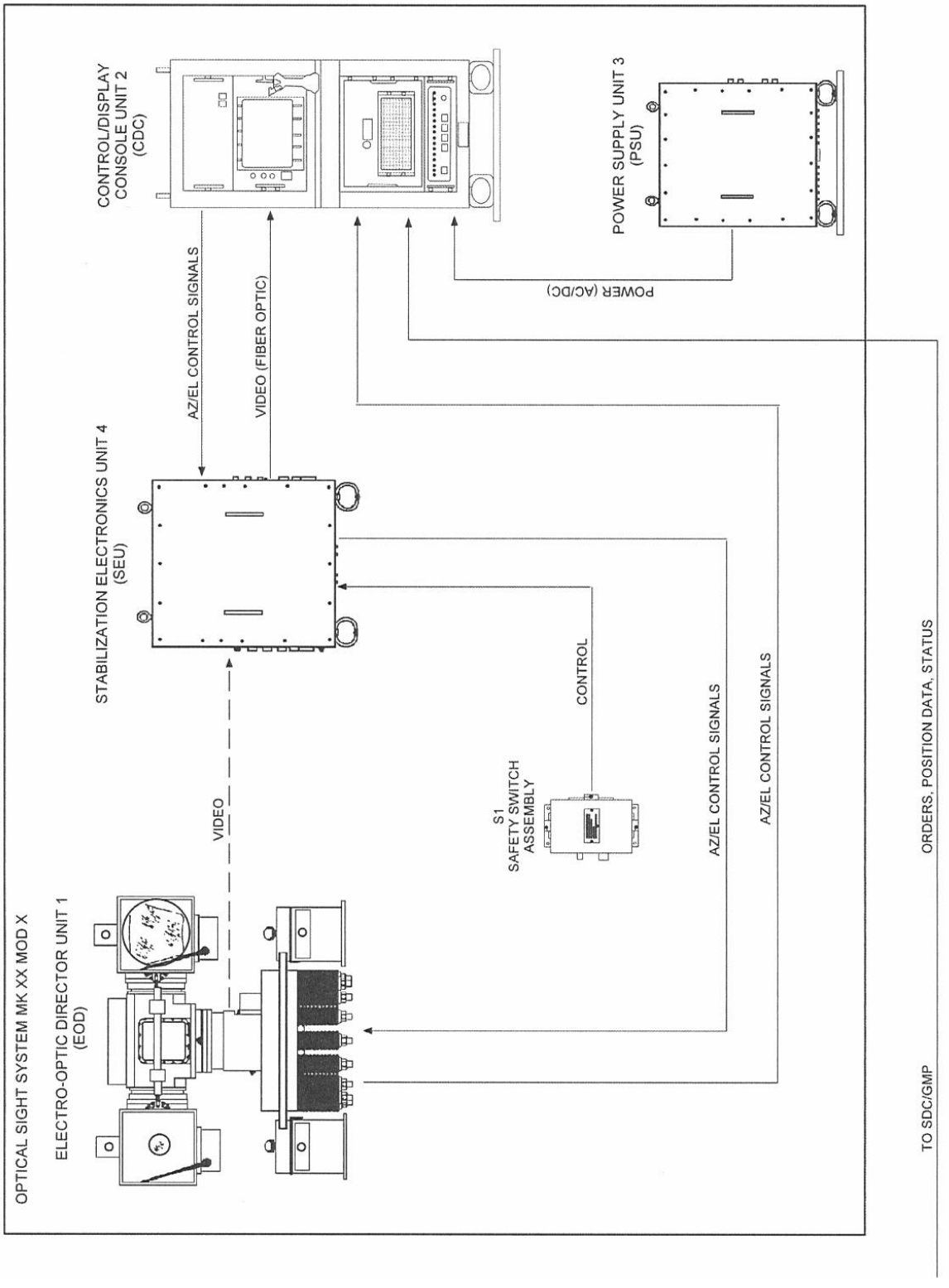
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 26. Support system (coolant, steam, and main fire) block diagram.

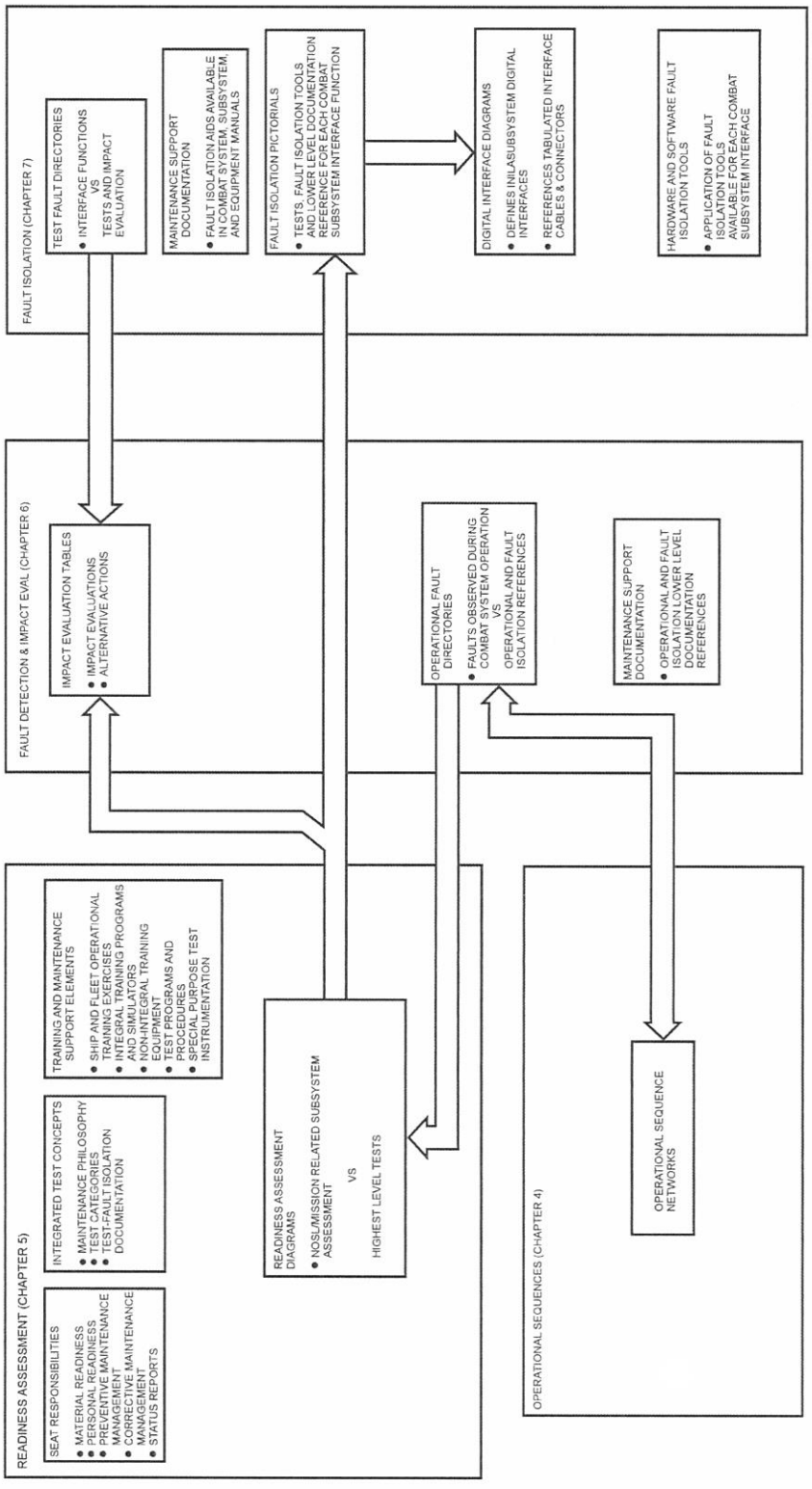
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 27. Functional flow pictorial, level 2.

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FIGURE 28. Readiness assessment and fault isolation tools overview.

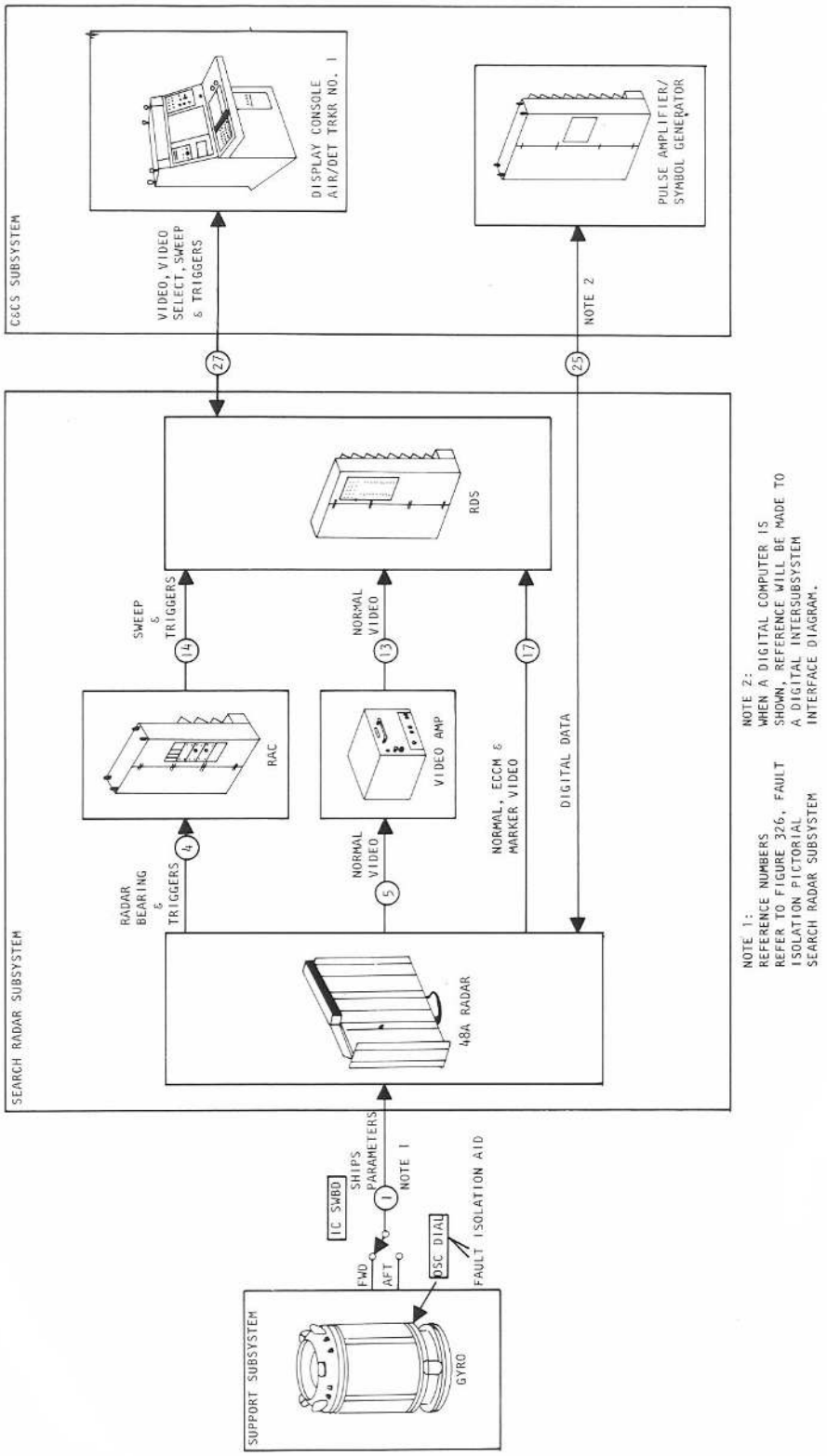
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Int	Function	From		To		Interface Reference
		Equipment	Test	Equipment	Test	
4	Ro	SPS-48A	R81/W-3	RAC	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
	Rmjx	SPS-48A	R81/W-3	RAC	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
	Rss	SPS-48A	R81/W-3	RAC	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
	Sln By 5	SPS-48A	R81/W-3	RAC	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
	Cos By 5	SPS-48A	R81/W-3	RAC	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
5	Yo	SPS-48A	SPS-48 Self Test	Video Amp No. 1	R25/Q-2	MS 0967-1P-014-1140 Vol 3/Pt 2
13	Yo	Video Amp No. 1	R25/Q-2	RDS	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
14	AX, AY	RAC	TD141/R1Q	RDS	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
	Sign AX, AY	RAC	TD141/R1Q	RDS	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
	uiu	RAC	TD141/R1Q	RDS	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2
	End Sweep	RAC	TD141/R1Q	RDS	TD141/R-1Q	MS 0967-1P-014-1140 Vol 3/Pt 2

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 29. Table of interface reference for fault isolation pictorial.

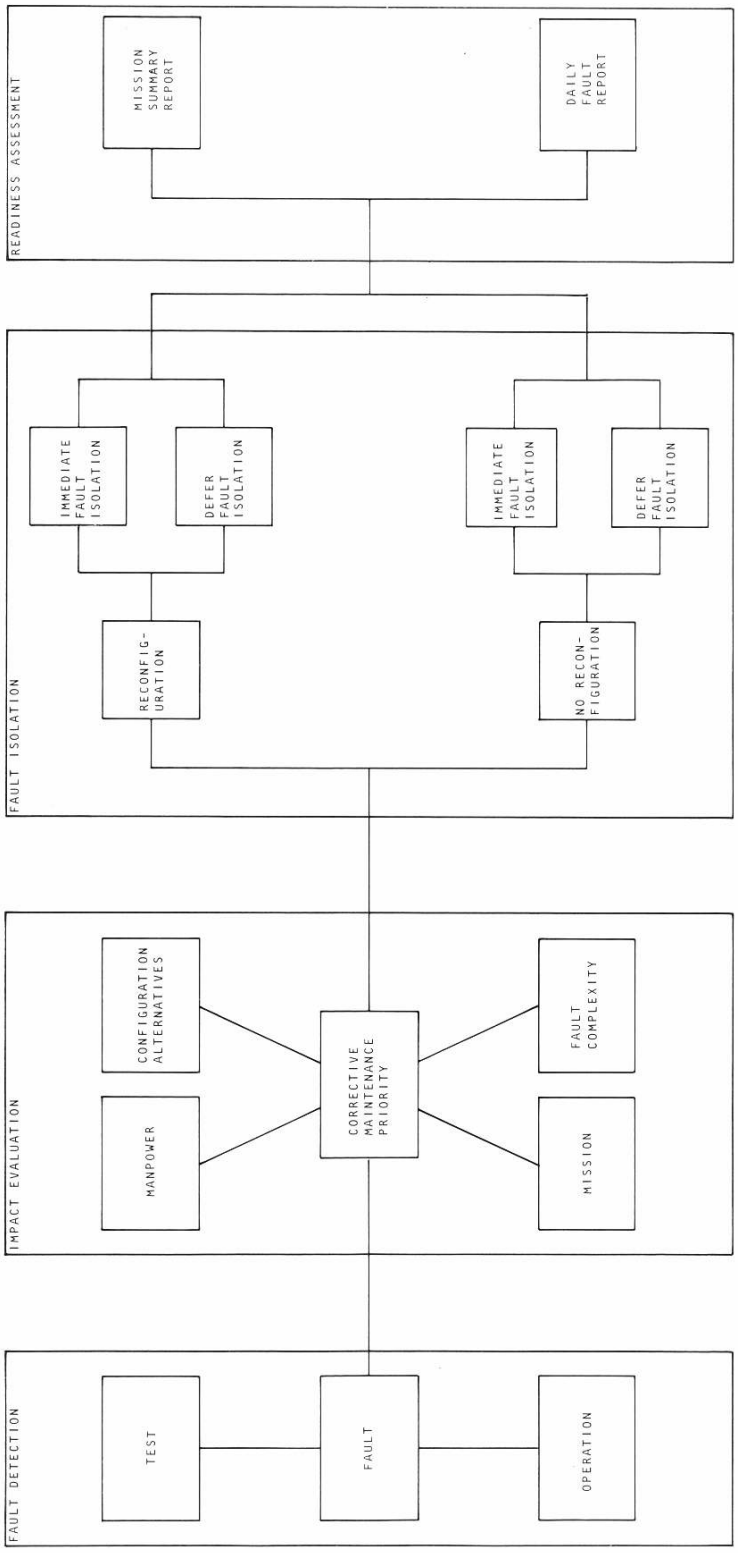
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 30. Fault isolation pictorial.

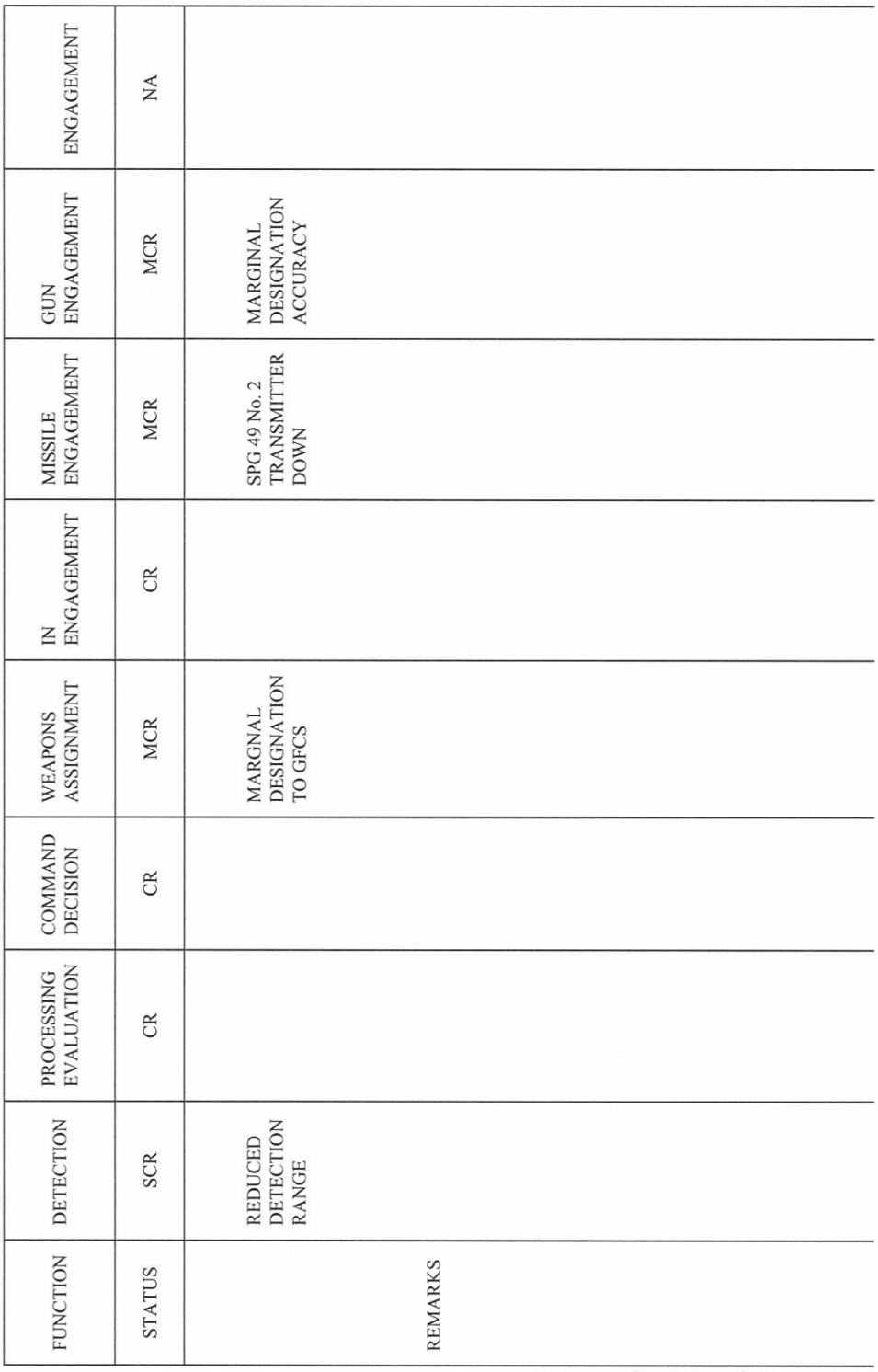
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Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 31. Readiness assessment process diagram.

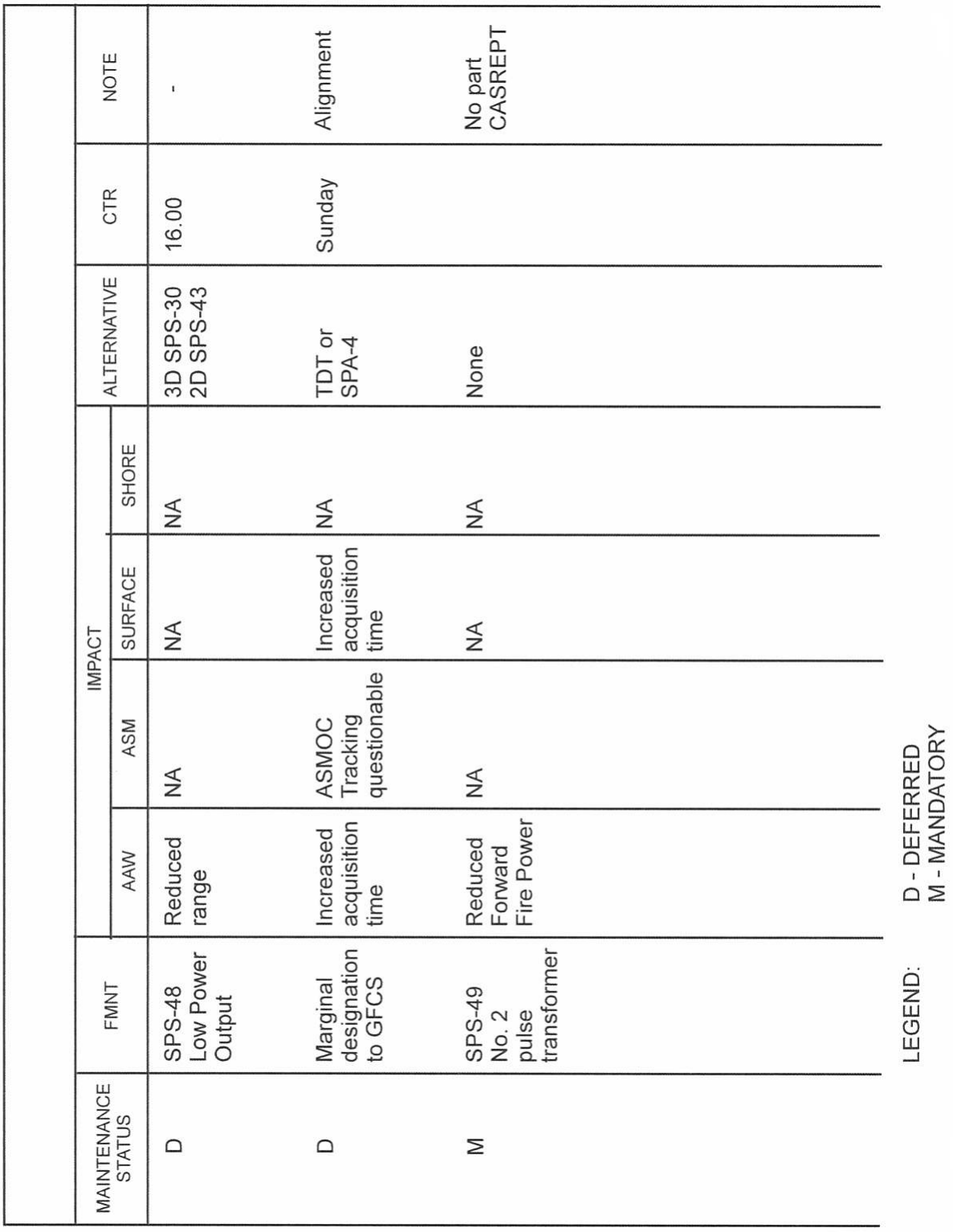
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 32. Mission summary report.

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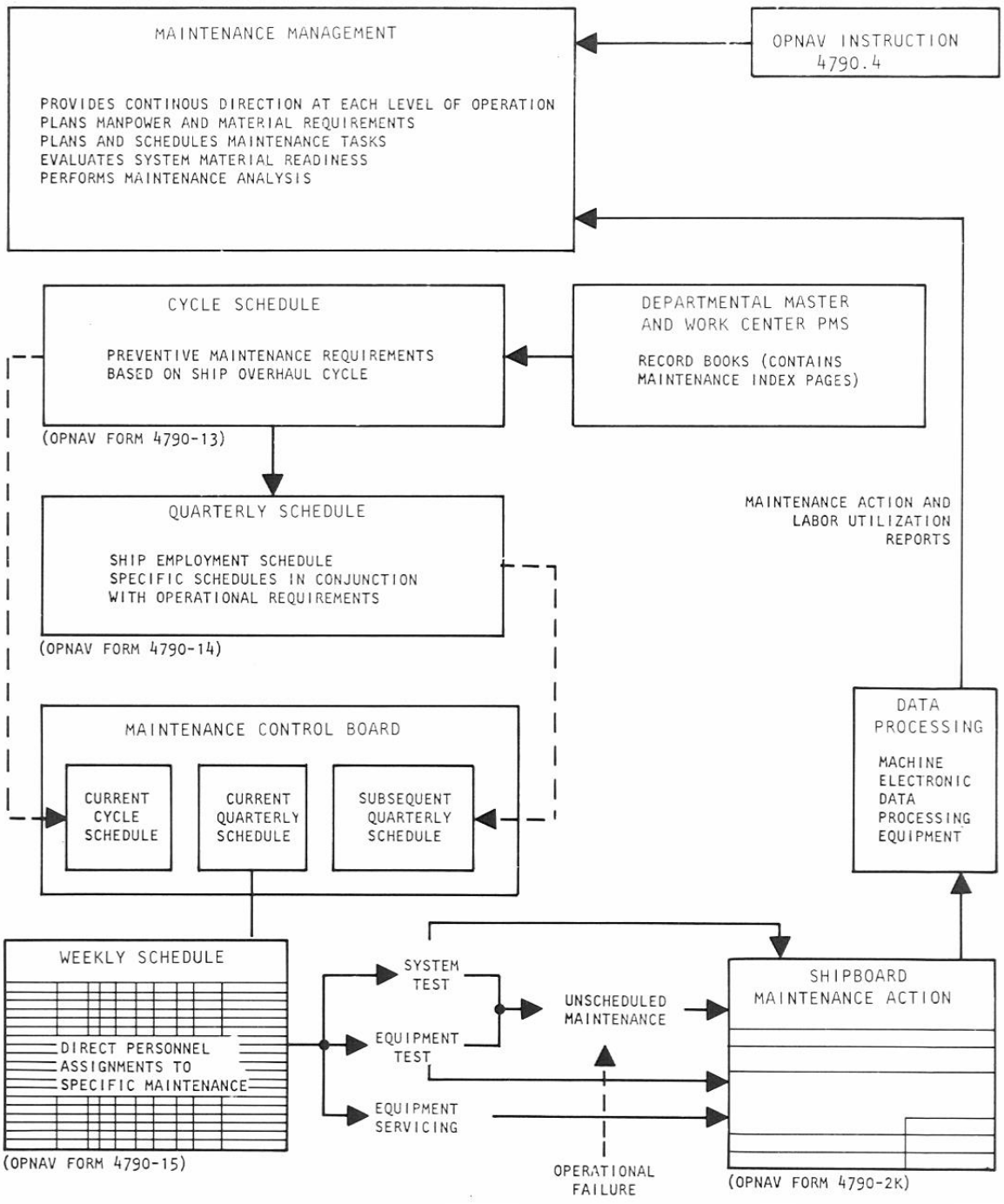
MAINTENANCE STATUS	FMNT	IMPACT				ALTERNATIVE	CTR	NOTE
		AAW	ASM	SURFACE	SHORE			
D	SPS-48 Low Power Output	Reduced range	NA	NA	NA	3D SPS-30 2D SPS-43	16.00	-
D	Marginal designation to GFCS	Increased acquisition time	ASMOC Tracking questionable	Increased acquisition time	NA	TDT or SPA-4	Sunday	Alignment
M	SPS-49 No. 2 pulse transformer	Reduced Forward Fire Power	NA	NA	NA	None		No part CASREPT

LEGEND: D - DEFERRED
M - MANDATORY

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 33. Combat system daily faults report.

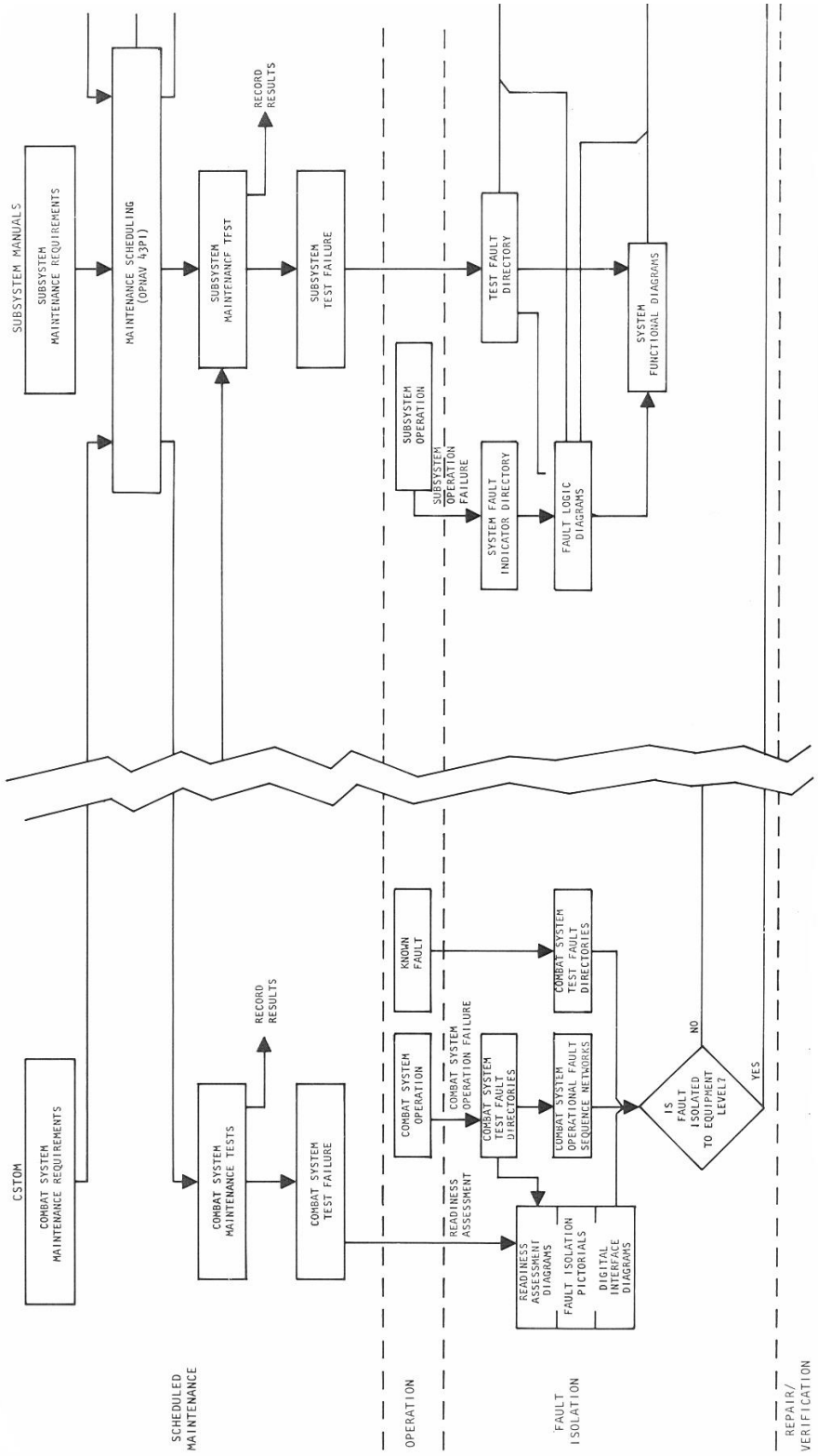
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:34Z
Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 34. Planned maintenance system block diagram.

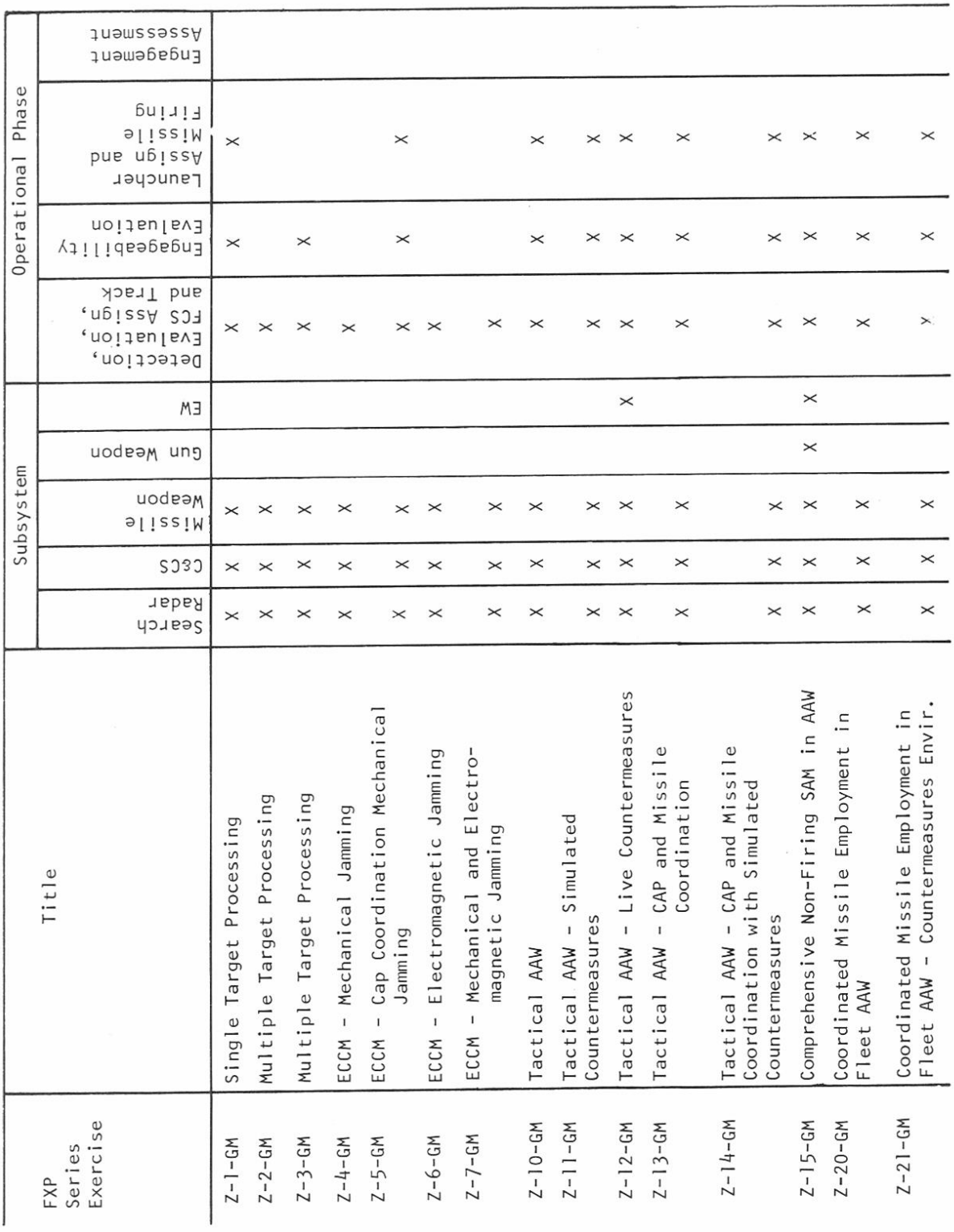
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T02:34Z
Check the source to verify that this is the current version before use.



NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 35. Integrated maintenance concept block diagram.

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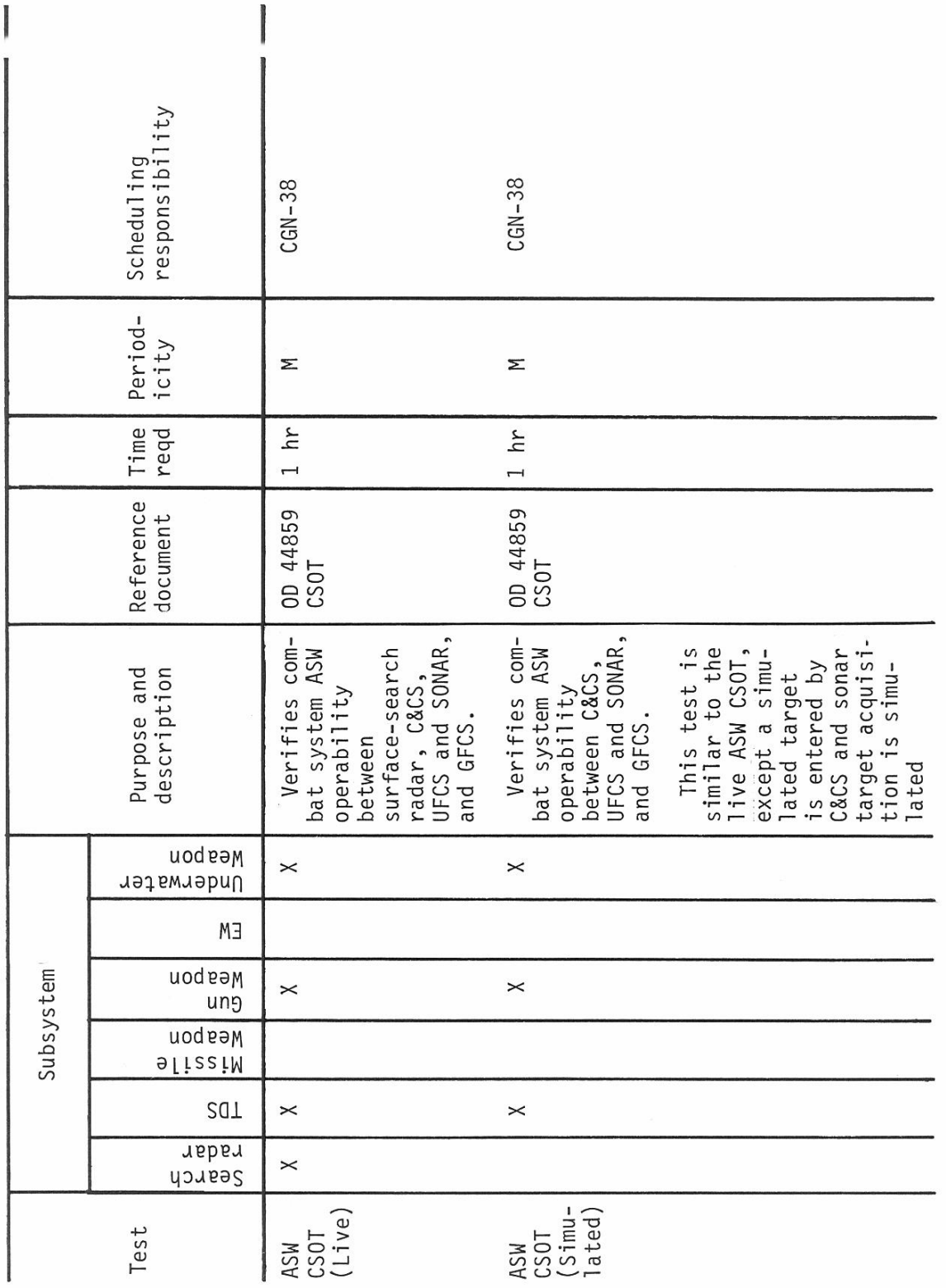


FXP Series Exercise	Title	Subsystem					Operational Phase			
		Search Radar	CECS	Missile Weapon	Gun Weapon	EW	Detection, Evaluation, FCS Assign, and Track	Engageability Evaluation	Launcher Assign and Missile Firing	Engagement Assessment
Z-1-GM	Single Target Processing	X	X	X			X	X	X	
Z-2-GM	Multiple Target Processing	X	X	X			X			
Z-3-GM	Multiple Target Processing	X	X	X			X	X		
Z-4-GM	ECCM - Mechanical Jamming	X	X	X			X			
Z-5-GM	ECCM - Cap Coordination Mechanical Jamming	X	X	X			X	X	X	
Z-6-GM	ECCM - Electromagnetic Jamming	X	X	X			X			
Z-7-GM	ECCM - Mechanical and Electro-magnetic Jamming	X	X	X			X			
Z-10-GM	Tactical AAW	X	X	X			X	X	X	
Z-11-GM	Tactical AAW - Simulated Countermeasures	X	X	X			X	X	X	
Z-12-GM	Tactical AAW - Live Countermeasures	X	X	X		X	X	X	X	
Z-13-GM	Tactical AAW - CAP and Missile Coordination	X	X	X			X	X	X	
Z-14-GM	Tactical AAW - CAP and Missile Coordination with Simulated Countermeasures	X	X	X			X	X	X	
Z-15-GM	Comprehensive Non-Firing SAM in AAW	X	X	X	X	X	X	X	X	
Z-20-GM	Coordinated Missile Employment in Fleet AAW	X	X	X			X	X	X	
Z-21-GM	Coordinated Missile Employment in Fleet AAW - Countermeasures Envir.	X	X	X			X	X	X	

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 36. Table of Fleet training exercises subsystem applicability.

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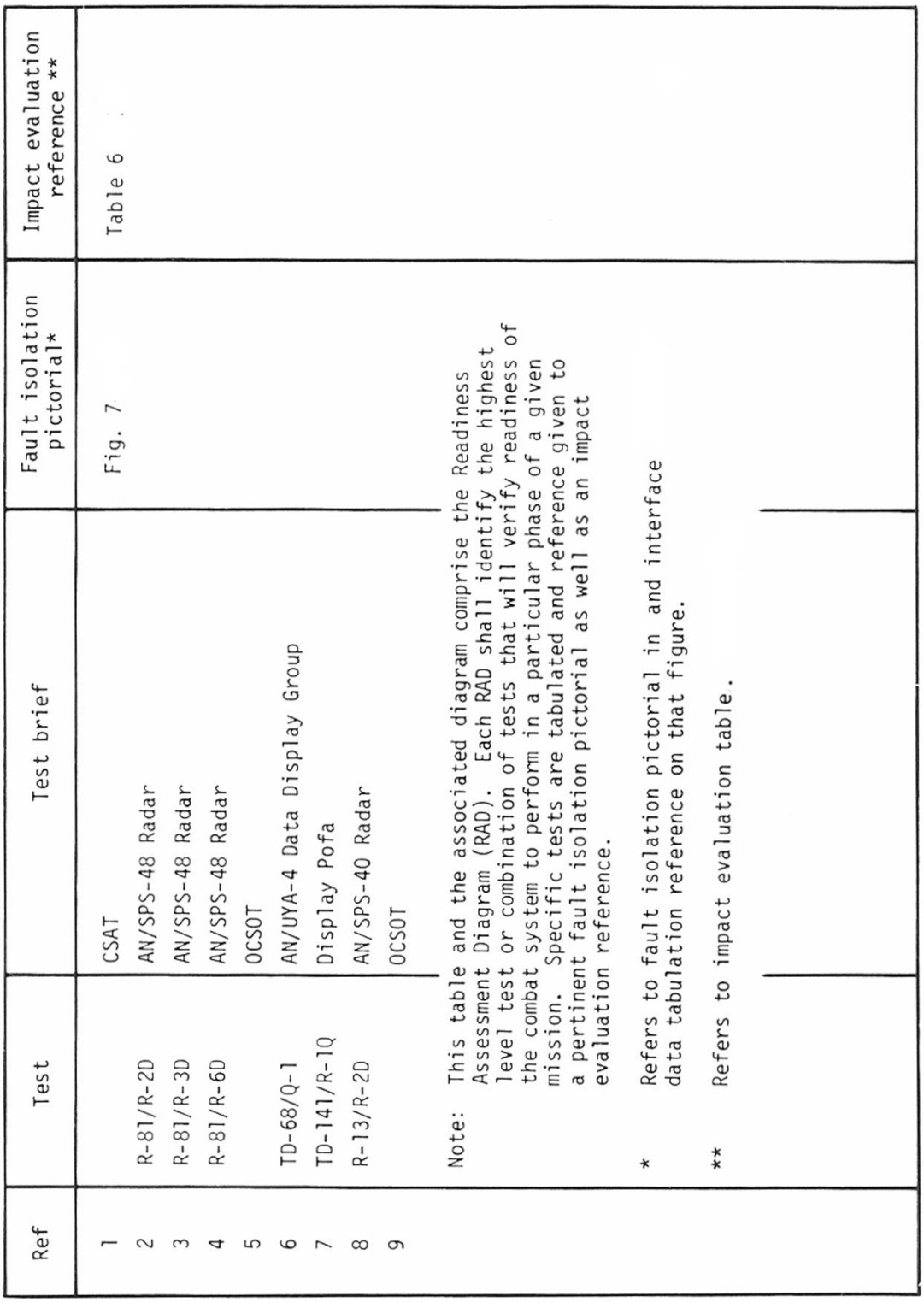


Test	Subsystem: Search radar	TDS	Missile Weapon	Gun Weapon	EW	Underwater Weapon	Purpose and description	Reference document	Time reqd	Periodicity	Scheduling responsibility
ASW CSOT (Live)	X	X		X		X	Verifies combat system ASW operability between surface-search radar, C&CS, UFCS and SONAR, and GFCS.	OD 44859 CSOT	1 hr	M	CGN-38
ASW CSOT (Simulated)		X		X		X	Verifies combat system ASW operability between C&CS, UFCS and SONAR, and GFCS. This test is similar to the live ASW CSOT, except a simulated target is entered by C&CS and sonar target acquisition is simulated	OD 44859 CSOT	1 hr	M	CGN-38

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 37. Table of operational test program.

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Ref	Test	Test brief	Fault isolation pictorial*	Impact evaluation reference **
1		CSAT	Fig. 7	Table 6
2	R-81/R-2D	AN/SPS-48 Radar		
3	R-81/R-3D	AN/SPS-48 Radar		
4	R-81/R-6D	AN/SPS-48 Radar		
5		OCSOT		
6	TD-68/Q-1	AN/UYA-4 Data Display Group		
7	TD-141/R-1Q	Display Pofa		
8	R-13/R-2D	AN/SPS-40 Radar		
9		OCSOT		

Note: This table and the associated diagram comprise the Readiness Assessment Diagram (RAD). Each RAD shall identify the highest level test or combination of tests that will verify readiness of the combat system to perform in a particular phase of a given mission. Specific tests are tabulated and reference given to a pertinent fault isolation pictorial as well as an impact evaluation reference.

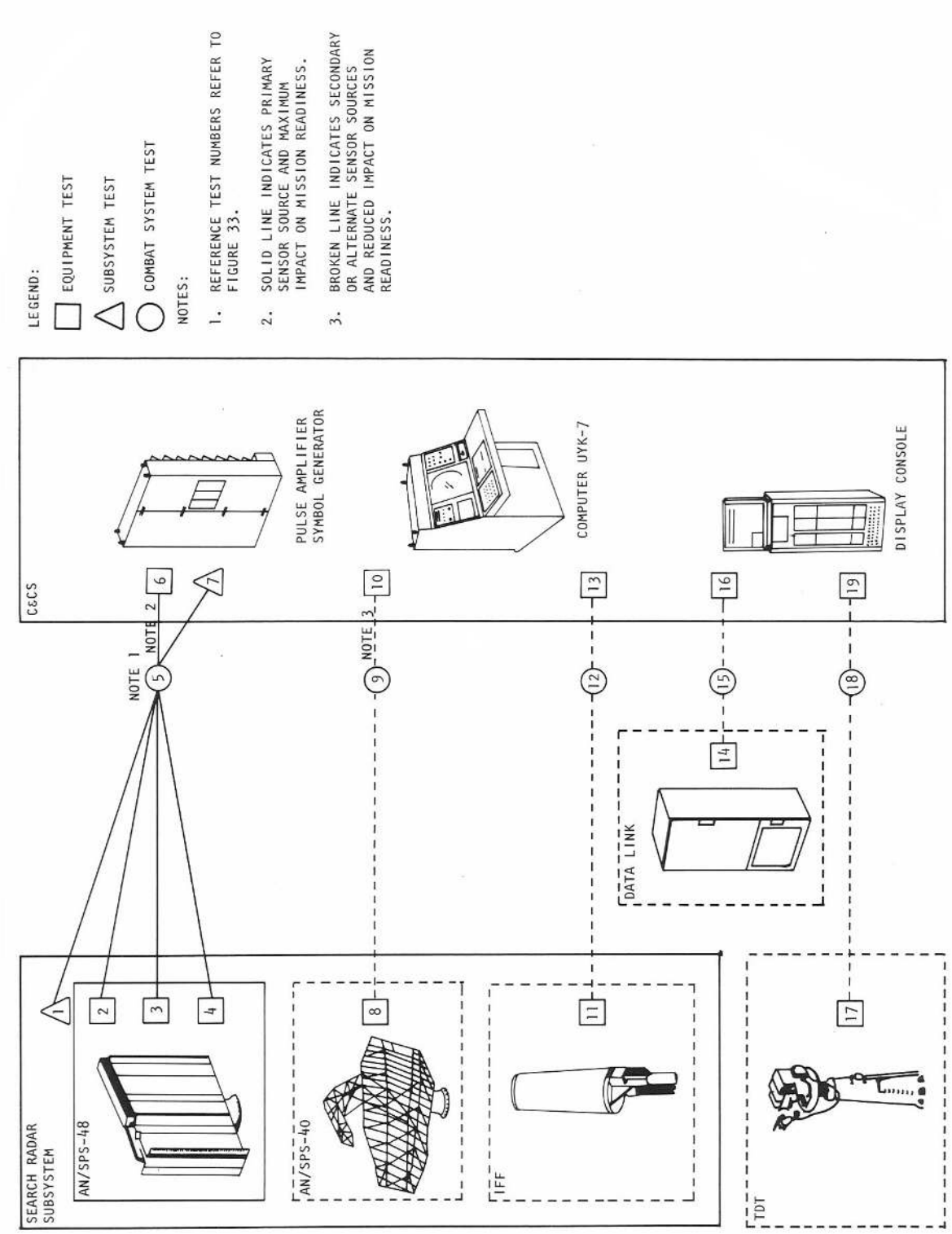
* Refers to fault isolation pictorial in and interface data tabulation reference on that figure.

** Refers to impact evaluation table.

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 38. Table of test and fault impact evaluation references.

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FIGURE 39. Readiness assessment diagram (RAD).

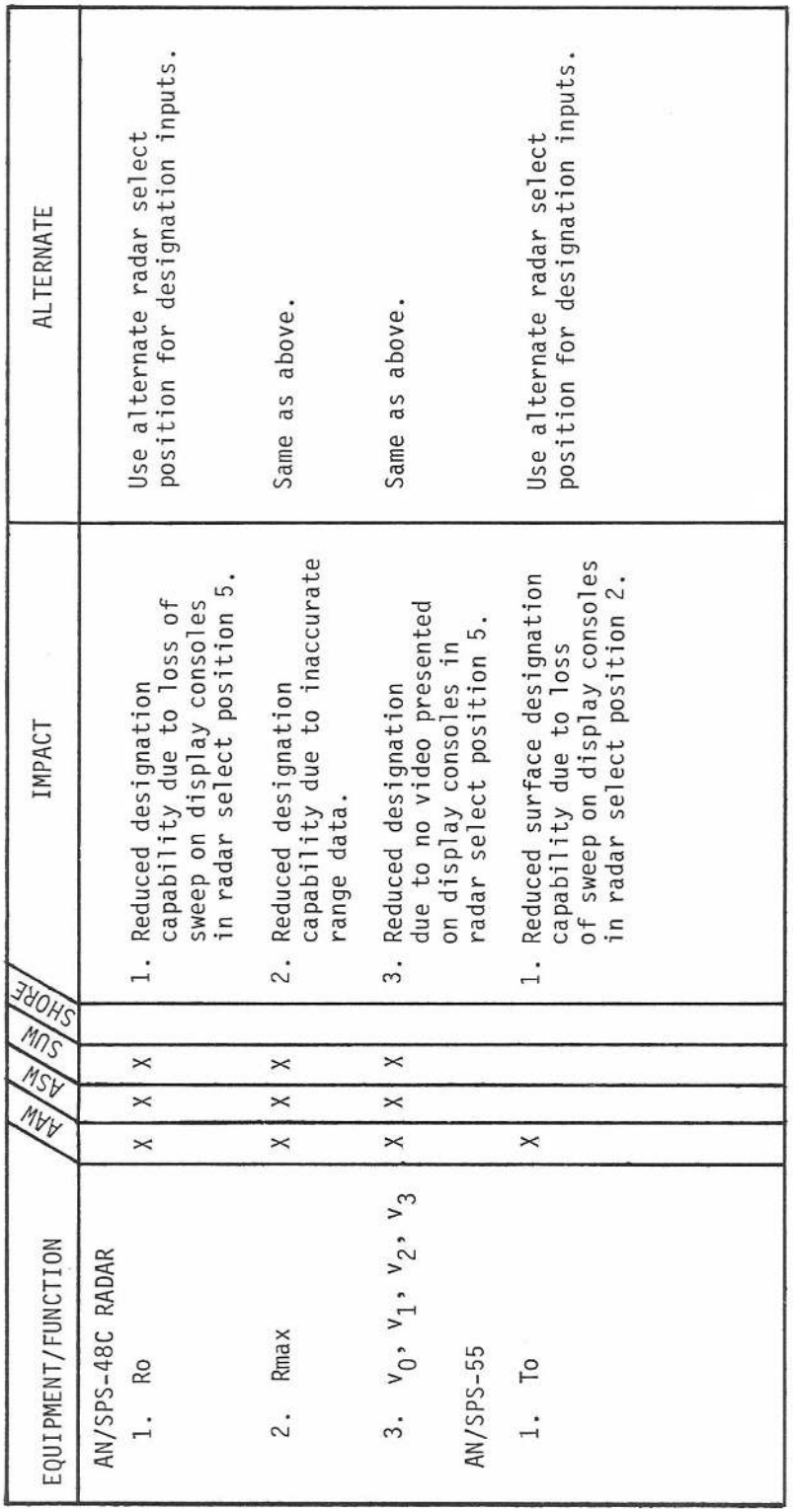
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Fault Symptom	Operational reference				Fault isolation reference						
Note:* F = Figure	*	OSN Vol 2. F4-1.1	OSN Vol 3. F4-1.2	NS0967-LP-014-1160	RAD. F5-5.8	NS0967-441-9010	NS0967-238-7010				
SPS-48C: DETECTOR TRACKER No radar video displayed in RADAR 5 positions.	X		X		X		X				
SPS-48-C: DETECTOR TRACKER No symbology displayed when new track is entered.	X		X		X		X				
SPS-40B: DETECTOR TRACKER No radar video displeyed in RADAR 4 positions.		X	X		X	X	X				
SPS-40B: DETECTOR TRACKER No symbology displayed when new track is entered.		X	X		X	X					
TRACK SUPERVISOR No radar video displayed in RADAR 5 positions.	X		X		X	X					
TRACK SUPERVISOR No symbology displayed when new track is entered.	X	X	X		X						
SPS-48C: RADAR SET CONSOLE Track not on SPS-48C RSC sequence list.	X				X						
TRACK SUPERVISOR No late detect alert on a late detect target.	X	X	X		X	X					

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 40. Operational fault directory.

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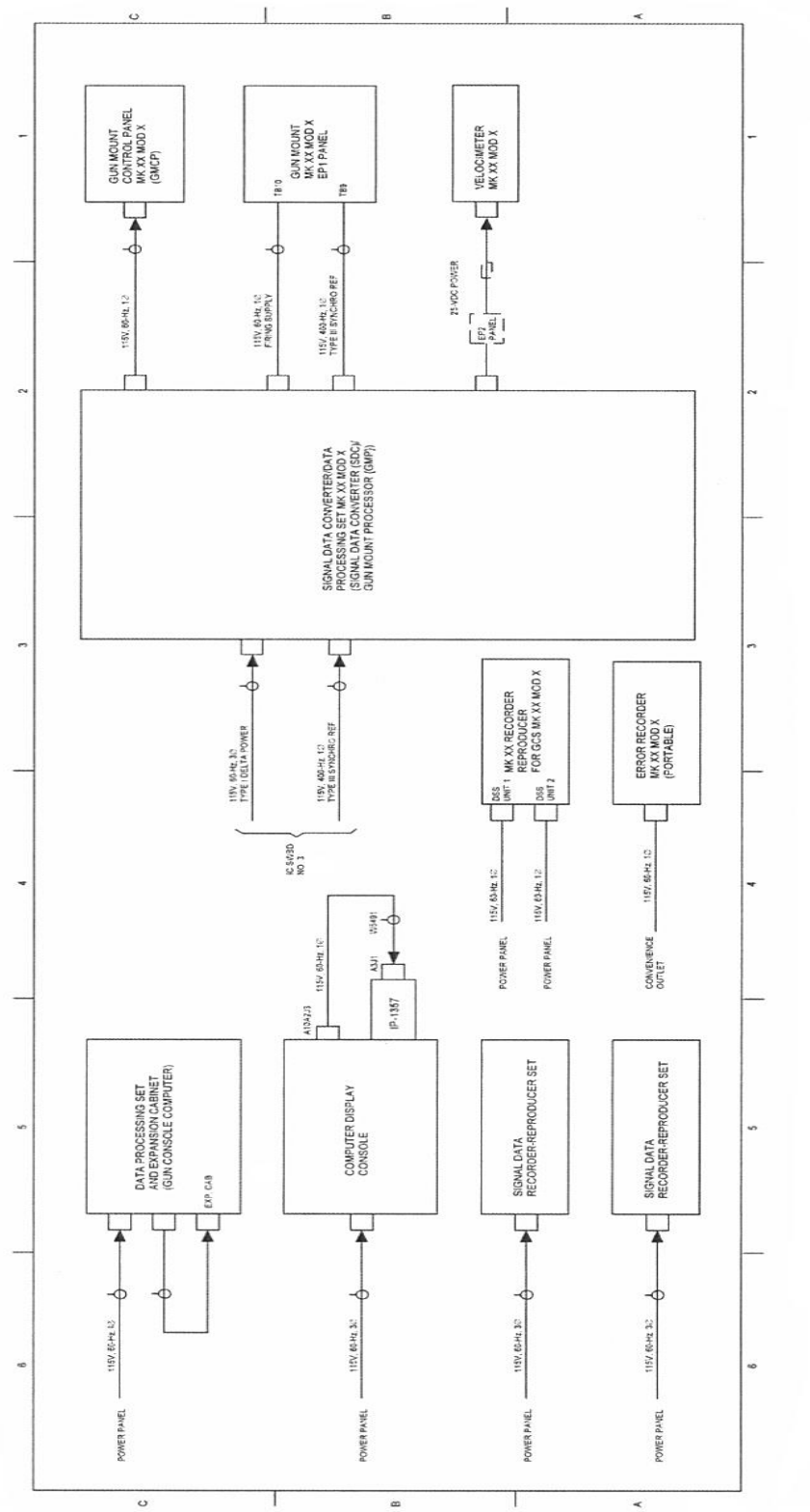


EQUIPMENT/FUNCTION	AAW	ASW	SUW	SHORE	IMPACT	ALTERNATE
AN/SPS-48C RADAR						
1. Ro	X	X X			1. Reduced designation capability due to loss of sweep on display consoles in radar select position 5.	Use alternate radar select position for designation inputs.
2. Rmax	X	X X			2. Reduced designation capability due to inaccurate range data.	Same as above.
3. V_0, V_1, V_2, V_3	X	X X			3. Reduced designation due to no video presented on display consoles in radar select position 5.	Same as above.
AN/SPS-55						
1. To	X				1. Reduced surface designation capability due to loss of sweep on display consoles in radar select position 2.	Use alternate radar select position for designation inputs.

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 41. System impact evaluation table.

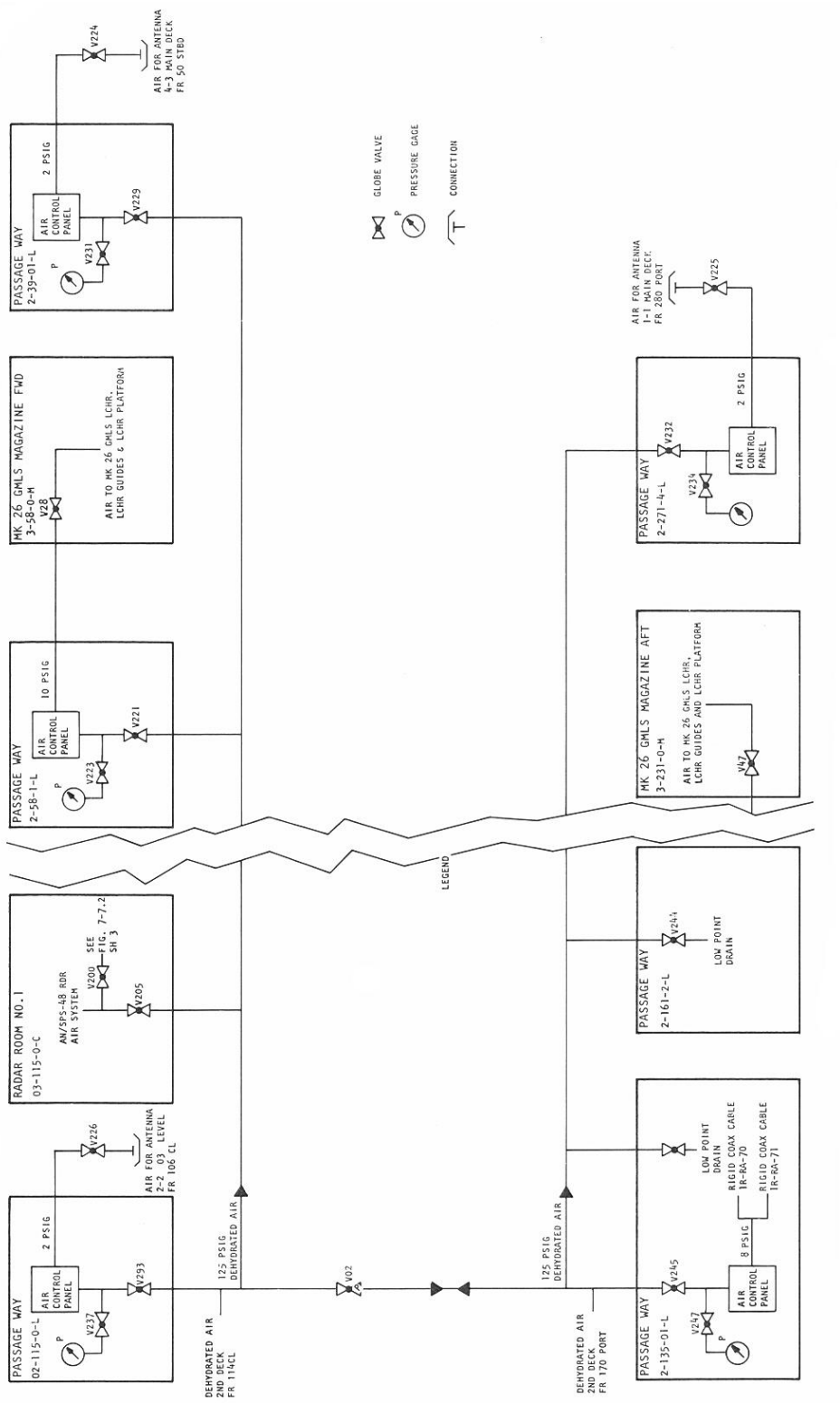
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 42. Support system electrical power distribution diagram.

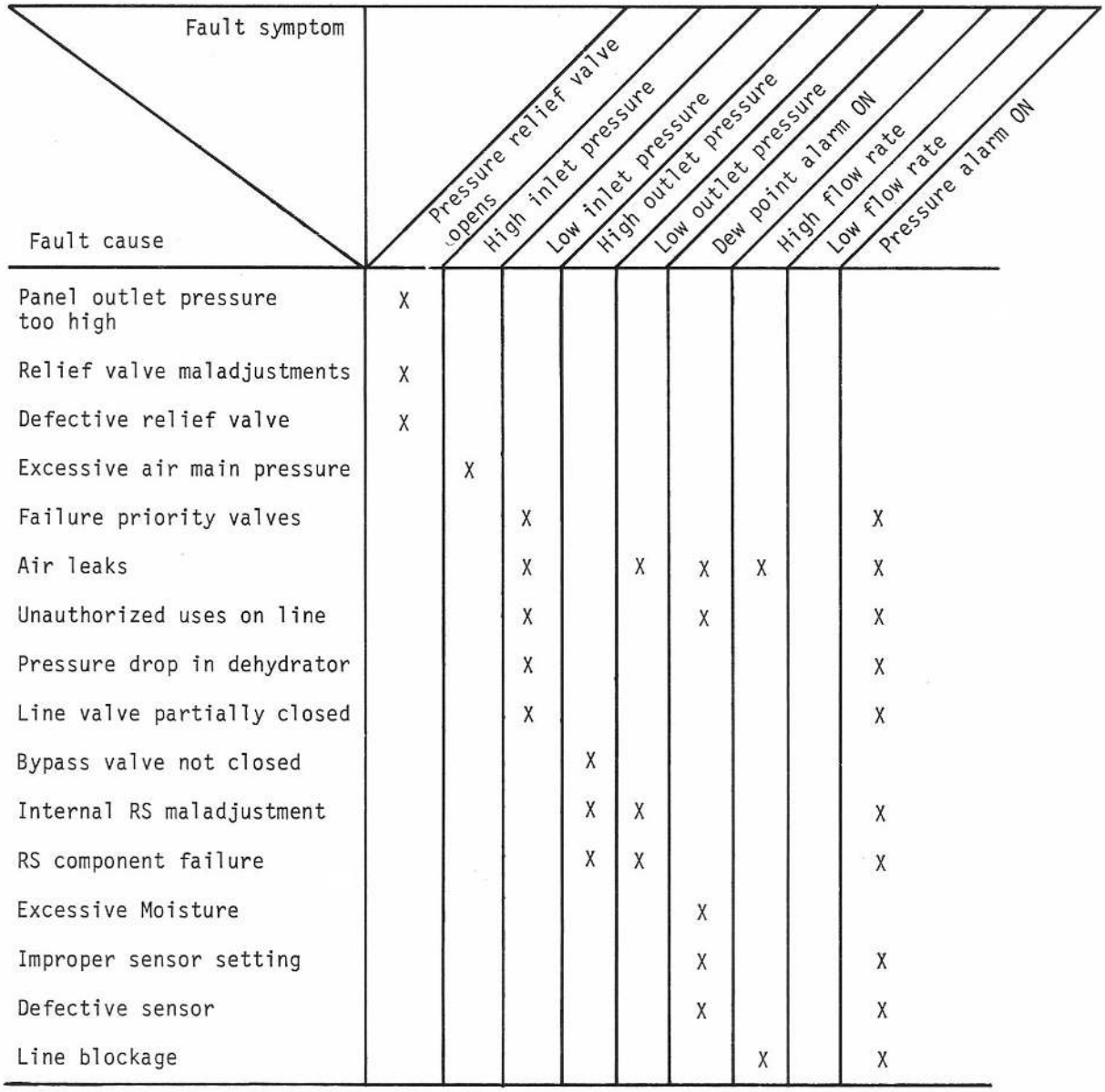
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NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 43. Support system dry air system distribution diagram.

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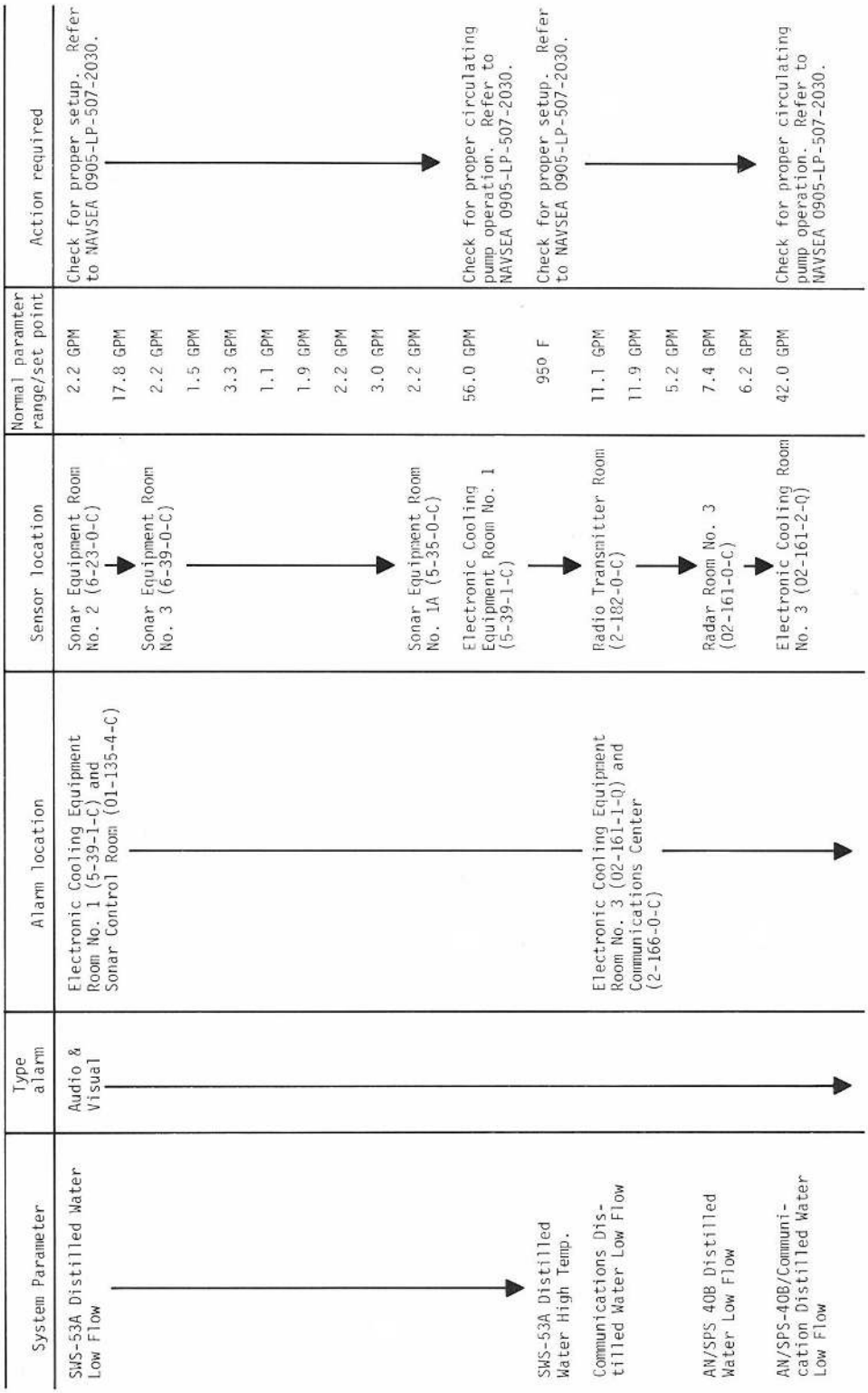


Fault cause \ Fault symptom	Pressure relief valve opens	High inlet pressure	Low inlet pressure	High outlet pressure	Low outlet pressure	Dew point alarm ON	High flow rate	Low flow rate	Pressure alarm ON
Panel outlet pressure too high	X								
Relief valve maladjustments	X								
Defective relief valve	X								
Excessive air main pressure		X							
Failure priority valves			X						X
Air leaks			X		X	X	X		X
Unauthorized uses on line			X			X			X
Pressure drop in dehydrator			X						X
Line valve partially closed			X						X
Bypass valve not closed				X					
Internal RS maladjustment				X	X				X
RS component failure				X	X				X
Excessive Moisture						X			
Improper sensor setting						X			X
Defective sensor						X			X
Line blockage							X		X

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 44. Support system symptom/cause fault directory.

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System Parameter	Type alarm	Alarm location	Sensor location	Normal paramter range/set point	Action required
SWS-53A Distilled Water Low Flow	Audio & Visual	Electronic Cooling Equipment Room No. 1 (5-39-1-C) and Sonar Control Room (01-135-4-C)	Sonar Equipment Room No. 2 (6-23-0-C)	2.2 GPM	Check for proper setup. Refer to NAVSEA 0905-LP-507-2030.
				17.8 GPM	
			Sonar Equipment Room No. 3 (6-39-0-C)	2.2 GPM	
				1.5 GPM	
				3.3 GPM	
				1.1 GPM	
				1.9 GPM	
			Sonar Equipment Room No. 1A (5-35-0-C)	2.2 GPM	
				3.0 GPM	
			Electronic Cooling Equipment Room No. 1 (5-39-1-C)	56.0 GPM	Check for proper circulating pump operation. Refer to NAVSEA 0905-LP-507-2030.
SWS-53A Distilled Water High Temp.				950 F	Check for proper setup. Refer to NAVSEA 0905-LP-507-2030.
Communications Distilled Water Low Flow		Electronic Cooling Equipment Room No. 3 (02-161-1-Q) and Communications Center (2-166-0-C)	Radio Transmitter Room (2-182-0-C)	11.1 GPM	
				11.9 GPM	
			Radar Room No. 3 (02-161-0-C)	5.2 GPM	
				7.4 GPM	
AN/SPS 40B Distilled Water Low Flow				6.2 GPM	
AN/SPS-40B/Communication Distilled Water Low Flow			Electronic Cooling Room No. 3 (02-161-2-Q)	42.0 GPM	Check for proper circulating pump operation. Refer to NAVSEA 0905-LP-507-2030.

NOTE: Sample arrangement only. Size and legibility requirements do not necessarily conform to minimum specification requirements.

FIGURE 45. Table of audio-visual alarms.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
Navy – EC

Preparing activity:
Navy – SH
(Project TMSS-2017-009)

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